Principles of Systematics

# Module# 1

# Principles of Systematics

## Basic Terminologies

# Basic Terminologies

**Taxonomy**

Theory and practice of classification of organisms

- Taxonomy is derived from Greek words "Taxis" which means arrangements and "Nomos" means laws /methods

# Basic Terminologies



Carolus Linnaeus (1707-1778)

## Taxonomy

- Carolus Linnaeus, the Swedish botanist, Zoologist and Physician
- Regarded as the father of modern taxonomy.

# Basic Terminologies

## Contribution of Carolus Linnaeus

His two important contributions to taxonomy were:

- A hierarchical classification system
- The system of binomial nomenclature

# Basic Terminologies

## Carolus Linnaeus

He proposed that there are three broad groups called Kingdoms into which whole nature could fit.

These kingdoms were:

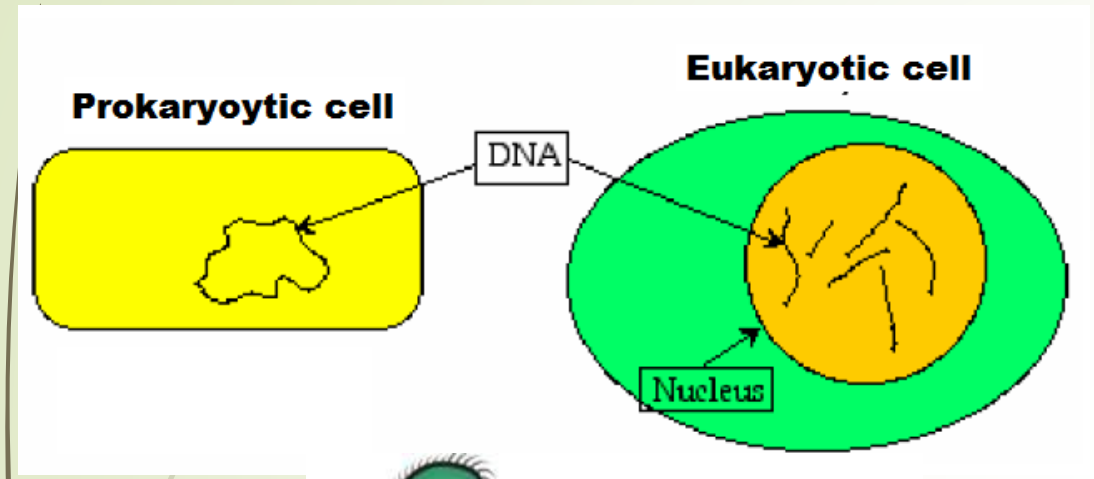
- Animals
- Plants and
- Minerals

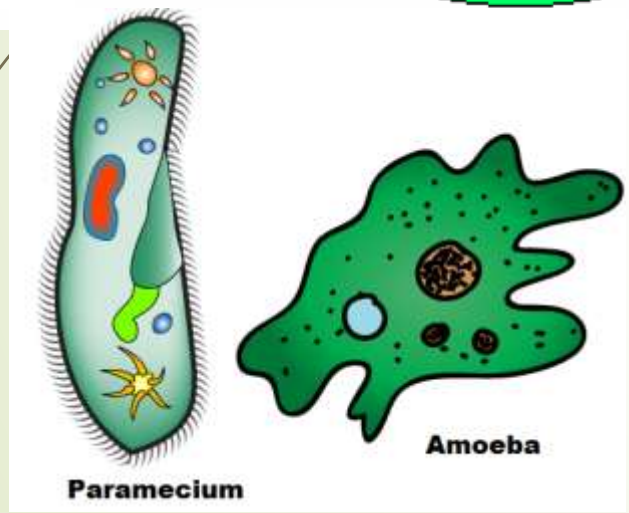
# Basic Terminologies

## Classification

- It is a systematic method of arranging organisms into different groups and subgroups, based on their similarities and their differences.
- It makes the study of organisms very easy and convenient.

# Basic Terminologies




## Classification

- Organisms can be classified on the basis of several different factors:

- Cell nature –Are cells Prokaryotic or Eukaryotic?
- Cells – Do they occur Singly or in Clusters?
- Mode of Nutrition
- Level of Body Organization

# Basic Terminologies



Aristotle
Greek Philosopher
384-322 BC

## Aristotle-Father of Biological Classification

- To understand anything, one must classify it according to its parts
- He established numerous collective categories or genera using differentiating characteristics

# Basic Terminologies



Aristotle
Greek Philosopher
384-322 BC

## Aristotle-Father of Biological Classification

- Blooded and bloodless
- Two-footed vs four footed
- Hairy vs feathered
- With or without an outer shell
- Note: He failed to provide orderly and fully consistent classification of animals.

# Principles of Systematics

## Module# 2

# Basic Terminologies

# Basic Terminologies



Simpson

(1902-1884)

## Systematics

According to Simpson (1961)

- Scientific study of kinds and diversity of organisms and of any or all relationship among them
- Broad term which include Taxonomy, Nomenclature and Classification etc.
- Professor of Biology at San Diego State University, Colombia

# Basic Terminologies

## Aims of systematics

- Invent worlds kinds of organisms (flora and fauna)
- Provide method for identification and communication
- To produce coherent and universal system of classification and
- Demonstrate evolutionary implications of biodiversity

# Basic Terminologies

## Component fields of systematics

- Component fields of systematics are:
  - Biodiversity
  - Taxonomy
  - Classification
  - Nomenclature
  - Biogeography
  - Evolutionary biology

# Basic Terminologies

## Taxon

- According to Simpson (1961) taxon is a group of real organisms recognized as formal unit at any level of hierarchic classification
- Blue birds, song birds, vertebrate refer to groups of organisms.

# Basic Terminologies

## Taxon

- Taxon always refer to concrete Zoological object
- Species of Robins is not a taxon but category
- Robin is a Taxon

END

# Principles of Systematics

## Module# 3

# Contribution of Systematics in Biology

# Contribution of Systematics in Biology

## Systematics contribution

The contribution of systematic to biology can be studied under two heads:-

- Theoretical Biology
- Applied Biology

# Contribution of Systematics in Biology

## Theoratical Biology

- Evolution
- Ecology
- Molecular Biology
- Evolutionary Biology

# Contribution of Systematics in Biology

## Evolution

- Evolution is a process by which plants and animals that exist today have been evolved from plants and animal of past.
- Before the rise of genetics, the study of evolution was carried out entirely by taxonomists.
- Virtually all major evolutionary problems were first pointed out and often

# Contribution of Systematics in Biology

## Evolution

- Taxonomists helps us categorize organisms so we can more easily communicate biological information.
- No other branch of biology has made greater contribution to our understanding  than systematics.

# Contribution of Systematics in Biology

## Ecology

- No through ecological surveys can be conducted without the identification of all species that are of ecological significance.
- Elton (1947) acknowledged the progress in ecology depends upon accurate identifications of species.

# Contribution of Systematics in Biology

## Molecular Biology

- Molecular biologist use genetic data for establishing relationships among different organisms.
- However, they need assistance of traditional taxonomist for accurate results.
- Examples: DNA barcoding and taxonomist

END

# Principles of Systematics

## Module# 4

# Contribution of Systematics in Biology

# Contribution of Systematics in Biology

## Applied Biology

- Agriculture and forestry
- Biological control
- Public health
- Quarantine
- Wild life management

# Contribution of Systematics in Biology

## Applied Biology

- Mineral prospecting
- National defense
- Environmental problem
- Soil fertility
- In commerce

# Contribution of Systematics in Biology

## Agriculture and Forestry

- In order to control and minimize pest attack on crops and diseases caused by vector
- It is necessary to know about correct names of pests
- Taxonomists give us correct identification of pests
- Identification is vital for control

# Contribution of Systematics in Biology



**Biological Control**

- Involved in designing biological control programs of pests
- Determination of exact country of origin and their parasite and parasitoids is compulsory.

# Contribution of Systematics in Biology



*Asian Citrus psylid* *Tamaria radiata*

**(Hoddle & Pandey, 2014)**

- Asian citrus psylid is serious pest of citrus orchards
- Its origin is Asia
- *Tamarixia radiata,* a parasitic wasp was introduced from Pakistan to California to control this pest.
- Best example of classical biological control.

# Contribution of Systematics in Biology

## Public Health

- Number of diseases are spread by Arthropods which are controlled by targeting  specific species.
- For example: All *Anopheles maculipennis* are not responsible for transmitting malaria.
- Out six siblings some are involved in transmission of malaria.

# Contribution of Systematics in Biology

## Public Health

- Although the species are virtually indistinguishable morphologically, they are isolated reproductively, in part because they breed in different habitats.
- Taxonomists help to identify this particular sibling species for effective control at minimum cost.

END

# Principles of Systematics

## Module# 5

# Contribution of Systematics in Biology

# Contribution of Systematics in Biology

## Quarantine

- Many diseases are spread from one country to other via transportation.
- Respective Governments established quarantine labs at aerodromes and ports etc
- Taxonomists help in prompt identification of these

# Contribution of Systematics in Biology

## Wild Life management

- The indiscriminate killing and felling of trees create disturbance in natural environment.
- It help environmental protectors by identifying economically and ecologically important wild life.
- It is important for preservation and

# Contribution of Systematics in Biology

## Mineral Prospecting

- The identification of fauna and flora by paleontologists in sedimentary rocks show the sequence of geological events
- Which help in search for fuels and mineral deposits.
- Example: It attain great success in American industrialization

# Contribution of Systematics in Biology



**National Defense**

- During World War II (On November 3, 1944), **Japan** released **balloon bombs**, into the Pacific jet stream
- Created havoc in the forest of north east America.
- Balloons covered with sand contained a large number of shells of micro-organisms.
- Bombing of this beach area destroyed the balloon launching site.

# Contribution of Systematics in Biology

## Detection of Environmental Problems

- Phytoplankton and zooplankton are sensitive to changes in water quality.
- They are good indicators for changes in water quality as they respond quickly to changes in nutrient input.
- Taxonomists help to detect such problem by identifying specific species.

# Contribution of Systematics in Biology




**Soil Fertility**

- Taxonomist help to detect organisms that enhance soil fertility
- For example bacteria, earthworms etc.

# Contribution of Systematics in Biology

## In commerce

- Systematics play role in increasing and improving the qualities of commercially used animal products such as:
  - Honey
  - Silk
  - Lac
  - Dyes etc

END

# Principles of Systematics

## Module# 6

# Principles of Systematics

## Binomial Nomenclature

# Binomial Nomenclature

- Scientific names of species is a combination of two names:
  - Generic name
  - Species name

# Binomial Nomenclature



- Spider *Oxyopes javanus* is found in agro-ecosystems.
- The first part of a scientific name, *Oxyopes, is called the genus.*
- The second part of a scientific name, *javanus* is in this example, is the species name

# Binomial Nomenclature



- *Hottentotta tamulus*, the Indian red scorpion also known as the eastern Indian scorpion, is a species of scorpion of the family Buthidae.
- It occurs in most of India, eastern Pakistan and the eastern lowlands of Nepal, and recently from Sri Lanka.

# Binomial Nomenclature



https://en.wikipedia.org/wiki/Hoplobatrachus_tigerinus#/media/File:HoplobatrachusTigerinus.jpg

- *Rana tagrina* is an amphibian which is solitary and nocturnal in nature.
- They inhabit holes and bushes near water source.
- They do not stay in water for longer time.
- They belong to order anura.

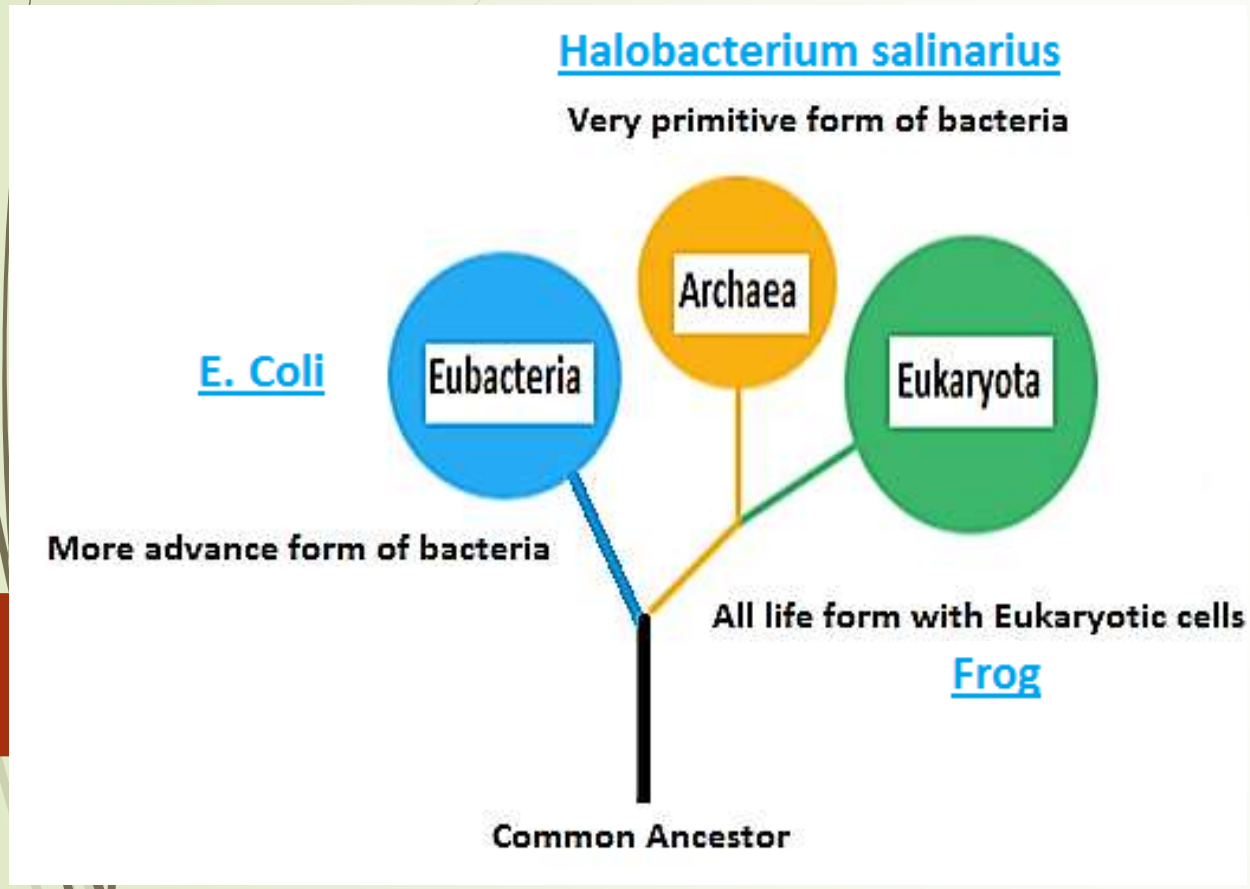
# Principles of Systematics

## Module# 7

# Three Domain system

**Archaea**
**Eubacteria**
**Eukaryota**

# Three Domain system




- A domain is a broadest taxonomic category.
- All organism belong to one of the three domains, depending upon their characteristics.
- A single domain can contain one or more kingdoms.
- Introduced by Charl Woese et al. In 1990.

# Three Domain system

## Domain of Viruses

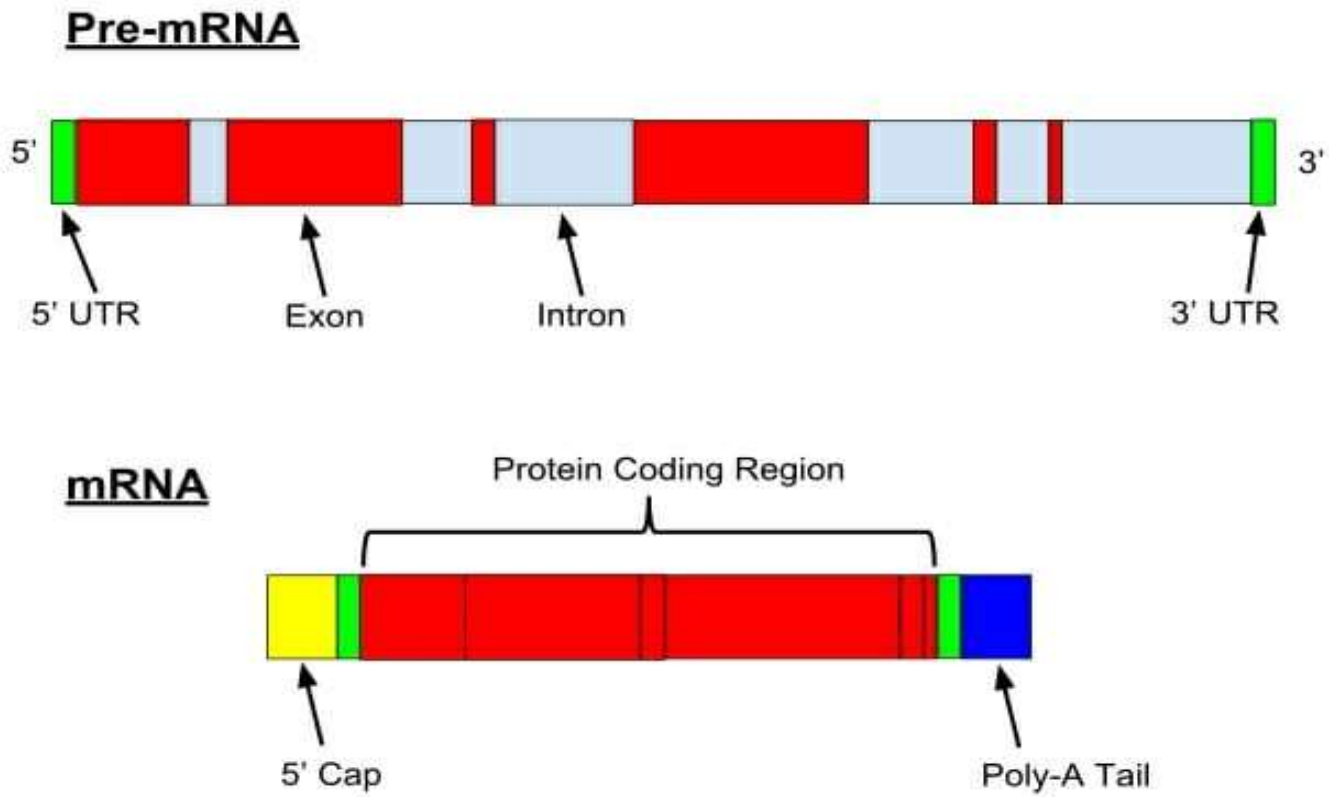
- Viruses are DNA and RNA shielded by protein coat called capsid.
- Viruses do not have a domain

# Three Domain system

Similarities and differences of Eubacteria and Archaea

| Characteristics | Eubacteria | Archaea |
|---|---|---|
| Cell type | Prokaryotic | Prokaryotic |
| Cell wall | Contain peptidoglycan | Carbohydrates, proteins or No cell wall |
| Interons | No interons | Interons present |
| Ribosomal proteins | No ribosomal proteins | Ribosomal proteins present |
| Diseases | Can be parasitic and cause diseases | None are parasitic |
| Gene resemblance | No resemblance | Eukaryotes |

# Three Domain system




https://biologydictionary.net/intron/

# Principles of Systematics

## Module# 8

# Principles of Systematics

## Five Kingdom Classification

# Five Kingdom Classification

## Two Kingdom classification

- Plants (Autotrophs)
- Animals (Hetrotrophs )

## Problem

- Euglena

## Solution by Ernst Hackel (1866)

- Protista

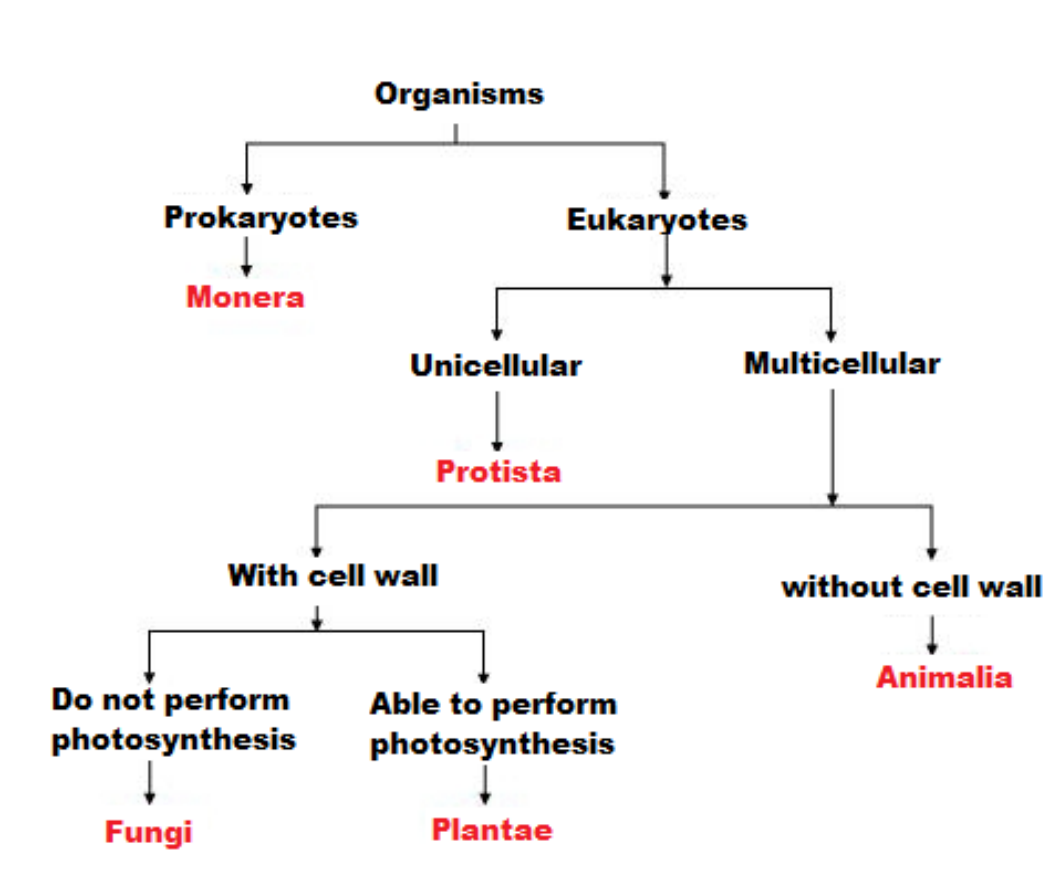
# Five Kingdom Classification

- E- Chatton in 1937 introduced the term "Procariotique" to describe bacteria and "Eucariotique" to describe animals and plant cells.
- Five kingdom classification was proposed by Robert Whittaker in 1967.

# Five Kingdom Classification

- Morgulis and Schwatz in 1988 modified R. Whittaker classification.
- They considered genetics along with cellular organization and mode of nutrition in classification.

# Five Kingdom Classification




- Five kingdoms proposed by Robert Whittaker include:
  - Monera
  - Protista
  - Fungi
  - Plantae
  - Animalia

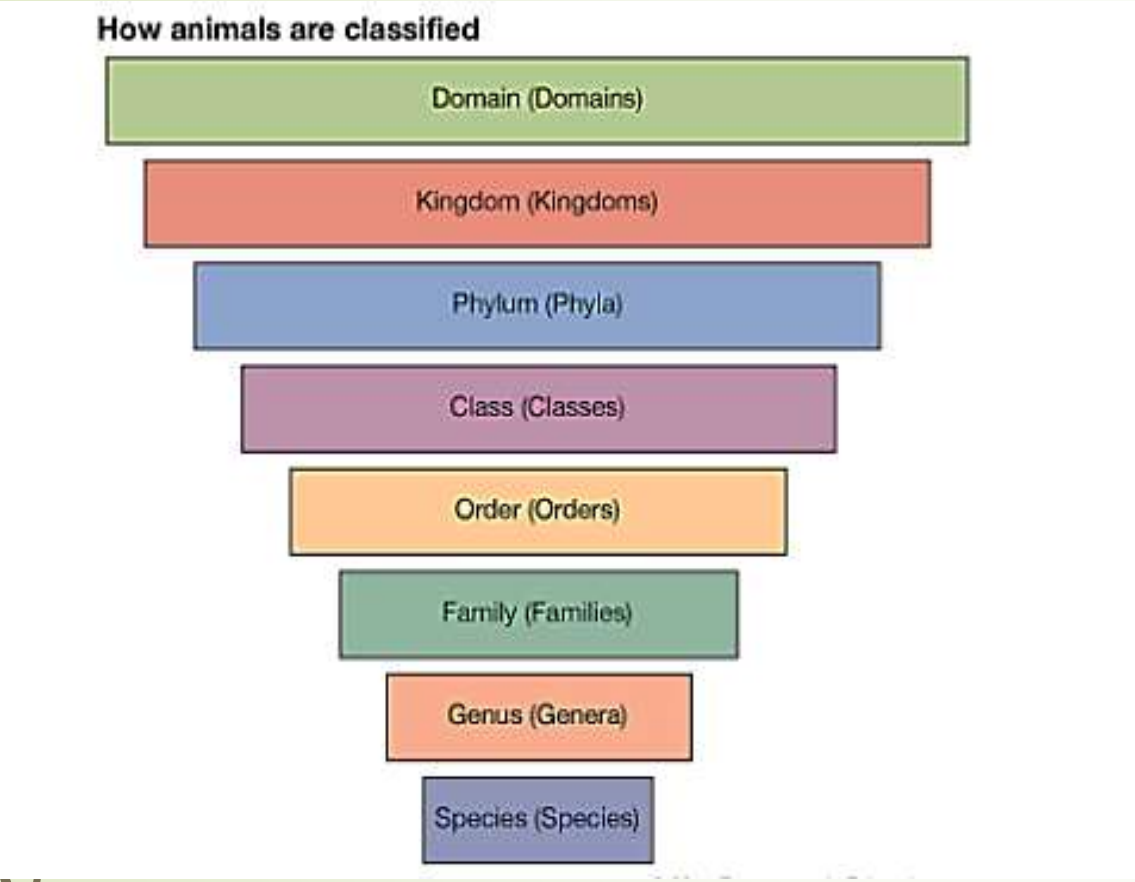
# Principles of Systematics

## Module# 9

# Principles of Systematics

## History of Taxonomy

# History of Taxonomy



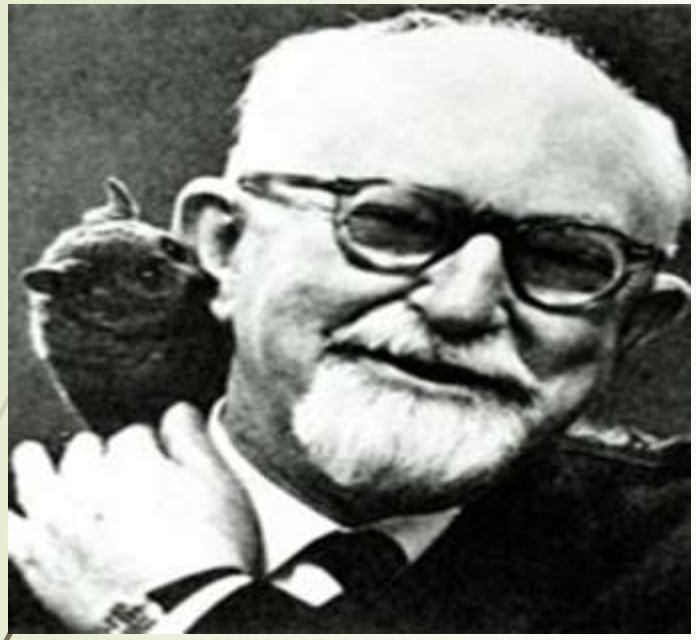

https://Britannica.com/science/taxonomy

- Taxonomy is as old as the language skill of mankind.
- Taxonomy is the science of naming, describing and classifying organisms including all plants, animals and microorganisms of the world.
- Taxonomy identifies and enumerates the components of biological diversity.

# History of Taxonomy

- Native tribes used to call the species on the basis of their morphological characters
- Development of scientific theory of classification ….a remarkably recent phenomenon

# History of Taxonomy



**George Gaylord Simpson (1902-1984)**

https://www.google.com/search?q=george+gaylord+simpson+pic&sxsrf=ALeKk016UQ5865EIGFEM0Awryp5IHsxm-Q:1613054410489&source=lnms&tbm=isch&sa=X&ved=2ahUKEwilsI_5hLuAhUJnlkKHYD5DkgQ_AUoAXoECA8QAw&biw=1366&bih=625#imgrc=siNionRQ1ItxWM

- Simpson in 1961 gives a valuable survey of History of Taxonomy.
- Simpson was the most influential paleontologists of the twentieth century.
- He published extensively on the taxonomy of fossil and extinct mammals.

# History of Taxonomy



**Frederick Smith 1805-1879**

https://www.google.com/search?q=frederick+smith+pic+pic+taxonomy&tbm=isch&ved=2ahUKEwiSPCJiuLuAhUP_BoKHfOBCwsQ2cCegQIABAA&oq=frederick+smith+pic+pic+taxonomy&gs_lcp=CgNpbWcQA1CQEFj1KWCxLWgAcAB4AIABjQOIAegXkgEFMi0xLjiYAQCgAQGqAQtnd3Mtd2l6LWItZ8ABAQ&sclient=img&ei=BkQIYJLdHo_4a_ODrlg&bih=625&biw=1366#imgrc=k331uXCLcv1lsM

- Frederick Smith worked in the zoology department of British Museum from 1849.
- He published from 1851 to 1870 on ant taxonomy.
- During the 28 years of his activity he describe 25 genera, 702 species and subspecies of whic hare.

# History of Taxonomy



**Thomas E. Mittler (1928-2012)**

https://www.google.com/search?q=thomas+E.Mittler+pic+pic+taxonomy&tbm=isch&ved=2ahUKEwiSPCJiuLuAhUP_BoKHfOBCwsQ2cCegQIABAA&oq=frederick+smith+pic+pic+taxonomy&gs_lcp=CgNpbWcQA1CQEFj1KWCxLWgAcAB4AIABjQOIAegXkgEFMi0xLjiYAQCgAQGqAQtnd3Mtd2I6LWItZ8ABAQ&sclient=img&ei=BkQIYJLdHo_4a_ODrlg&bih=625&biw=1366#imgrc=k331uXCLcv1lsM

- Thomas E.Mittler was an Austrian entomologist.
- He worked on history of higher taxonomy and also on the classification of insects

# History of Taxonomy



**Ernst Mayr (1904- 2005)**

https://www.google.com/search?q=ernst+mayr+pic+taxonomy+&tbm=isch&ved=2ahUKEwjs6e2JjuLuAhXBwIUKHfhfAU0Q2cCegQIABAA&oq=ernst+mayr+pic+taxonomy+&gs_lcp=CgNpbWcQA1D6IlibLmD8MmgAcAB4AIABjgSIAcgXkgEHMy0zLjMuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=OEglYOztC8GBlwT4v4XoBA&bih=625&biw=1366#imgrc=PdnA5xdNdrOdwM//

- Mayr in 1982 provided a details of various periods of History of Taxonomy
- His work contributed to the conceptual revolution that led to modern evolutionary synthesis of taxonomy and systematics

# Principles of Systematics

## Module# 10

# Principles of Systematics

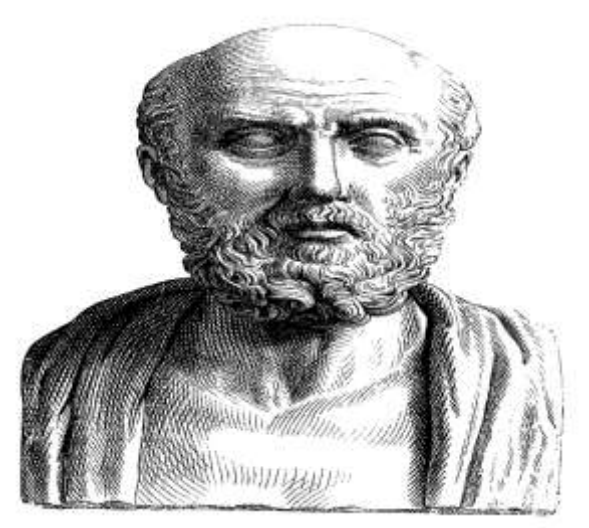
## History of Taxonomy

# History of Taxonomy

- Mayr described the History of Taxonomy into four periods:

- 1st Period
- 2nd Period
- 3rd Period
- 4th Period

# History of Taxonomy



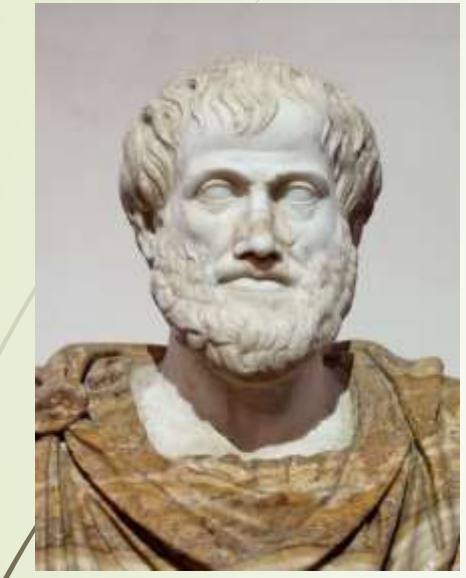
**Hippocrates (460-377 B.C.)**

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1st Period (Study of local fauna)

- Several early Greek scholars notably Hippocrates enumerated the types of animals But he did not classify.

# History of Taxonomy



**Aristotle (384-322 B.C.)**

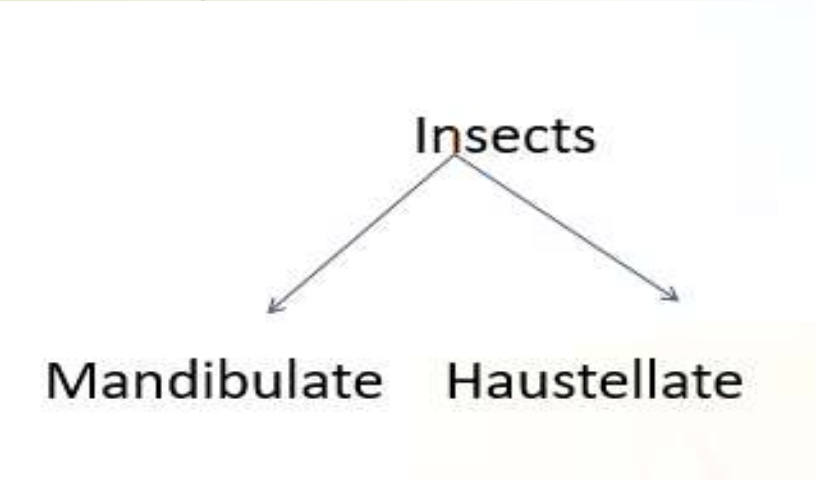
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1st Period (Study of local fauna)

- Aristotle known as father of biological classification, referred to the major groups of animals.

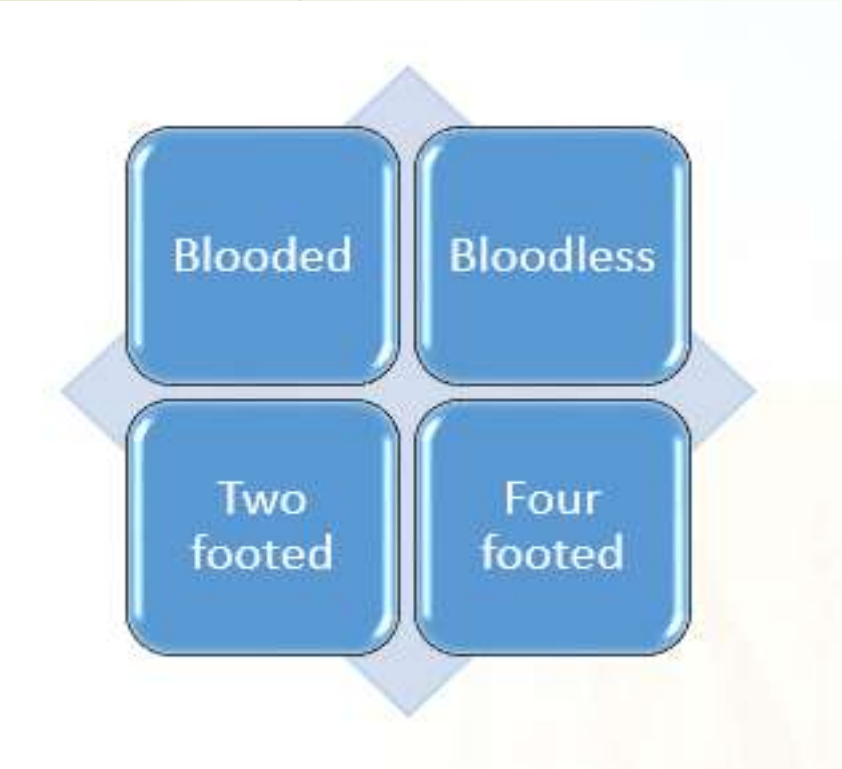


# History of Taxonomy



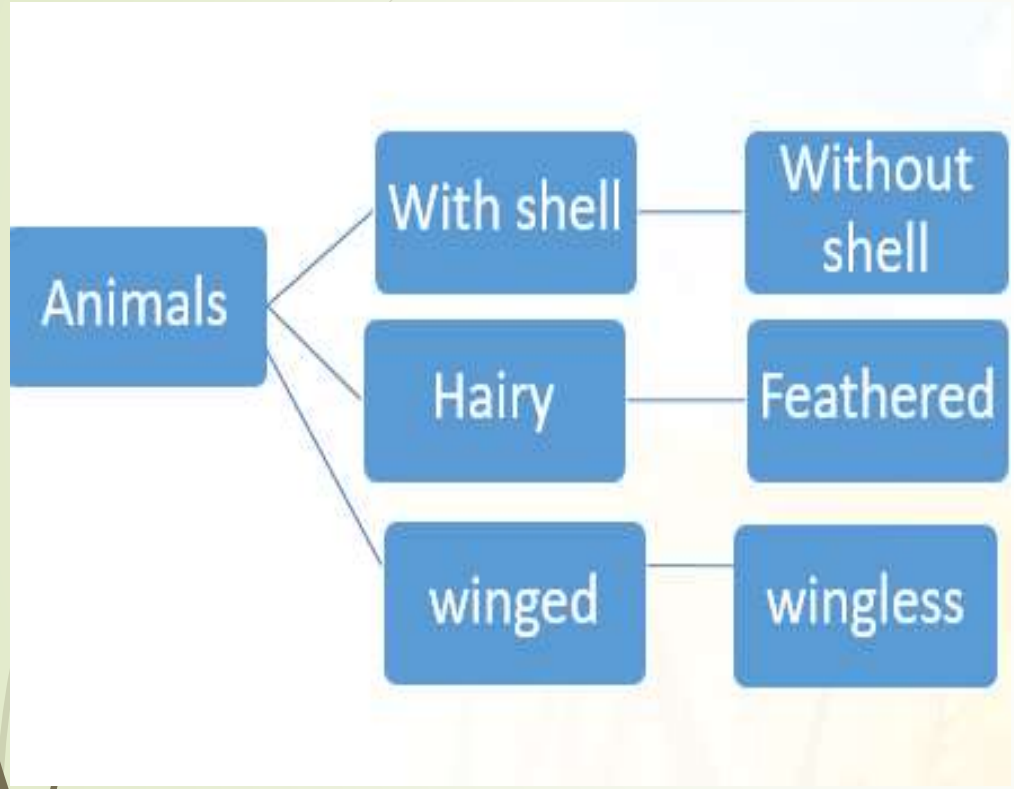

- Aristotle also made distinction in insects
- He first used the following terms: Coleoptera and Diptera

# History of Taxonomy




- He established numerous collective categories, or genera using differentiating characters

# History of Taxonomy




- Aristotle made further categories of animals
- His thinking dominated for next two thousand years
- After his death no system of classification were recognized

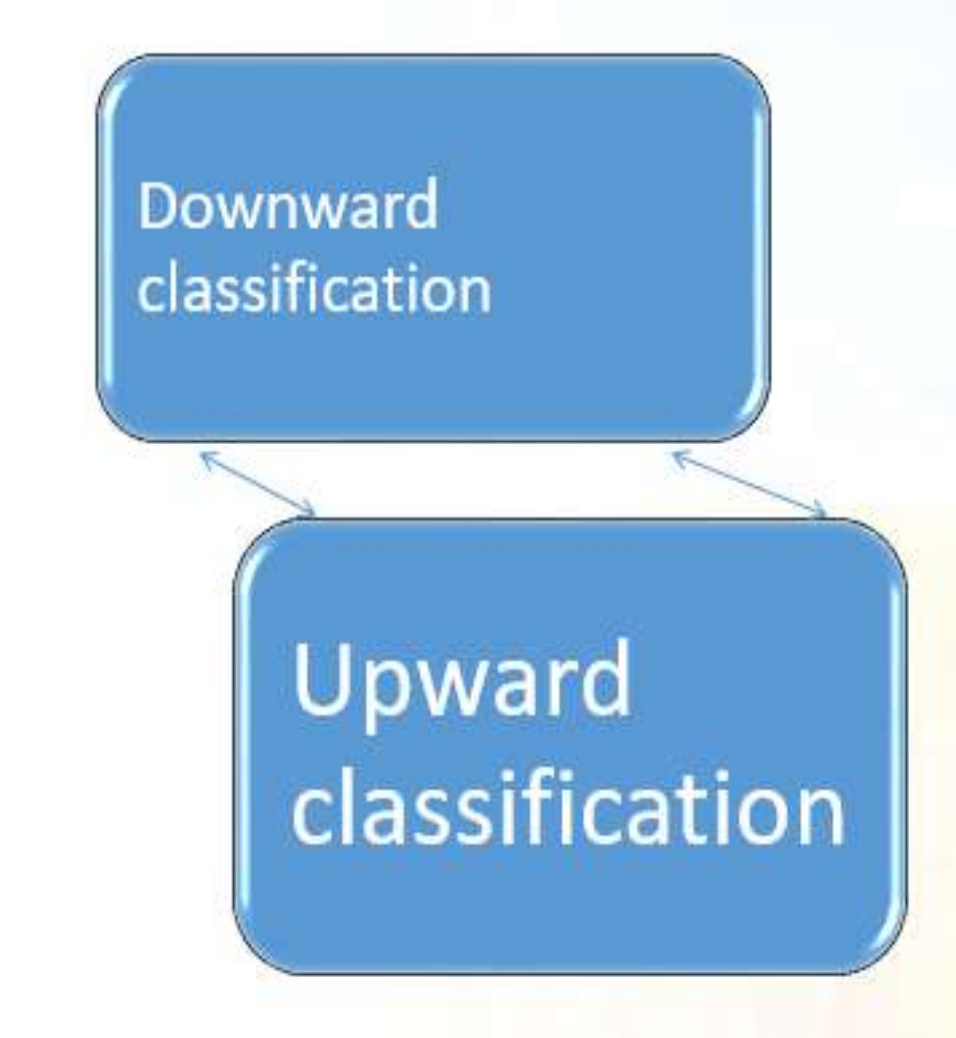
# Principles of Systematics

## Module# 11

# Principles of Systematics

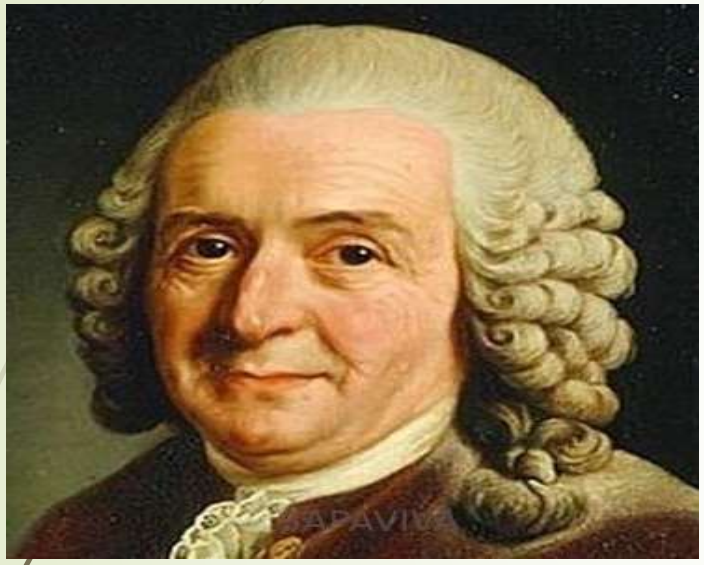
## History of Taxonomy

# History of Taxonomy




- From 1551 rapid progress started in knowledge about animals.
- There are two major forms of classification in first period

# History of Taxonomy

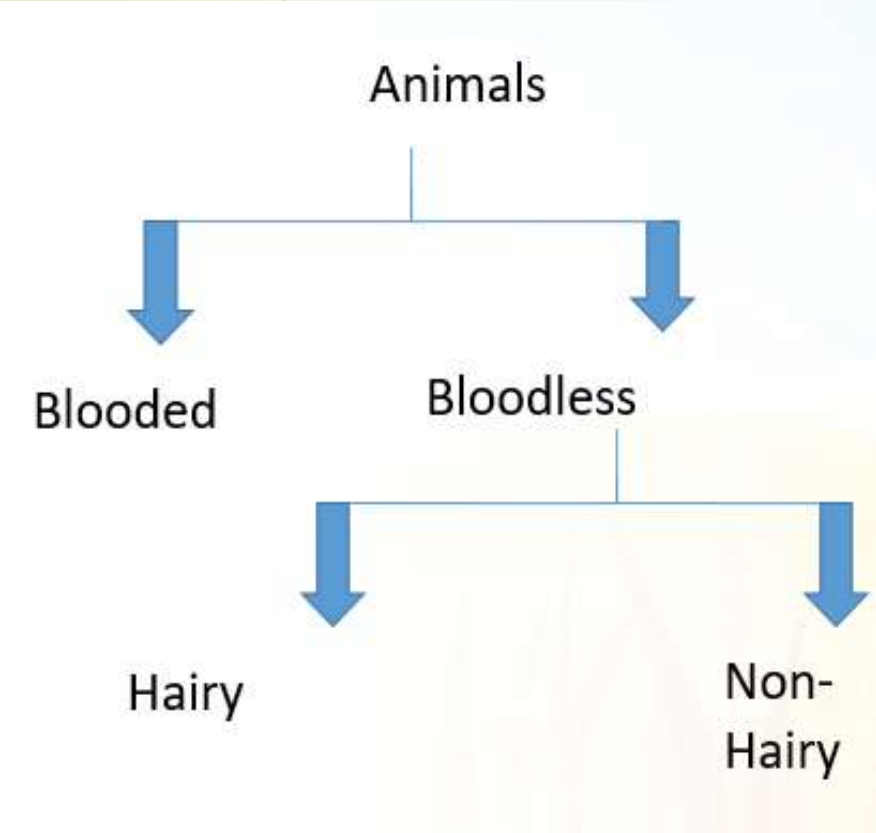


**Carl Linnaeus (1707-1778)**

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- Taxonomy based on local fauna reached its peak by the work of Linnaeus (1707-1778)

# History of Taxonomy




- Downward classification are the principles of logical division which consist of dividing the larger groups based on dichotomy into smaller groups. e.g,

  Animals

# History of Taxonomy

- Linnaeus adopted the principle of downward classification in 15th edition of his book (Systema naturae)
- He for the first time applied binomial nomenclature to animal kingdom.

# History of Taxonomy

- Linnaeus's idea was largely acceptable for the classification of insects.
- Not acceptable for the classification of birds, amphibians, and lower vertebrates (Vermes)

# History of Taxonomy



**Comte de Buffon (1707-1778)**

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- Buffon prepared the way for the biological species concept, by using sterility barriers instead of degree of morphological differences as the species criterion

# Principles of Systematics

## Module# 12

# Principles of Systematics

## History of Taxonomy

# History of Taxonomy

- By the mid of 18$^{th}$ century shortcomings of downward classification system were recognized.
- Gradually replaced by upward classification system

# History of Taxonomy

- It consists of assembling species by inspection into groups of similar and related species and forming hierarchy by again grouping of similar taxa of next lower rank.

# History of Taxonomy



**Comte de Buffon (1707-1778)**

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- Buffon in 1749 supported this system.
- He said that the only way to design a method is to group together the things that resemble each other and separate the things that differ from each other

# History of Taxonomy



**James Anderson (1738– 1809)**

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- Anderson 1763 (Botanist) applied this design for the classification of plants

# History of Taxonomy

- Cuvier recognized five phyla



**Georges Cuvier (1769-1832)**

https://www.google.com/search?q=george+cuvier+1769+to+1832&tbm=isch&ved=2ahUKEwjQx9O-kOLuAhUMEhoKHf1eCH4Q2-cCegQIABAA&oq=george+cuvier+1769+to+1832&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoHCCMQ6gIQJzoFCAAQsQM6AggAOggIABCxAxCDAToECAAQAzoGCAAQBRAeOgQIABAeUNkRWJ6PAWCskgFoAXAAeAKAAbsDiAHCbpIBBjMtMzQuNpgBAKABAaoBC2d3cy13aXotaW1nsAEKwAEB&sclient=img&ei=v0olYJDsL4ykaP29ofAH&bih=625&biw=1366#imgrc=_rEsvqaiDKV5CM

# History of Taxonomy




- Van Baer, Owen divided all species into limited no. of groups each representing a distinct type

https://www.www.google.com/search?q=karl+richaed+-oven-1769+to+1832&tbm=isch&ved=2ahUKEwjQx9OkOLuAhUMEhoKHf1eCH4Q2cCegQIABAA&oq=george+cuvier+1769+to+1832&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoHCCMQ6gIQJzoFCAAQsQM6AggAOggIABCxAxCDAToECAAQAzoGCAAQBRAeOgQIABAeUNkRWJ6PAWCskgFoAXAAeACAAbsDiAHCbpIBBjMtMzQuNpgBAKABAaoBC2d3cy13aXotaW1nsAEKwAEB&sclient=img&ei=v0olYJDsL4ykaP29ofAH&bih=625&biw=1366#imgrc=_rEsvqaiDKV5CM

# History of Taxonomy

## Developments during Linnaeus and Darwin Period

1. Specialization became more pronounced. Many authors identified birds, beetles, or butterflies
2. Classification became more hierarchical and soon the categories family and phylum were added
3. Philosophical guidelines were rejected
4. Search for a natural system was intensified

# History of Taxonomy

- This thought was applied by a botanist in 1873 and later by nearly all zoologists.
- This shift from downward classification to upward classification was a philosophical change

# History of Taxonomy

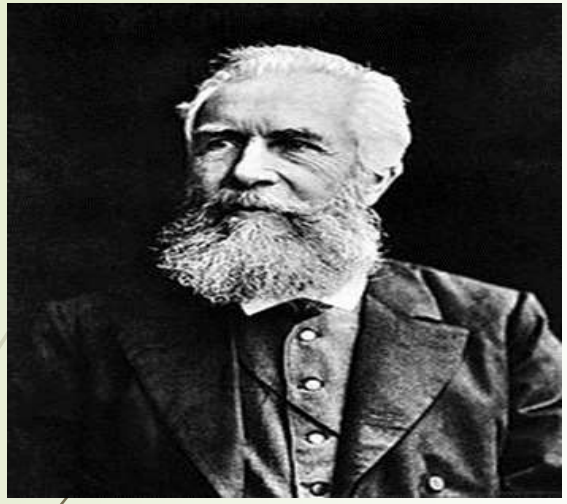


**Charles Darwin (1809-1882)**

https://www.google.com/search?q=charles+darwin&tbm=isch&ved=2ahUKEwi5vpykneLuAhUN-YUKHSv2Dr4Q2-cCegQIABAA&oq=charles+darwin&gs_lcp=CgNpbWcQAzlECAAQQzlCCAAyAggAMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADoHCCMQ6gIQJzoECCMQJzoFCAAQsQM6CAgAELEDEIMBOgcIABCxAxBDULLLGJFid-SRgzvskaAFwAHgDgAHWBYgB4kOSAQkzLTYuNi42LjGYAQCgAQGqAQtnd3Mtd2l6LWItZ7ABCsABAQ&sclient=img&ei=KlglYLmBAY3ylwSr7LvwCw&bih=625&biw=1366#imgrc=mrcMmxqef7ryCM

- Evolutionary thought was widespread in 18th century
- Darwin believed in fixity of species but after observing natural fauna in Galapagos island, he begin to believe in the plasticity of species

# History of Taxonomy



**Ernest Haeckel (1834-1919)**

https://https://www.google.com/search?q=ernst+hackel&tbm=isch&ved=2ahUKEwiH3uDFn-LuAhUt5IUKHQxDDdoQ2-cCegQIABAA&oq=ernst+hackel&gs_lcp=CgNpbWcQAzICCAAyBggAEAoQGDIGCAAQChAYMgYIABAKEBgyBggAEAoQGDIGCAAQChAYMgYIABAKEBg6BAgjECc6BAgAEEM6BwgjEOoCECc6BQgAELEDOgQIABAeOgQIABAYULWsDFjy5wxgpu0MaAFwAHgAgAGCBogBrUWSAQkzLTguMy42LjKYAQCgAQGqAQtnd3Mtd2l6LWItZ7ABCsABAQ&sclient=img&ei=iVolYMf1DK3IlwSMhrXQDQ&bih=625&biw=1366#imgrc=cIElbdaRJgSkeM

- Empirical taxonomists were greatly influenced by Darwin's idea.
- The phylogenetic tree of Ernest Haeckel also stimulated the empirical workers.
- Large number of new species were discovered and described.

# History of Taxonomy

- The greatest development of modern taxonomy started around 1930, when the workers realized that the Linnaeus species based on one or two specimens are not as perfect as those which are based on population

# History of Taxonomy

- Due to this Mayr in 1942 considered species as a group of interbreeding natural population.
- This idea of population taxonomy was useful in developing the polytypic concepts.
- New terms like new systematics and biosystematics were added to extend the taxonomic theory.

# History of Taxonomy

- The publication of the book ‘new systematics’
- Become a landmark in the History of Taxonomy.
- Thus taxonomy got a new label of biological taxonomy
- By 1955 taxonomy reach such a status that it was called taxonomic explosion

# History of Taxonomy

- New systematic deal exclusively with the species level.
- The present day taxonomic work include all available differences and similarities.

# History of Taxonomy

- Instead of morphology as an original base, phylogenetic adaptations, embryological patterns, biochemical variations, genetic variabilities and similarities, behavioral characteristics are all applicable to taxonomic study

END

# Principles of Systematics

## Module# 13

# Basic Definitions

# Basic Definitions

## Sibling Species

- Two or more than two closely related species which are morphologically alike but behaviorally or reproductively isolated from each other.
- Examples are *Drosophila persimilis* and *D. pseudoobscura*.

# Basic Definitions

## Sibling Species

- The mosquito *Anopheles maculipennis* complex consists of several sub-species, of which a few are vector of malaria and the rest are harmless.

# Basic Definitions

## Monotypic Species

- species with a single subspecies, called monotypic species
- *Example: Bardala labarda*

## Polytypic Species

- A species contains two or more subspecies
- Examples are tiger, *Panthera tigris* which has several subspecies

# Basic Definitions

## Endemic species

- The species which are found in a particular region, called endemic species.

## Example

- The Darwinian finches are the endemic species of Galapagos Islands
- Indus dolphin in Pakistan

END

# Principles of Systematics

## Module# 14

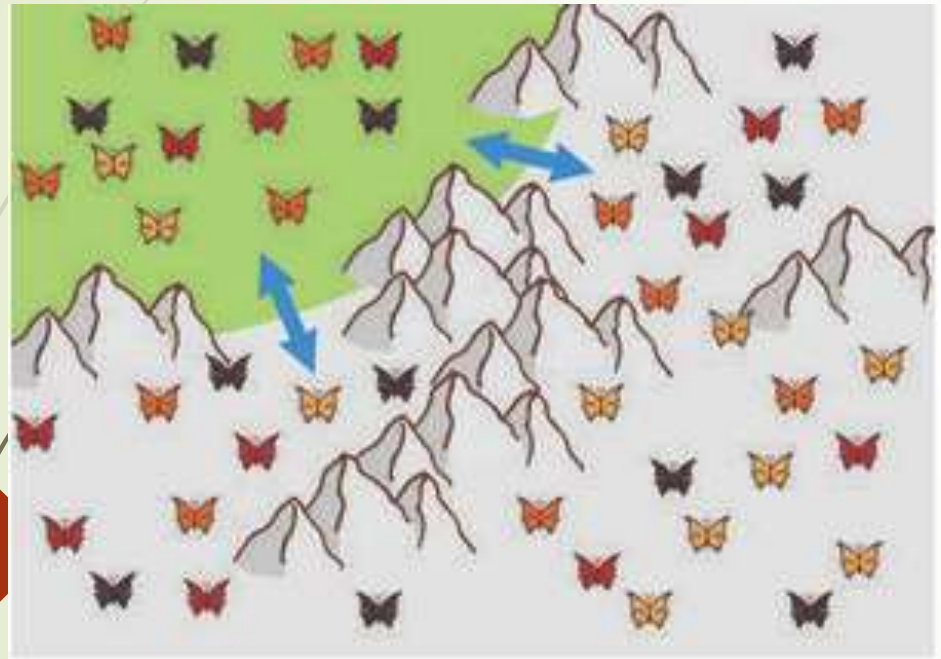
# Basic Definitions

# Basic Definitions

## Deme

- A deme is a sub-population (a smaller group within the population) that can freely interbreed.
- A population is usually made of several demes (groups) that are partially isolated from each other.

# Basic Definitions



https://www.pathwayz.org

## Deme

- Often a geographical barrier limits gene flow between demes, but not sufficiently to result in speciation.
- Enough gene flow occurs between demes ensuring that all of it's members remain the same species.

# Basic Definitions



https://www.pathwayz.org

## Cline

- A cline is the gradual change in the phenotypes and genotypes of a species across a geographical gradient
- 
- Allele frequencies and traits might vary slightly across the geographical gradient because of different selection pressures.

# Basic Definitions

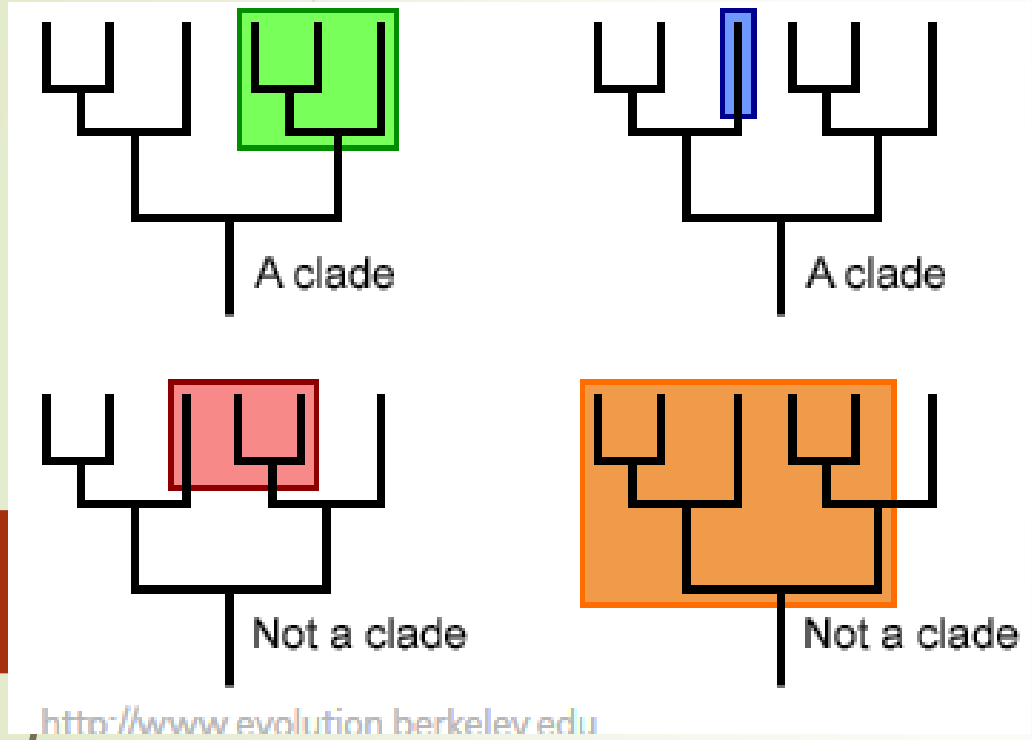



https://www.pathwayz.org

## Types of cline

- Smooth cline
- Step cline

# Basic Definitions




## Clade

- A clade is a group of organisms that evolved from a common ancestor
- Eutheria is one mammalian clade and the other is Metatheria, which includes marsupials.
- Another example of a clade could be birds: they all also descended from a

# Principles of Systematics

## Module# 15

# Concept of Speciation and Causes

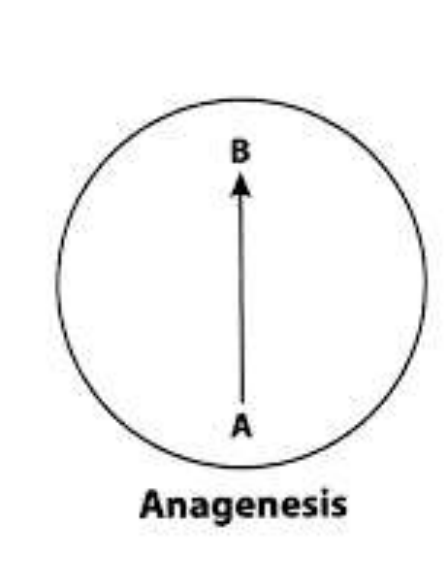
# Concept of Speciation and Causes

## Speciation

Speciation refers to the creation a new species.

Through this process, the earliest groups of similar organisms were able branch out and populate the world with millions of different varieties of life

# Basic Definitions

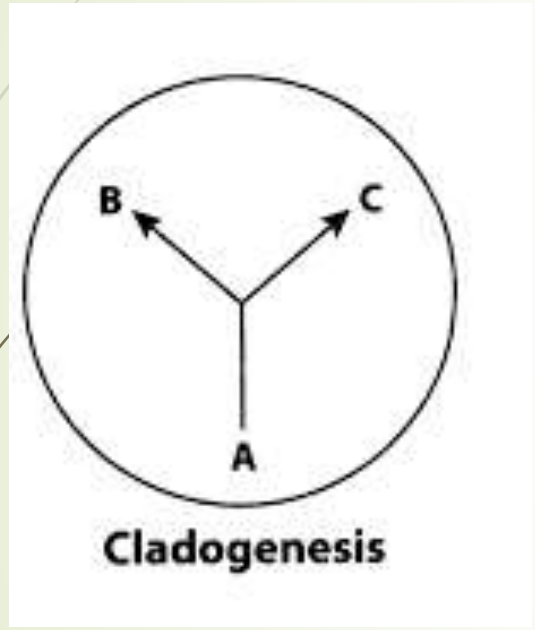



## Types of speciation

### Anagenesis

- A single species may change over time into a new form that is different enough to be considered a new species.
- One species replaces other
- No net change in number of species

# Basic Definitions




## Types of speciation

### Cladogenesis

- A species may become split into two groups that no longer share the same gene pool.
- Net increase in number of species
- Increase in diversity

# Concept of Speciation and Causes

## Cusses of speciation

- Geographical Isolation
- Behavioural isolation
- Reproductive isolation

END

# Principles of Systematics

## Module# 16

# Concept of Speciation and Causes

## Types of speciation

# Concept of Speciation and Causes

## Types of speciation

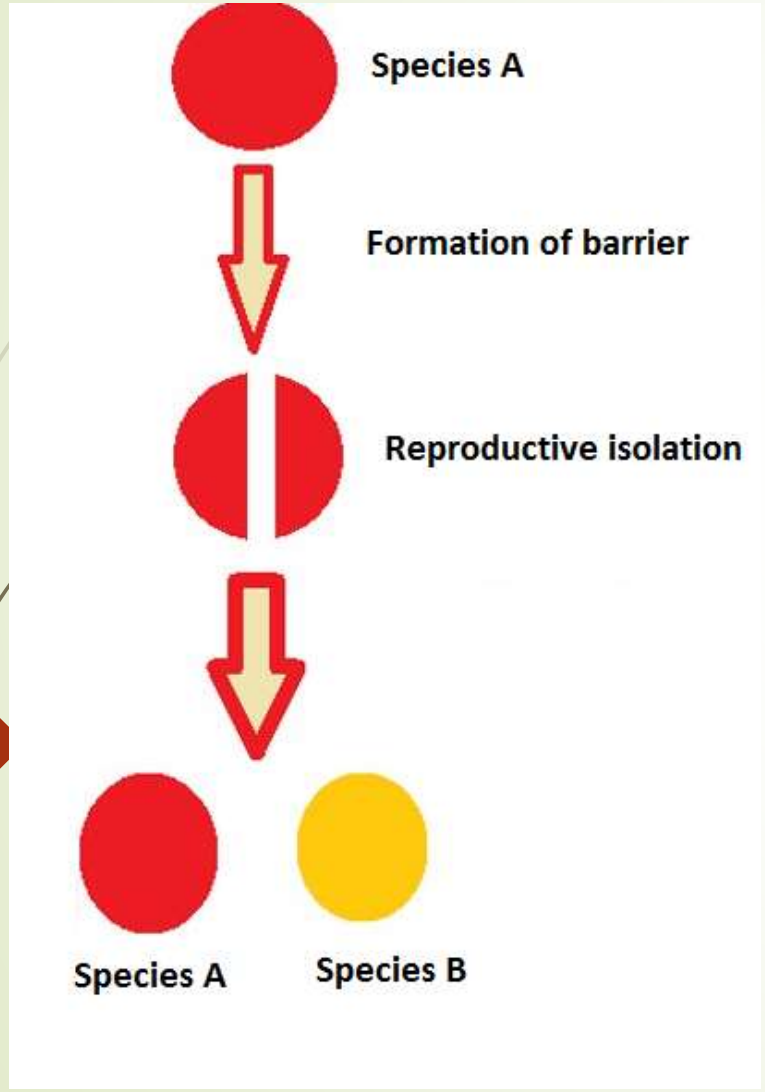
- Allopatric speciation
- Peripatric speciation
- Sympatric speciation
- Parapatric speciation

# Concept of Speciation and Causes

## Allopatric Speciation

- The most common form of speciation
- Occurs when populations of a species become geographically isolated.

# Concept of Speciation and Causes




## Allopatric Speciation

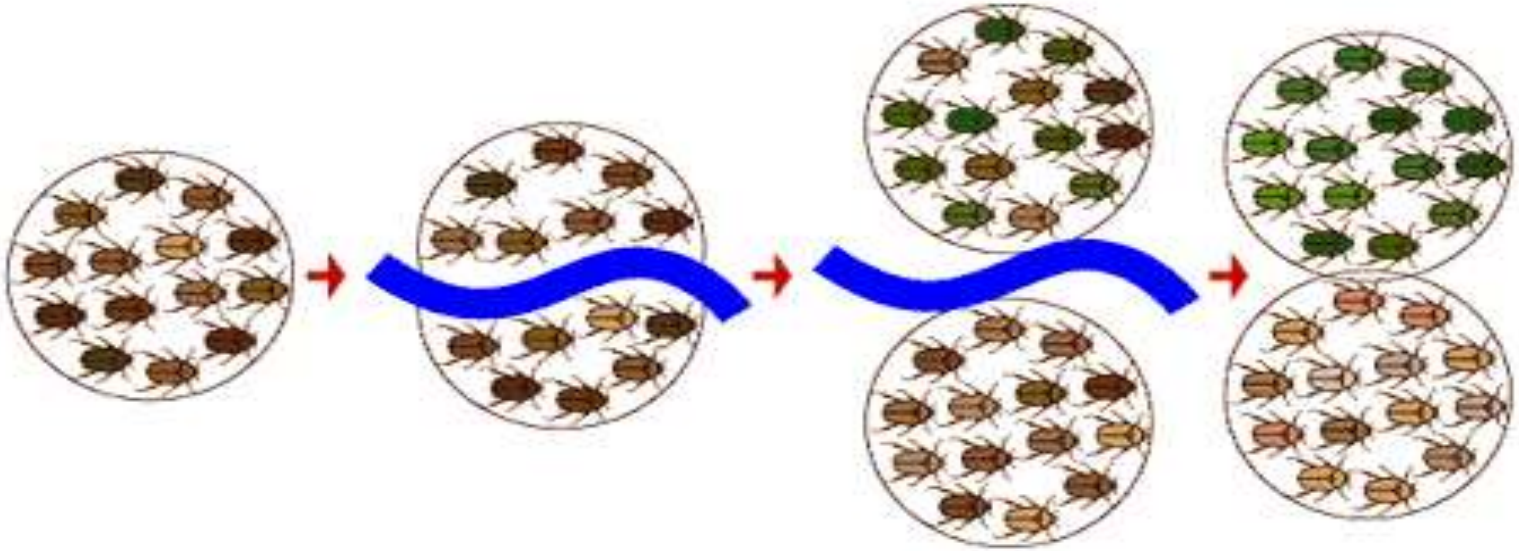
- When populations become separated, gene flow between them ceases.
- Over time, the populations may become genetically different

# Concept of Speciation and Causes

## Examples

- Speciation is the Galápagos finch
- Different species of these birds live on different islands.
- The finches are isolated from one another by the ocean.
- Over millions of years, each species of finch developed a unique beak that is especially adapted to the

# Concept of Speciation and Causes



https://evolution.berkeley.edu

# Principles of Systematics

## Module# 17

# Sympatric Speciation

# Sympatric Speciation

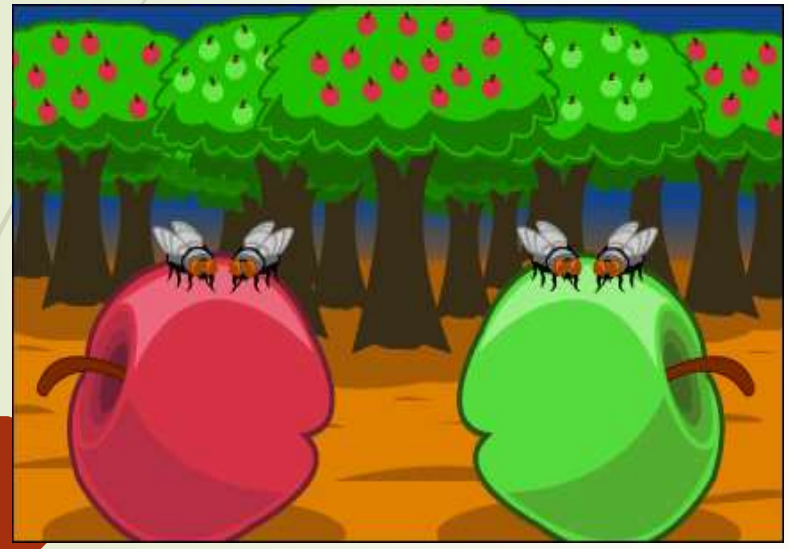
- A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range.
- Share the same habitat but become reproductively isolated from each other.
- More common in plants

# Sympatric Speciation

## Popyploidy in plants

- Plants produce offspring that are polyploid. Hence, the offspring live in the same environment as their parents but are reproductively isolated.

# Sympatric Speciation



https://images.app.goo.gl/5Q15k3p4fVkKcktA7

## Example

- 200 year ago ancestor of maggot flies were used to lay eggs on hawthorn.
- Now they lay eggs on hawthorn (native to America) and domestic apples (introduced)
- Host shift/genetic differences

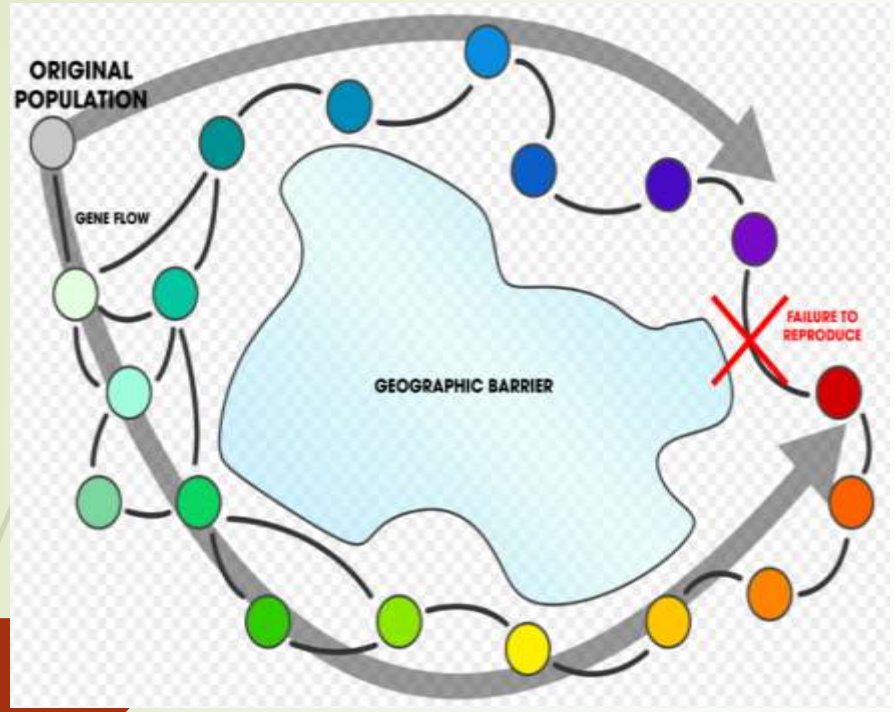
(Coyne, 2007)

# Principles of Systematics

## Module# 18

# Parapatric Speciation

# Parapatric Speciation




https://en.wikipedia.org/wiki/Parapatric_speciation#/media/File:Ring_Species_(gene_flow_around_a_barrier).png

## Parapatric speciation

- It occurs when populations are separated not by a geographical barrier, such as a body of water, but by an extreme change in habitat.

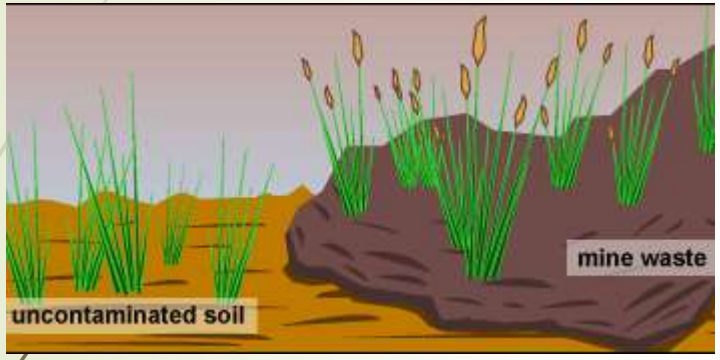
- Populations in these area may interbreed.
- Charles Darwin was the first to propose this mode of speciation.

# Parapatric Speciation

## Parapatric speciation

- Although interbreed but reduced gene flow.
- They often develop distinct characteristics and lifestyles.
- Reproductive isolation is rather temporal or behavioral.

# Parapatric Speciation




https://images.app.goo.gl/iMY6DE1PcER2Dx1k7

## Example

- Population of grass plant is present near mining area
- Soil near mining area is infused with metals due to which plants grow slowly than those that grows away from mines.
- As a result, two populations of grasses arise having different flowering time.

# Parapatric Speciation



https://www.google.com/search?q=cave+snails&tbm

## Example

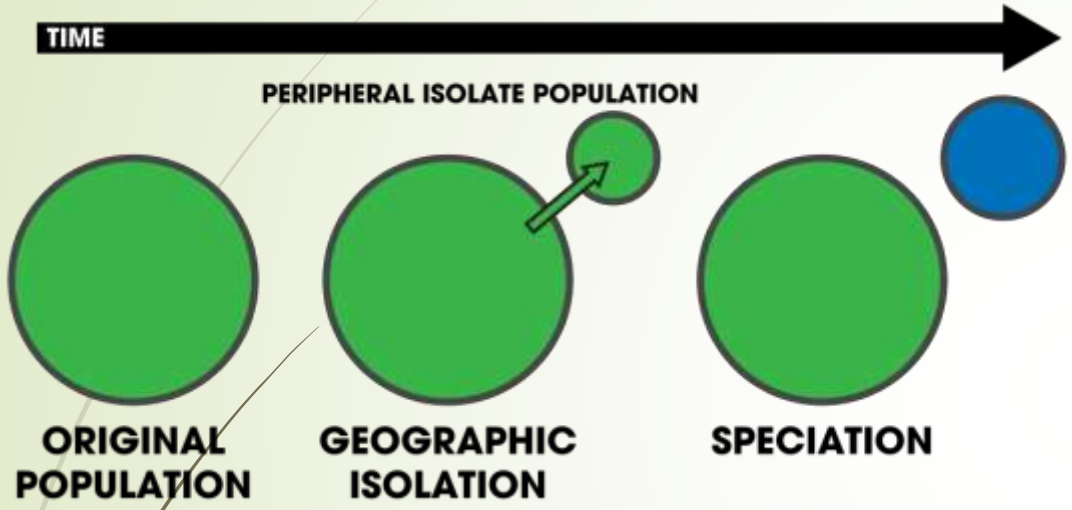
- A study of tropical cave snails *(Georissa saulae)* found that cave-dwelling population descended from the above-ground population, likely speciating in parapatry

# Principles of Systematics

## Module# 19

# Paripatric Speciation

# Paripatric Speciation

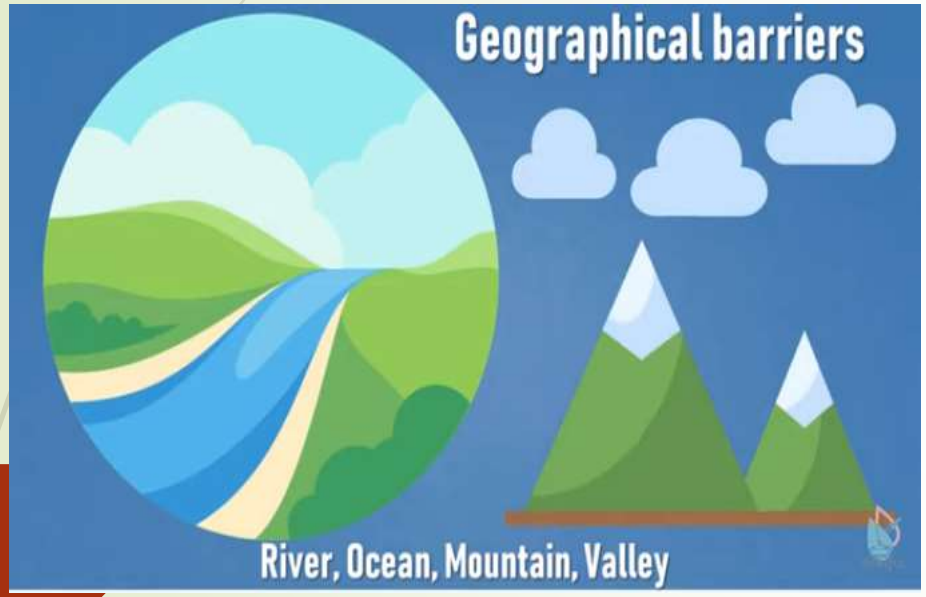



https://en.wikipedia.org/wiki/Peripatric_speciation#/media/File:Peripatric_Speciation_Schematic.svg

## Paripatric speciation

- Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.
- Resemble alloprtraic speciation/barriers
- Isolating population is small

# Paripatric Speciation




https://www.youtube.com/watch?v=7TNN-WGYWks

## Paripatric speciation

- Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.
- Resemble alloprtraic speciation/barriers
- Isolating population is small

# Paripatric Speciation



https://www.google.com/search?q=london+underground+mosquito&hl

## Example

- An **example** of this is the London Underground mosquito, a variant of the mosquito *Culex pipiens*, which entered in the London Undgerground in 19th century.

- It is found in the London Underground railway system

# Paripatric Speciation

## Example

- Evidence for its **speciation** includes genetic divergence, behavioral differences, and difficulty in mating.

END

# Principles of Systematics

## Module# 20

# Typological Species Concept

# Typological Species Concept

- Species is the basic unit of classification according to the International Commission of Zoological Nomenclature.
- Survey of taxonomic literature shows that there are more than 20 species concepts.

# Typological Species Concept

## Typological species concept

- This concept was proposed by Aristotle and Plato

**According to this concept**

"A certain pattern of features is found in each species of organisms and all members of species show maximum equality in this format"

**Example:** certain pattern in dogs

# Typological Species Concept

## Typological species concept

- Monotypic species
- Polytypic species

# Typological Species Concept

## Postulates of Typological species concept

- Species consist of similar individuals sharing the same essence.
- Each species is separated from all others by sharp discontinuity.
- Each species is completely constant through time.
- There are strict possible limits to variations within any one species.

# Typological Species Concept



*Ara ararauna*



*Ara macao*



sterile hybrid
*Catalina macaw*

## Example

- Two macaw species, blue-and-yellow macaw ***Ara ararauna*** and the scarlet macaw ***Ara macao*** are considered different species due to different coloration.
- In nature, their habitats overlap but they not produce hybrids except in captivity.

# Typological Species Concept



***Anchura substriata***

https://www.google.com/search?q=macaw&tbm=isch&ved=2ahUKEwiQyL

## Strengths

- Applicable to living or extinct species.
- Doesn't depend on complex genetic analysis.
- Morphology can be readily observed, without harming the organisms.

# Typological Species Concept

## Problems with Typological species concept

- Polymorphism within populations.
- Sibling or cryptic species.
- Geographical variations among populations

END

# Principles of Systematics

## Module# 21

# Biological Species Concept

# Biological Species Concept

- According to Biological species concept:

- Species are groups of actually or potentially interbreeding natural populations, which are reproductively isolated from other such groups." Ernst Mayr(1942,1963)

# Biological Species Concept

- Can successfully interbreed and produce fertile offspring.
- A species' integrity is maintained by interbreeding within a species.

# Biological Species Concept

## Why members of species resemble

- This concept explains why members of species resemble and differ from other species.
- When organisms breed they shuffle their gene pool which gives the species its identity.

# Biological Species Concept

## Biological species concept

Members of species constitute:

- Ecological unit
- Reproductive unit
- Genetic unit

END

# Principles of Systematics

## Module# 22

# Biological Species Concept

# Biological Species Concept

## Biological species concept

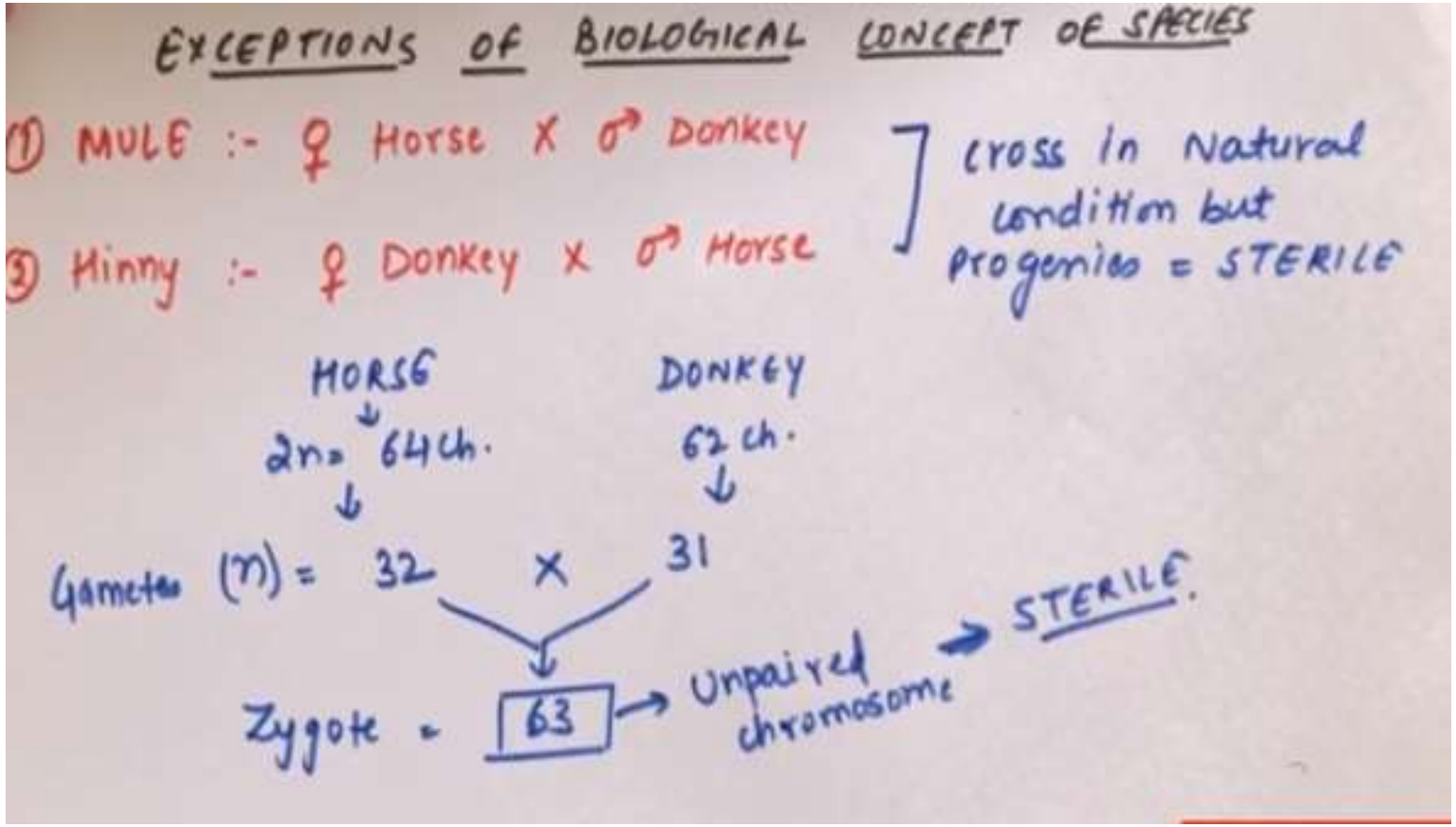
- Ecological unit
- Reproductive unit
- Genetic unit
- ." Ernst Mayr (1942, Darwin of 20th century)

# Biological Species Concept

## Biological species concept

- Most widely accepted species concept.
- Dobzhansky (1951) and Hanson (1981) accepted the definition of biological Species concept because of two reasons:
- Gene pool
- Reproductive isolation

# Biological Species Concept




https://www.youtube.com/watch?v=3Chz-XJHvNE

# Principles of Systematics

## Module# 23

# Drawbacks of Biological Species Concept

# Drawbacks of Biological Species Concept

## Drawbacks

- Asexual reproducing organisms.
- Fossils
- Sibling or cryptic species
- Distinguishing between species on the basis of reproductive separation is problematic because reproductive isolation is often incomplete .

# Drawbacks of Biological Species Concept

**Asexual reproduction/Uniparental reproduction**

- Not applicable on apomictic species
- Do not fulfill interbreeding criterion **(Most important in BSC)**
- Apomictic groups (parthenogenesis, apomixis, budding etc)

# Drawbacks of Biological Species Concept

## Asexual reproduction/Uniparental reproduction

- Uniparental reproduction is Common in lower invertebrates and lower vertebrate
- Decedents are called agamospecies, binoms or paraspecies
- According to Mayr (1988a) these are not species

# Drawbacks of Biological Species Concept

**Sibling species/Cryptic species**

- As they are morphological similar but reproductively isolated

**Incompleteness of Speciation/fossil record**

- Evolution is gradual and continuous process. However, fossil record is not

END

# Principles of Systematics

## Module# 24

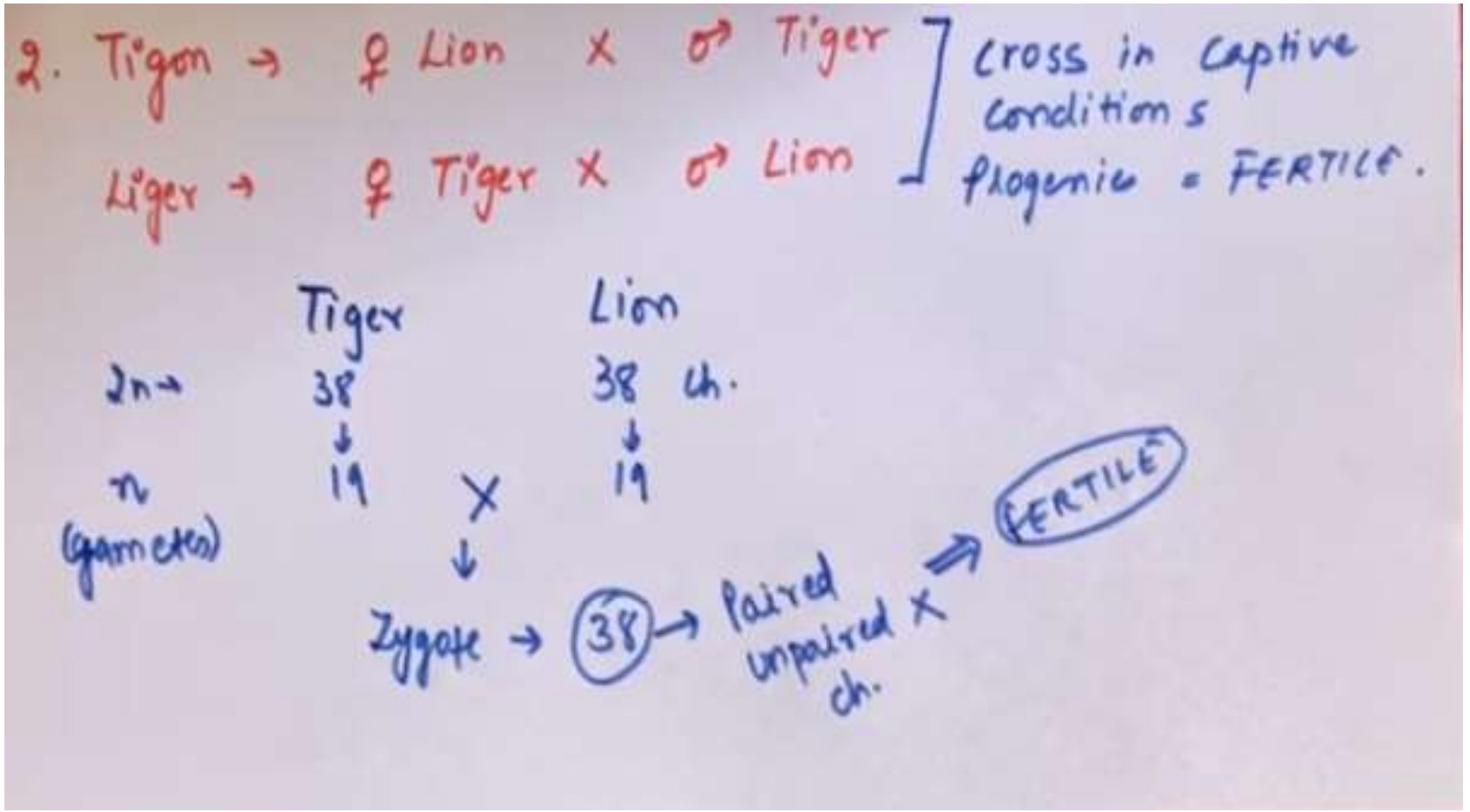
# Drawbacks of Biological Species Concept

# Drawbacks of Biological Species Concept

**Interbreeding between two different species**

- According to BSC two different species can not breed and produce fertile offsprings
- However, there are exceptions

# Drawbacks of Biological Species Concept




https://www.youtube.com/watch?v=3Chz-XJHvNE

# Principles of Systematics

## Module# 25

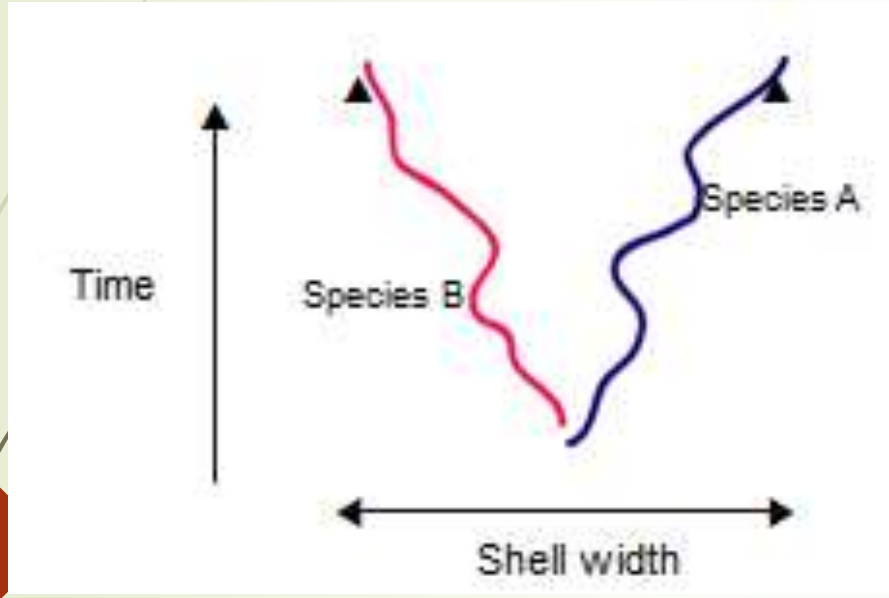
# Evolutionary Species Concept

# Evolutionary Species Concept

**Evolutionary species concept**

- Simpson (1940s)
- An **evolutionary species** "is a single lineage of ancestor-descendant populations of organisms which maintains its:
- Identity from other such lineages [in space and time]
- Own **evolutionary** tendenci es
- historical fate" (Wiley, 1981)

# Evolutionary Species Concept




## Evolutionary species concept

- Initially developed to define fossil species.
- Applicable to both sexually and asexually reproducing species and emphasizes common descent.

# Evolutionary Species Concept

## Drawbacks

- Arbitrary: how to define independent roles and tendencies.
- Does not provide mechanism.

END

# Principles of Systematics

## Module# 26

# Taxonomic Characters

# Taxonomic Characters

## Character

- It is any attribute of member of taxon by which it may differ from member of other taxon.
- A heritable trait possessed by an organism
- Characters are usually described in terms of character states.

# Taxonomic Characters

## Character state

- One of the variant conditions of a trait.
- **Binary states:** For example: "hair present" vs. "hair absent," where "hair" is the **character**, and "present" and "absent" are its **states**
- **Multistate characters:** Yellow, white or pink flowers or AGCT for nucleotide sequence data.

# Taxonomic Characters

## Qualities of key characters

- Apply to all member of taxon
- Fairly stable and durable
- Qualitative and absolute
- Detectable without special equipment's, dissection or histological preparations

# Taxonomic Characters

Taxonomic Characters are illustrated as:

- Morphological character
- Physiological character
- Behavioral character
- Ecological character
- Geographic character
- Molecular and biochemistry character
- Ontogenetic character

END

# Principles of Systematics

## Module# 27

# Morphological Characters

# Morphological Characters

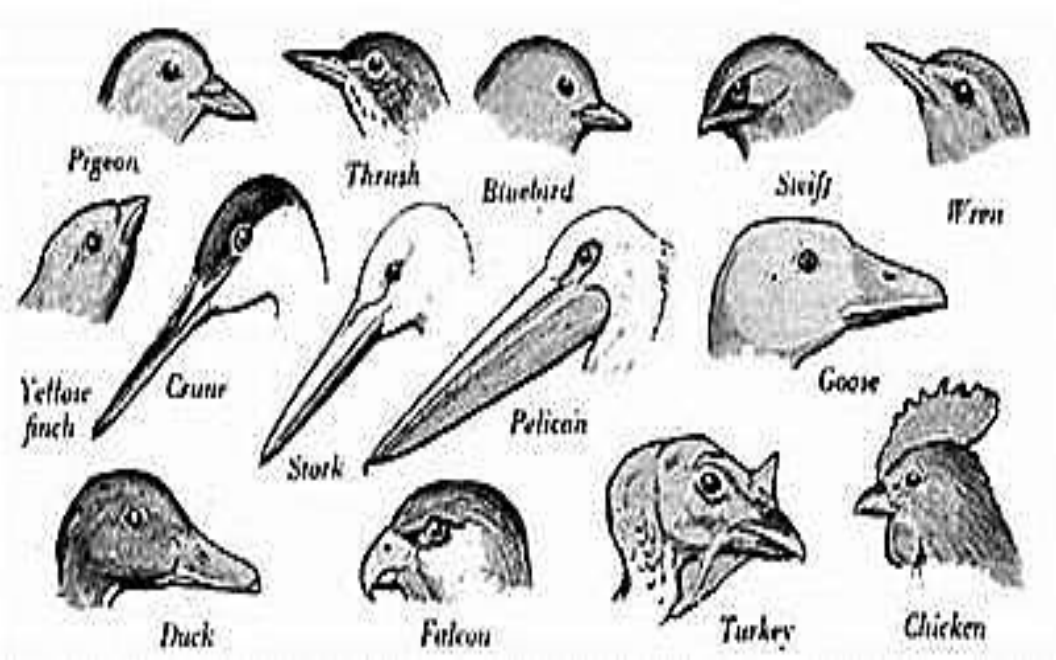
## Morphological Characters

Morphology describe the structural features.

Morphological characters include:

- External morphology
- Genetelic structure
- Internal morphology
- Embryology
- Karyology and other cytological differences

# Morphological Characters




https://tse2.mm.bing.net/th?id=OIP.k_dPDFg2fI7OVNpR7Y72-gHaDa&pid=Api&P=0&w=

**General external morphology**

The apparent features are easy to detect and observe.

**e.g.** Plumage of birds, scales of fish and reptiles, pelage of mammals, phylogenetically significan sutures and sclerites of arthropod body etc.

# Morphological Characters



https://www.researchgate.net/profile/Aidas_Saldaitis/publication/321343778/figure/fig2/AS:568147356336130@1512468293710/Male-genitalia-and-leg-structures-of-the-C-katsumii-holotype_Q320.jpg

## Genetalic structures

- The genital structure of each specie are different to depict the reproductive isolation.
- Used to delimit species
- Key and Lock model
- Highly effective insects
- Genetalic structures appear to be among the first to change in the course of speciation.

# Morphological Characters

## Coloration

- Color pattern and other aspects of coloration are convenient characters in certain group of animals.
- Bird and coloration except few genera with sibling species.
- Same is true or certain reef fishes and butterflies
- Many subspecies are indentified on the basis of colour

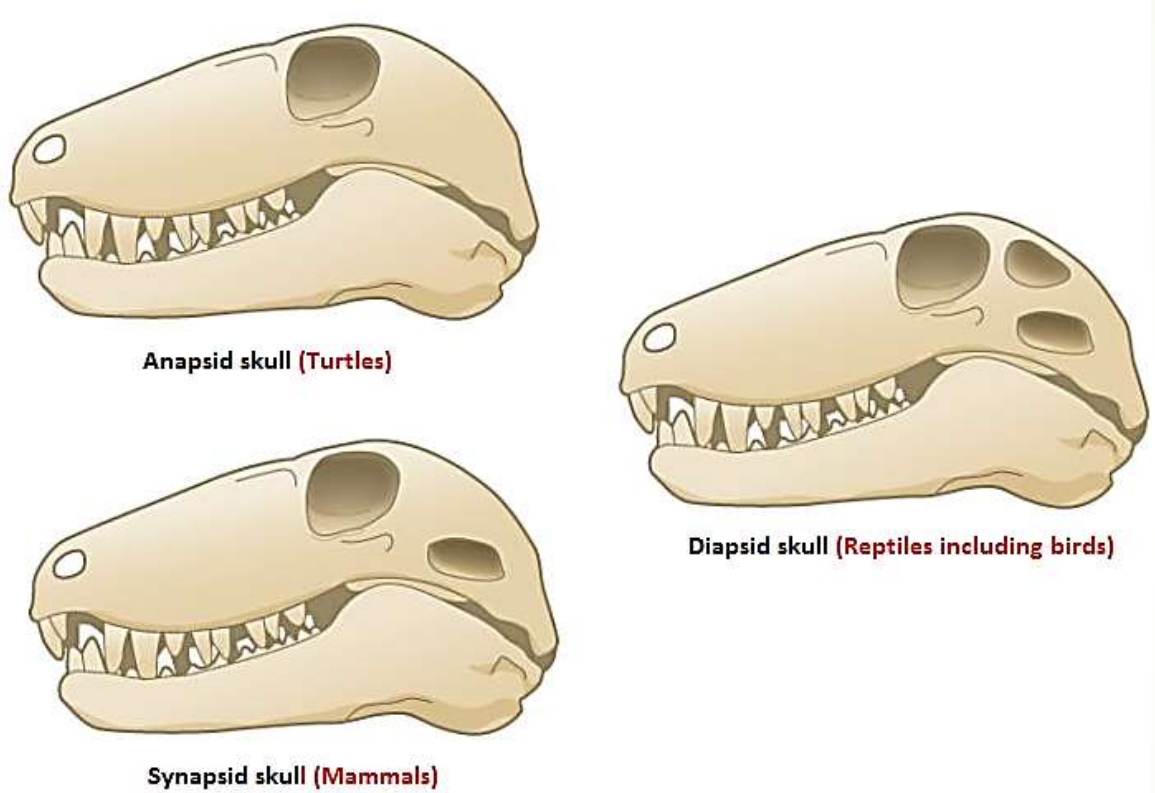
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# Principles of Systematics

## Module# 28

# Morphological Characters

# Morphological Characters

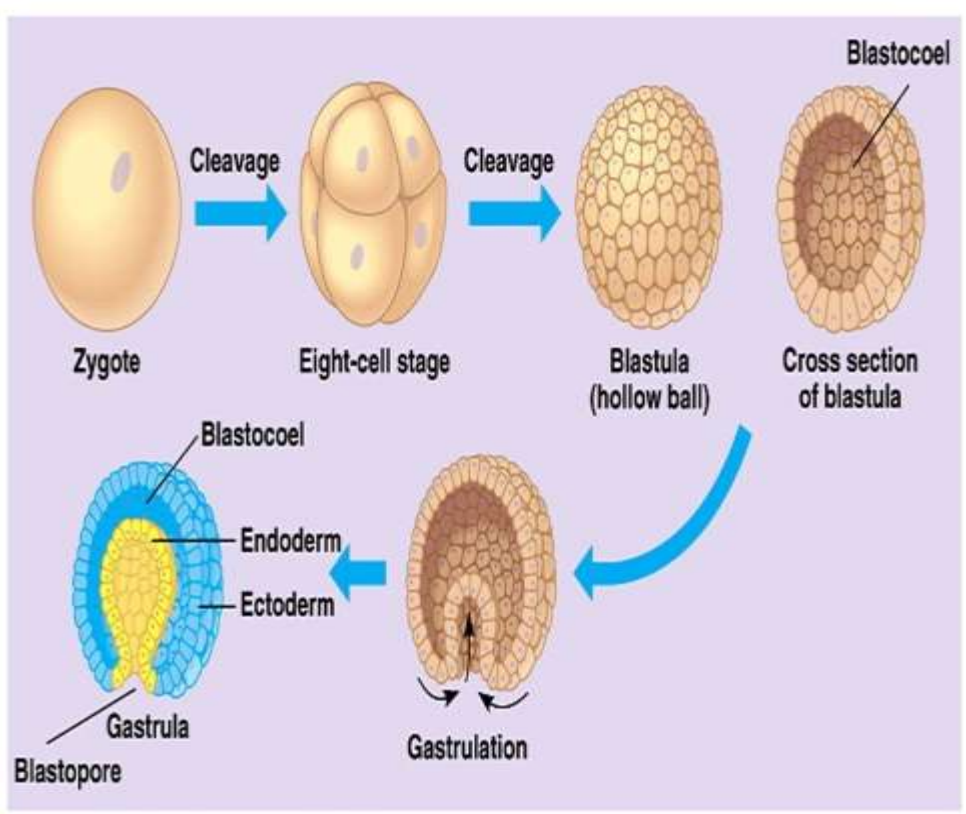



https://www.google.com/search?q=evolution+in+skulls+of+vertebrates

## Internal morphology

- Internal anatomy provides Taxonomic Characters for all groups of higher animals.
- The extend to use vary from group to group.
- The skull of routinely used in classification of higher mammals.
- Teeth and bones of fossils and other hard parts.

# Morphological Characters



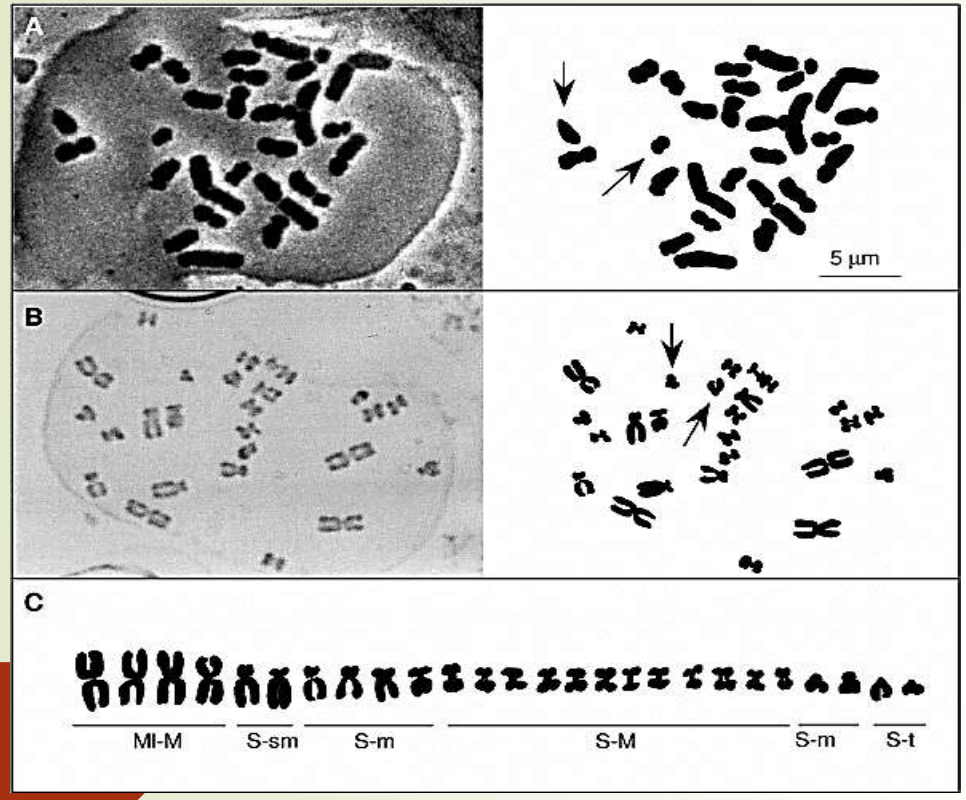

https://i1.wp.com/orbitbiotech.com/wp-content/uploads/2018/04/embryonic-development-Orbit-Biotech-Training.jpg?ssl=1

## Embryology

Various larval stages, embryology and even eggs may provide taxonomic information.

- Sibling species of *Anopheles maculipenis*------*Egg structure*
- Identification of white flies are based on pupae
- Comparative embyological stages like cleavage, blastulation, gastrulation etc are helpful

# Morphological Characters




https://www.liberaldictionary.com/wp-content/uploads/2019/02/karyology-4570.jpg

## Karyology and other cytological factors

- Karyology is description of size, shape and number of chromosomes
- The simplest cytological character is chromosomal number (species specific)
- Comparison of closely related species (sibling species)
- Establishing phyletic lineage

# Principles of Systematics

## Module# 29

# Physiological Characters

# Physiological Characters

## Physiological Characters

Physiological characters are low weight characters and are variable:

- All structures are product of physiological process and are thus physiological characters.
- This character can not be studied in preserved animals.

# Physiological Characters

## Physiological Characters

- Metabolic activities
- Body secretion
- Serological protein and other biochemical differences
- Genic sterility factor

# Physiological Characters

## Metabolic Factors

- Metabolic factors/enzymes help to find the relations in the closely related species

**e.g.** Gram +ive and Gram –ive bacteria

# Physiological Characters



https://thecotyledon.files.wordpress.com/2010/10/scale-honey.jpg

**Body secretion**

Body secretions also have the consistent pattern

- Links can be created by finding the number
- Size and
- Structure of the glands

**e.g.** waxy secretion of scale insects

# Principles of Systematics

## Module# 30

# Physiological Characters

# Physiological Characters

## Serological protein and other biochemical differences

- Serology is concerned with the nature and interaction of antigens and antibodies.
- Antigens are the substances capable for inducing the formation of antibodies when introduced into the blood stream of animals.

# Physiological Characters

- According to antibodies produced in response to the introduction of foreign antigen

**Bovden** (1943)

- Protein antigens are conservative hereditary trait.
- Proteins of one organism react more strongly with the antibodies of closely related organism than more distantly related.

# Physiological Characters

- Good precipitin technique are reveal the relative degree of biochemical similarity of protein antigens among organisms.

# Physiological Characters

## Genic sterility factors

- Reproductively incompatible species cannot reproduce new young one
- Only closely related species can hybridize successfully.
- As a result give genetically sterile species.
- Cross fertility in Dobkey and Hourse- Mule
- Cross fertility in ducts (species and genera)

END

# Principles of Systematics

## Module# 31

# Behavioral Characters

# Behavioral Characters

## Behavioral characters

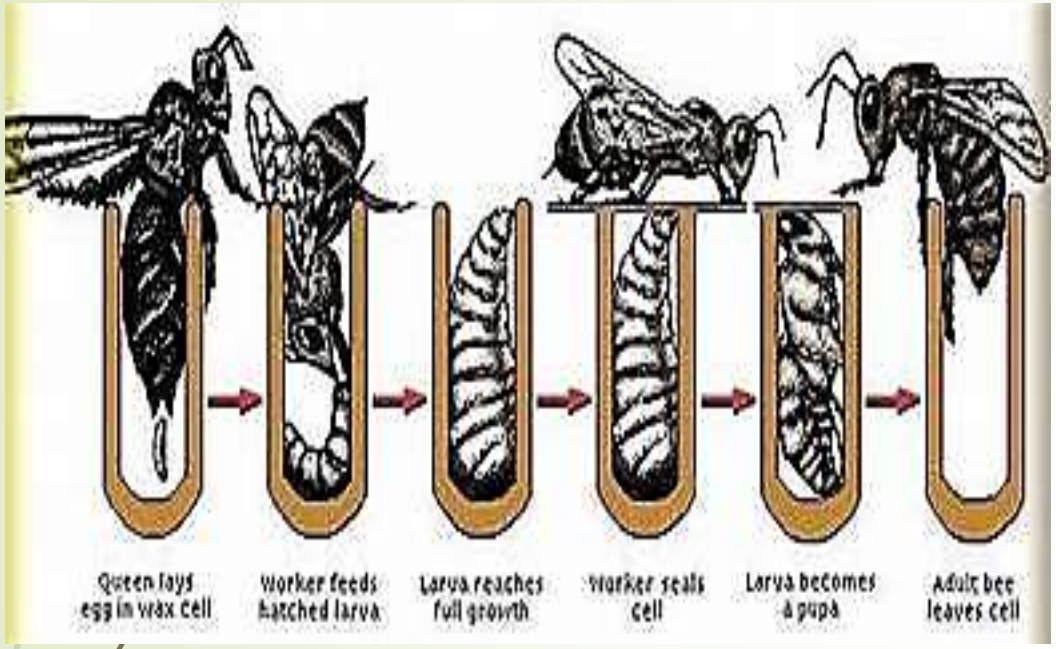
- Behaviour is one of the most important source of Taxonomic Characters.
- They are more superior than morphological characters in the study of closely related species.
- Especially in sibling species
- Mating behaviour, egg laying, nesting behaviour, drinking habit of birds etc.

# Behavioral Characters

## Behavioral characters

- Mating calls of frogs, toads and crickets are used in species delimitation.
- Webs pattern in spiders can be at various level in classification.

# Behavioral Characters




https://tse3.mm.bing.net/th?id=OIP.eubyf9OPir2yCdF7wsFukQHaDO&pid=Api&P=0&w=392&h=171

## Reproduction

Mating and reproduction behavior of species help in taxonomic classification and also help in determining reproductive isolation and speciation.

**E.g. slugs**

# Behavioral Characters



https://media.buzzle.com/media/images-en/photos/birds/peacocks/1200-77866817-peacock-drinking-water.jpg

## Other behavior pattern

The living, eating and nesting behavior of the animals also depicts the taxonomic character

**e.g.** drinking habit of pigeon and sand grouse, use of mud in nest building

# Behavioral Characters

**Drawbacks**

- Can not be studied in preserved material
- It is intermittent even in living animal.
- Some behaviours are seasonal
- Change within 24 hours

# Principles of Systematics

## Module# 32

# Ecological Character

# Ecological Character

## Ecological characters

- Each species has specific niche
- Types of niches
- Habitats and hosts environment
- Food sources
- Seasonal variation
- Parasite and host interaction reaction

# Ecological Character

## Habitats and host environment

- Habitat of the organisms is not same
- Species cannot coexist at same place or in same environment or partitioned their resource to avoid competition.

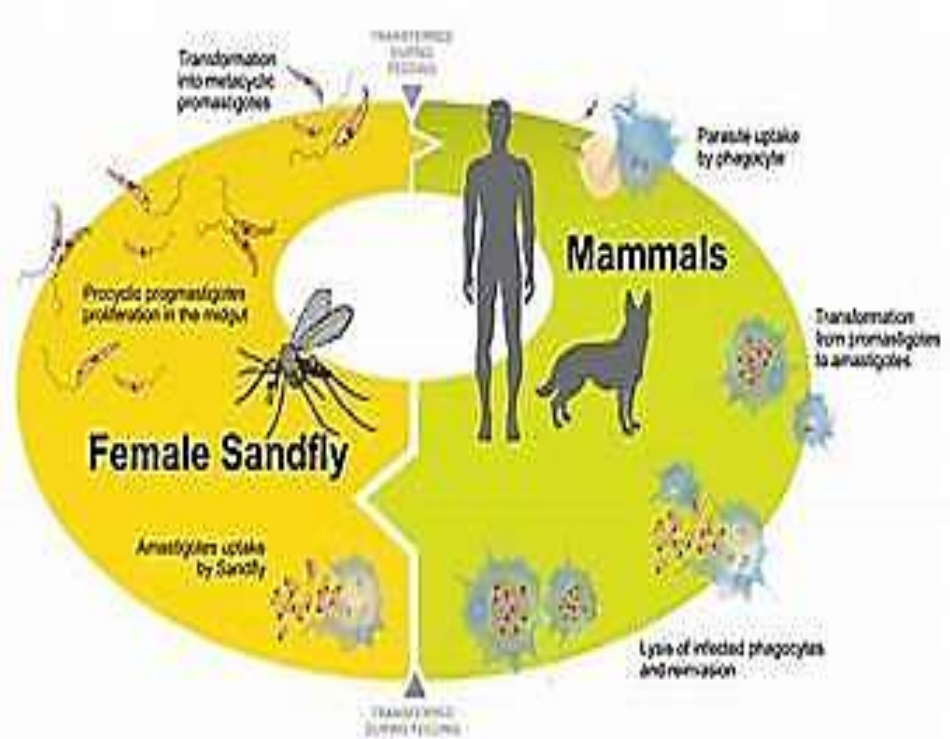
# Ecological Character

## Food and seasonal variation

- Food sources also varies like in parasites and many others.
- Characteristics of organism also vary depending on season

**e.g.** plumage of birds and breeding coloration

# Ecological Character




https://tse4.mm.bing.net/th?id=OIP.CDWG77aR8Ie8a2HGocriegHaEV&pid=Api&P=0&w=341&h=201

## Parasite and host interaction reaction

- Structural changes can occur in host after parasitic attack
- Various responses of body like distortion, swelling etc.

# Principles of Systematics

## Module# 33

# Ecological Character

# Ecological Character

## Ecological Character

- Every species has its own ecological niche in nature differing from its close relatives in (spider example)
- Food preference
- Breeding season
- Tolerance to physical factors
- Resistance to predators
- Compotators and pathogens

## Ecological Character

- Larvae of both *Drosophila mulleri* and *Drosophila aldriachi* live simultaneously in decaying pulp of the fruits of cactus *Opuntia lindheimeri*.
- Both species are specialized in using certain types of yeast and bacteria.

## Ecological Character

- Similarly tapeworms of man and rodents and Ascaris of man and pigs are host specific.
- Each species of Galapagos finch is also specialized according its habitat/ environments

END

# Principles of Systematics

## Module# 34

# Geographical Characters

## Geographical Character

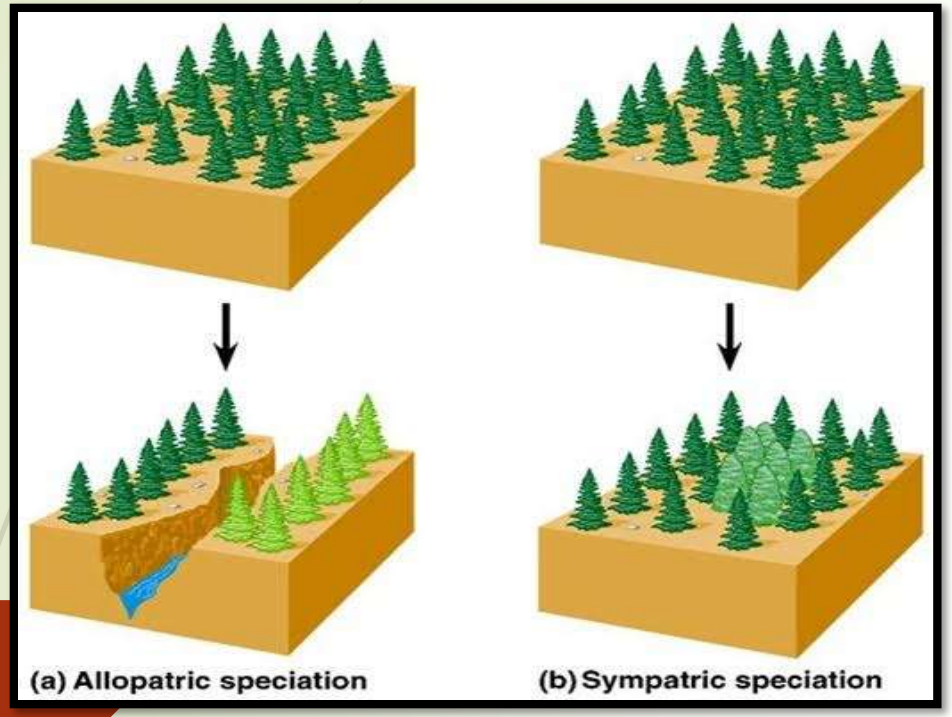
- Each taxon has a geographical range
- Range changes as the taxon evolve
- Distribution of taxa provide past history
- Geographic characters are important at two levels
- Microtaxonomy
- Macrotaxonomy

# Geographical Character

## Geographical Character

- Taxonomists are important in two types of geographical characters:
- General biogeogrphical patterns, which are especially useful in arrangement of higher Texa
- Allopatric-sympatric relationship, which is helpful in determining wether or not two species are conspecific.

# Geographical Character




https://s-media-cache-ak0.pinimg.com/736x/18/49/1c/18491ced4bf91ceac2f85a3fdee59a5f.jpg

**Sympatric-allopatric relationship of population**

These isolation pattern that occur in the population also determine the links in population

# Principles of Systematics

## Module# 35

# Molecular and biochemistry character

# Molecular and Biochemistry Character

## Molecular and biochemistry character

This character as taxonomic character is also an approach for the evolutionary links and relations determines if classical taxonomy is defined.

# Molecular and Biochemistry Character

## Molecular and biochemistry character

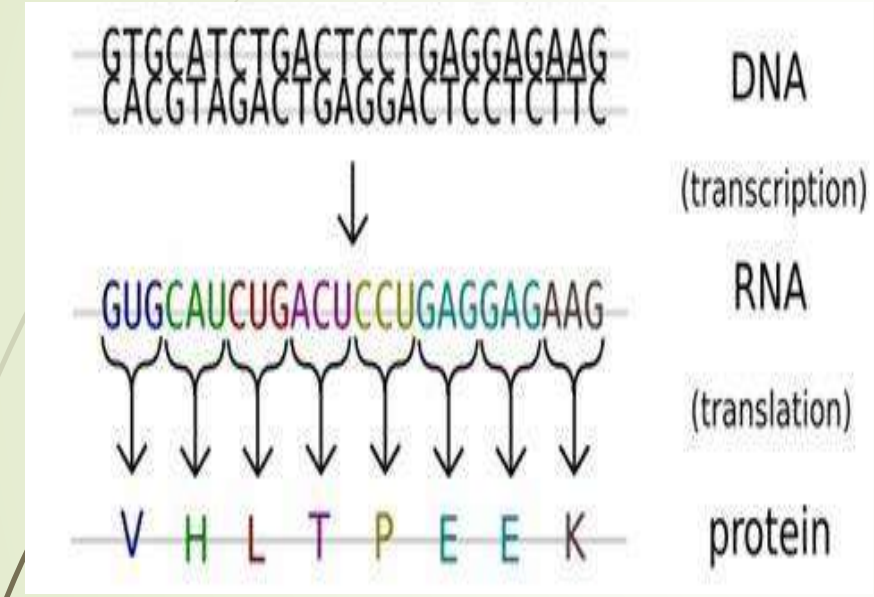
- Immunological distances
- Electrophoretic differences
- Amino acid sequence of protein
- DNA hybridization
- DNA and RNA sequence

# Molecular and Biochemistry Character

## Amino acid sequences

- Amino acid sequences help to determine the links.
- This approach as a taxononmic character is more absolute.

# Molecular and Biochemistry Character




https://tse2.mm.bing.net/th?id=OIP.Uo3B4dsPEz-PEzHf3uJ4OwHaCz&pid=Api&P=0&w=517&h=197

## DNA and RNA sequences

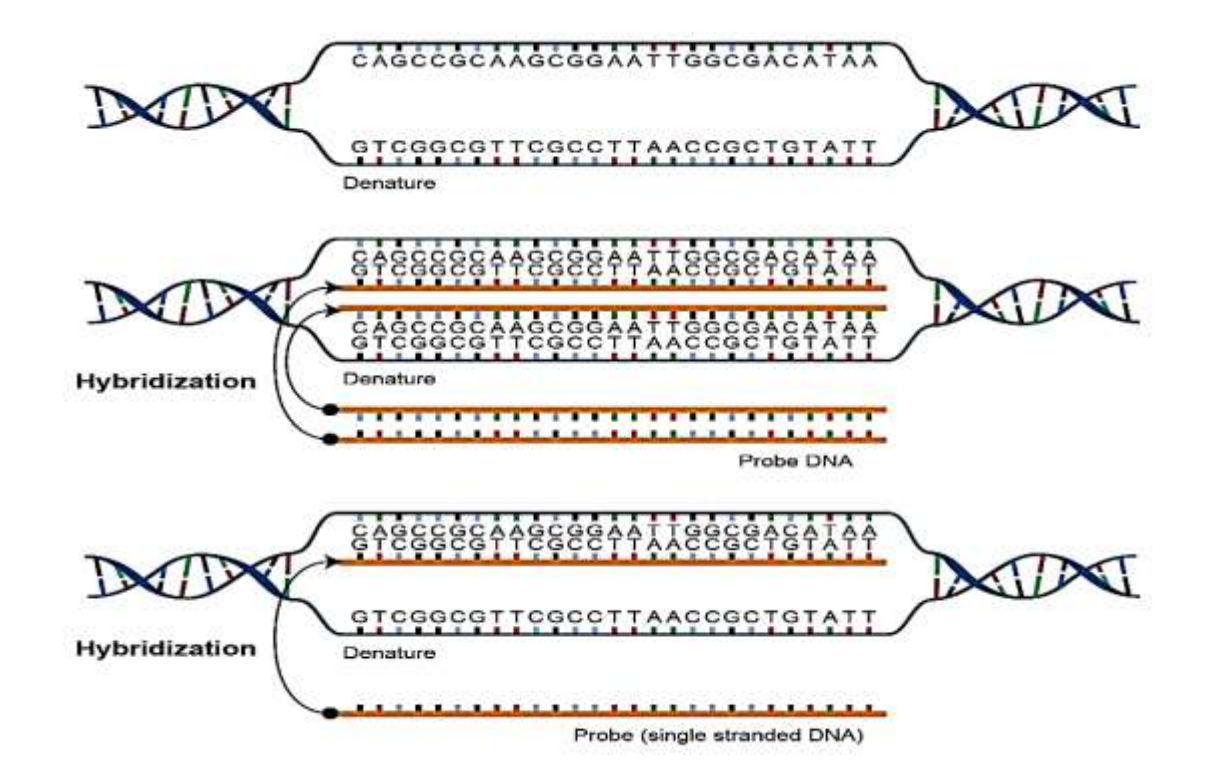
- These sequences help to determine the direct links on the basis of sequences
- But these sequences can also undergo some changes under the evolutionary pressure

# Molecular and Biochemistry Character

## Electrophoretic differences

- This method determine the difference in the isozyme and allozyme.
- Can help to determine share character among taxa.

# Molecular and Biochemistry Character




https://www.genome.gov/sites/default/files/tg/en/illustration/hybridization.jpg

## DNA hybridization

- This method is used to determine the genetic distance between two organisms
- Mainly measure the genetic similarity

# Principles of Systematics

## Module# 36

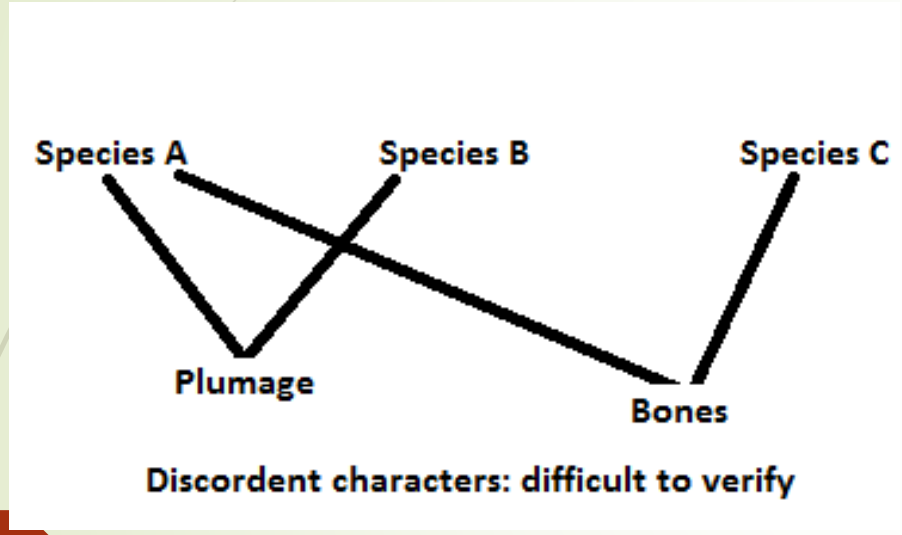
# Weighing of Characters

# Weighing of Characters

- Any attribute by which member of a taxon differ
- Raw material of classification
- Amount of phyletic information contents in a given character
- More phyletic information----high weigh

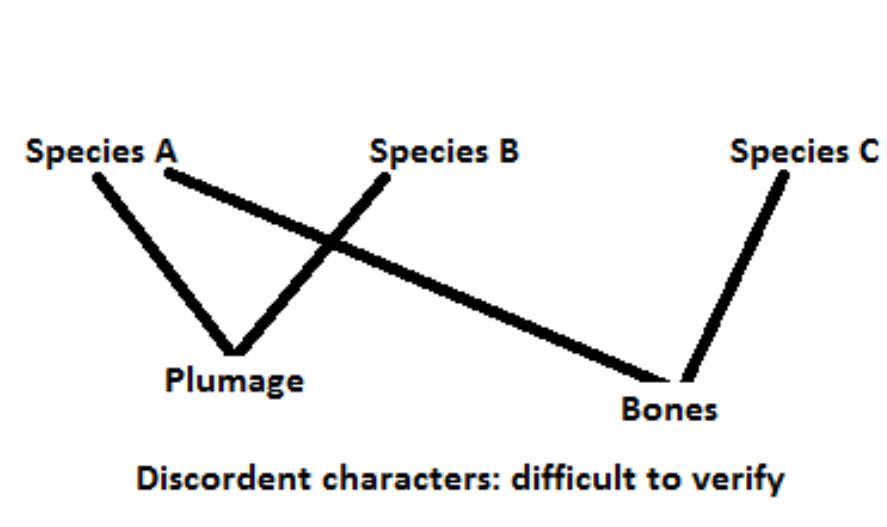
- Low phyletic information---low weight

# Weighing of Characters




- Animal character may of high weight ----mostly used by taxonomist
- Low weight---rarely used
- Also characters of intermediate weight
- Challenge for taxonomists which discordant character should be truested?

# Weighing of Characters




- Process to determine that which character is more reliable is called *weighting*.

# Weighing of Characters

1. **Priori (before/without phyletic information)**
   - Proposed by Artistotle

2. **Posteriori**
   - Proposed by Adenson (after phyletic information collection)
   - All classification is based on this model.

**END**

# Principles of Systematics

## Module# 37

# Principles of Systematics

## Correlated Characters

# Correlated Characters

- Characters which are functionally correlated have very low weight
- They have low phylogenetic content
- Do not deserve to be treated as separate character
- Analogues organs birds wings, bats and insects (analogus)/convergent evolution
- Redundant

# Correlated Characters

- Phyletic correlation
- High weight
- Higher genetic contents
- Diagnostic characters of Deuterostomes and Chordates (concordance)---phyletic correlation—evolutionary related

END

# Principles of Systematics

## Module# 38

# Principles of Systematics

## High Weight Characters

# High Weight Characters

## High weight characters

- Complexity
- Constancy
- Consistency
- Not affected by ecological shift
- Apomorphy
- Not a specific adaptation
- e.g. Morphological and behavioral character

# High Weight Characters

## Complexity

- Complexity-Genetalia
- Deep genetic character
- Higher complexity-higher weight age

## Constancy

- Character which is constant in a large group of species than variable character (Darwin 1859)
- According to Farris (1969) Low phenotypic variability be given high weight

# High Weight Characters

## Consistency

- A character which is consistently present in one group and consistently absent in related groups should have high weight

# High Weight Characters

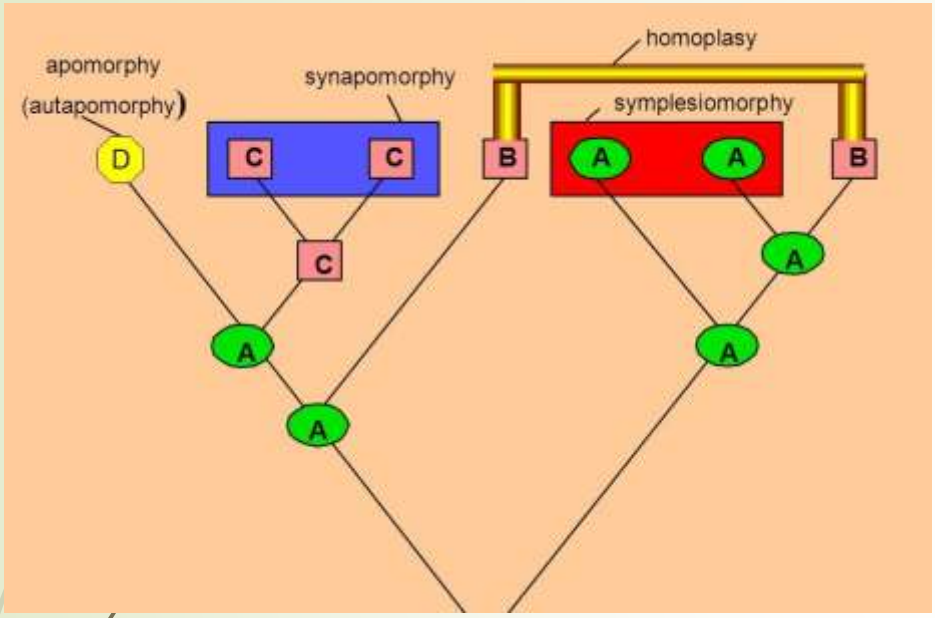


https://www.google.com/search?q=mergansers&sxsrf=ALeKk03jIyrLE0TeIgIBxO6mWo9hBiogvg:1603685582863&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjdtv2rstHsAhXvDmMBHZtJBOcQ_AUoAXoECCYQAw&biw=1360&bih=657

**Not affected by ecological shift**

- A character that no change with ecology
- A bird feeding on fish in one habitat continue to feed on fish in another habitat
- An example is adaptation of mergansers (Mergus) for fish eating

# High Weight Characters




https://www.google.com/search?q=apomorphy&sxsrf=ALeKk03Y_tzp73qJK0Em_Bfingu_YHOxiw:1603614144234&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjbjLGbqM_sAhWzShUIHdkiB1oQ_AUoAXoECCAQAw&biw=1360&bih=600#imgrc=GCYTyU7VcBY0LM

## Apomorphy

- A novel evolutionary trait that is unique to a particular species and all its descendants and which can be used as a defining character for a species or group in phylogenetic terms.
- Hence, the possession of feathers is unique to birds and defines all members of the class Aves.

# Principles of Systematics

## Module# 39

# Principles of Systematics

## Low Weight Characters

# Low Weight Characters

- Characters having less taxonomic importance
- Less phylogenetic content
- Not reliable
- Poor indicators of relationships

# Low Weight Characters

- Highly variable Characters (hairs, branching pattern of arteries)
- Monogenic and oligogenic characters (albinism) varies independently of other characters

# Low Weight Characters

- Regressive characters (loss of eyes, loss of wings, wing veins in insects, loss of teeth in mammals, loss of segments in segmented animals are unnatural.
- Loss may be due to environmental conditions (eye loss in cave, loss of digestive systems in parasitic tapeworms)
- Narrow specialization

  e.g. Physiological character

# Low Weight Characters

- Not constant
- Not consistent
- Change with ecology
- Less complex

  e.g. Physiological character

END

# Principles of Systematics

## Module# 40

# Non-Genetic Variations

# Non-Genetic Variations

## Types of Non-Genetic variations

- Individual variations in Time
- Social Variations
- Ecological Variations
- Traumatic variation
- Post-morten variations

# Non-Genetic Variations

## Individual variations in Time

- Age variations
- Seasonal variations of individuals
- Seasonal variations of generations

# Non-Genetic Variations



https://www.google.com/search?q=eel&sxsrf=ALeKk02QiXShlrI7phddXt7HPFtE0QOvmA:1603309550565&source=lnms&tbm=isch&sa=X&ved=2ahUKEwihuefBucbsAhWPTN8KHUuJCQsQ_AUoAXoECDUQAw&biw=1360&bih=600#imgrc=eWSxWZoTy-WYyM

## Age Variations

- Animals pass through large numbers of juvenile or larvae which are quite different from adult.
- Immature stage of eel (*Anguilla anguilla*) were originally described as *Leptocephalus brevirosteris in 1856*
- Larval stages different from adult (Caterpillar and butterfly)

# Non-Genetic Variations

## Seasonal Variations of individuals

- Same animal may have different appearance in different parts of the year
- In arctic and sub-arctic birds there may be change from white winter dress to normal colour summer dress.

# Non-Genetic Variations



https://www.google.com/search?q=ptarmigan&sxsrf=ALeKk015nxwTbGZOdxCypPYsK-TETw:1603517715139&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjH3bT-wMzsAhVoQEEAHabtBjEQ_AUoAXoECCcQAw&biw=1360&bih=600#imgrc=sK318okbBMjJj

## Rock ptarmigians (*Lagopus muta*)

- Gray and brown upper parts in summer
- White plumage in winter

# Principles of Systematics

## Module# 41

# Individual Variations

# Individual Variations



https://www.google.com/search?q=european+starling+in+october&hl=en&sxsrf=ALeKk00IEQwoUE0nvDEAwcEeS_aFlcXkJg:1603518950740&source=lnms&tbm=isch&sa=X&ved=2ahUKEwil7MvLxczsAhV4UhUIHTdZA0QQ_AUoAXoECBYQAw&biw=1360&bih=600#imgrc=nUnihAZz9S-NkM

## European Starling (seasonal variation)

- Freshly molted bird of October is covered with white spots and all feathers show whitish margins
- In winter edges of feathers wear off

# Individual Variations



https://www.google.com/search?q=european+starling+in+spring%2C+glossy+black&tbm=isch&ved=2ahUKEwi6xauzxszsAhVC-4UKHZh3BakQ2-cCegQIABAA&oq=european+starling+in

## European Starling

- In spring (beginning of breeding season) whole bird is glossy black without the mold of single feather

# Individual Variations



https://www.google.com/search?q=bicycles+anynana+in+wet+season&tbm=isch&ved=2ahUKEwjQsK3DxszsAhULcxoKHfjYAKMQ2-cCegQIABAA&oq=bicycles+anynana+in+wet+season&gs_lcp=CgNpbWcQAzoECCMQJzoHCAAQsQMQQzoECAAQQzoFCAAQsQM6AggAOgYIABAIEB5QyY0uWN6XL2CLmi9oAHAAeACAAegDiAGKRJIBCDItMjEuOC4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=4cKTX5C8Novmafixg5gK&bih=600&biw=1360&hl=en#imgrc=zxMXuCnjPwLPSM

**Seasonal variations of consecutive generations**

- Seasonal plasticity
- These variations are called polyhenism.
- African butterfly *Bicyclus anynana*
- Eyespots on margins (ventral side)---- in wet season
- Very small or No eyespot --- Dry season

# Principles of Systematics

## Module# 42

# Social Variations

# Social Variations

- In social insects there are definite groups of individuals in a colony.
- Each type of individual in a colony perform specific function.
- For examples in a colony of Honey bee:

- Queen, males (drones) and workers, soldiers

# Social Variations

END

## Social variations

- **workers:** All workers are females, product of fertilized eggs.
- They are in majority in the colony.
- May be from 10000 to 50,000 per colony.
- development.
- Projection of colony, food collection, feeding of queen, take care of young ones, cleaning of hive etc

# Principles of Systematics

## Module# 43

# Social Variations

# Social Variations

## Social variation in ants

- Ants live in complex social colonies.
- **Queen:** founder and leader of the colony.
- Primary goal is to populate colony .
- Lay thousands of eggs
- Queen chambers are deep within the colony for protection.
- Long life in some species upto 30 years.

# Social Variations

## Social variation in ants

- **Drones:** Only function is to fertilize female
- Die after fertilization
- Rarely encountered outside the colony
- **Workers:** Foraging and protection
- Sometime: queen lay eggs, winged male and females- **Alates or reproductive.**

# Social Variations

## Social variation in ants

- After maturation leave the colony in swarms.
- Establish new colony

END

# Principles of Systematics

## Module# 44

# Ecological Variations

# Ecological Variations

A. Habitat Variation
B. Variations induced by climatic conditions
C. Host determined variations
D. Density dependent variations
E. Allometric variations
F. Neurogenic color variations

# Ecological Variations

## A. Habitat variation

- Populations of single species in different habitat are visibly different.

**Oyster in still water**

- Round or broader

**Oyster in strong currents**

- Narrow, elongated and usually straight

# Ecological Variations

## A. Habitat variation

**Oyster in clean water on pebble**

- Lower valve is deep and more or less sharply ribbed to acquire strength
- Not needed attachment is with perfectly flat surface

**Oyster in muddy area**

- Valves are less commonly ribbed

END

# Principles of Systematics

## Module# 45

# Ecological Variations

# Ecological Variations

**B. Variations induced by climatic conditions**

- Animals with phenotypic plasticity
- Variations with temperature, draught, cold etc.

# Ecological Variations



https://www.google.com/search?q=apenteles+flaviconchae&tbm=isch&ved=2ahUKEwjkg7Ob_czsAhUD8IUKHQs4BLsQ2-

## C. Host determined variations

- Braconid wasp, *Apanteles flaviconchae* spin white cocoons if reared from blue green caterpillars of *Choliase philodice*

- Golden cocoons if reared from yellow green caterpillars of same species.

# Principles of Systematics

## Module# 46

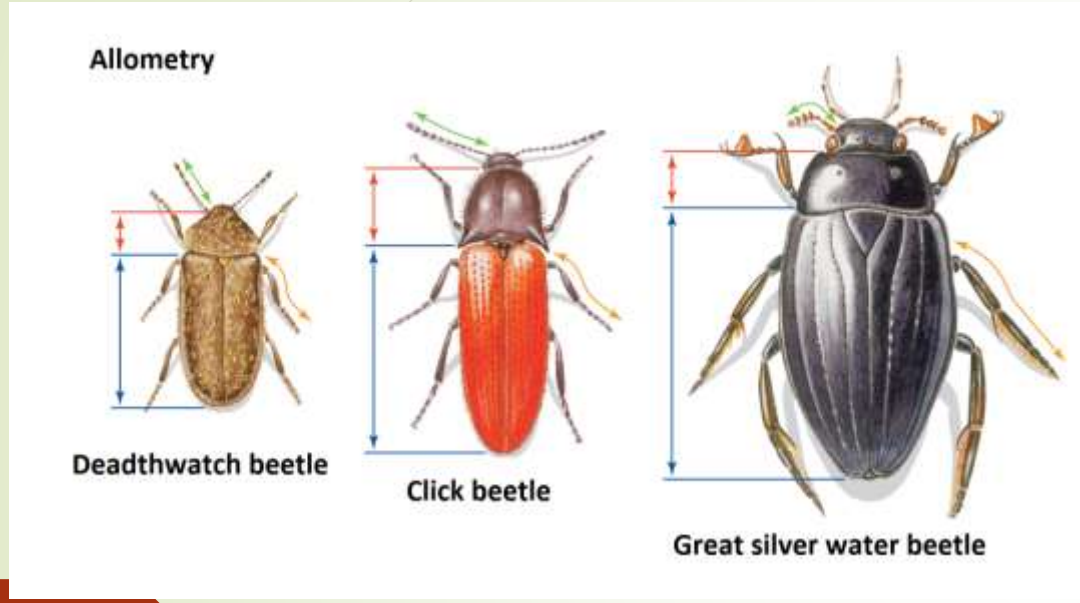
# Ecological Variations

# Ecological Variations

## D. Density-Dependent variations

- When newly hatched nymphs of locust are reared under crowded conditions----Gregarious phase
- Less crowded condition---Transitional phase
- Separate rearing---Solitary phase
- These phase differ in anatomy, colour and behavioral characteristics---

# Ecological Variations




https://www.google.com/search?q=allometric+variations&tbm=isch&ved=2ahUKEwjjrbyKg83sAhXC4YUKHRKVBOwQ2-cCegQIABAA&oq=allometric+variations&gs

## E. Allometric variation

- Allometry results in misappropriate size of some structure in relation to rest of the body.
- This results in animals of different sizes----Allometric variability

# Ecological Variations



https://www.google.com/search?q=chameleon&sxsrf=ALeKk03w9hngHVfxaApti0b4bszoZ9g-

## F. Neurogenic variation

- It is colour change in response to environment.
- Chameleons have Chromatophores in three layers below their transparent skin.
- Xenthophors (Yellow) and erythrophores (red) in upper layer
- Iridophores or guanophores (blue or white) 2nd layer
- Melanophores (dark colour) in deep layer

# Ecological Variations



https://www.google.com/search?q=chameleon&sxsrf=ALeKk03w9hngHVfxaApti0b4bszoZ9g-

## F. Neurogenic variation

- Chameleons change their colours in response to temperature and to chamouflage.

# Principles of Systematics

## Module# 47

# Traumatic Variations

# Traumatic Variations

- **Teratology** is the study of abnormalities of physical development.
- **Taratogen** are the toxic substances which cause abnormal development.
- Mercury, alcohol, lead, thalidomide
- Teratological variations are common in animals.

# Traumatic Variations

- In those forms which undergo metamorphosis injury to an earlier stage may produce abnormalities which are not easily recognized.
- Important if abnormality is in characters of taxonomic importance.
- Modification in wing pattern

END

# Principles of Systematics

## Module# 48

# Principles of Systematics

## Post-Mortem Variations

# Post-Mortem Variations



https://www.google.com/search?q=twelve+wire+bird+of+paradise&sxsrf=ALeKk00Z6PMYPqESiQ3ZTFbrIC6T6hsR6Q:1603611950298&tbm=isch&source

- It is impossible to prevent post-mortem changes in some animals.
- Deep yellow plumage of twelve wire bird of paradise----white

# Post-Mortem Variations



https://www.google.com/search?q=Chinese+jay+bird%2C+green&tbm=isch&ved=2ahUKEwiMgpaQoc_sAhUCYBoKHRVPDI0Q2-

- Chinese jay plumage changes from green to blue in collection
- Wasps over-expressed to cyanide turn bright red
- Insects
- Spiders

# Principles of Systematics

## Module# 49

# Principles of Systematics

## Genetic Variation

# Genetic Variations

- Difference in the individual members of a population
- Genetic variations are important for survival and adaptation of species
- Evolution
- Natural selection

**Causes of evolution**

- Mutations
- Genetic recombination
- Genetic drift
- Migration

# Genetic Variations

- Population s with small variation
- Population with large variability among individual
- Variability in characters
- Variability  biological fitness

**Major causes variability**

- Mutations
- Genetic recombination
- Mutations are not always bad (Sickle cell disease—present Malaria)  END

END

# Principles of Systematics

## Module# 50

# Principles of Systematics

## Types of Genetic Variation

# Type of Genetic Variation

**Sexual Dimorphism**

Many of the variants in a population are sex-associated.

- They may be sex limited (express themselves in one sex only) or otherwise associated with one sex or the others
- **Primary Sex differences (PSD)** Difference in primary sex organs such as gonads genetalia etc.)

# Type of Genetic Variation

**Sexual Dimorphism**

- **Secondly Sex differences (SSD)**
- Striking differences in male in females e.g bird of paradise, humming bird and ducks
- Due to differences different sexes were considered different species until painstaking works by naturalist established their relationship.

# Type of Genetic Variation



https://www.google.com/search?q=King+parrot+Eclectus+roratus+,+male+female&sxsrf=ALeKk02h0vVHJDTAOjHTEAE5PQBiERJ9_Q:1604040706059&source=lnms&tbm=isch&sa=X&ved=2ahUKEwirmfWj3dvsAhXBDGMBHZgZCBEQ_AUoAXoECBkQAw&biw=1360&bih=657#imgrc=kLDkWlLeXUZ6cM

**King parrot *Eclectus roratus*** (Muller) of Papuan region

- Male is green with orange bill
- Female is red and blue with black bill.
- Described in 1776, two considered different for nearly 100 years
- Until naturalist proved conclusively in 1873 that they belonged togather.

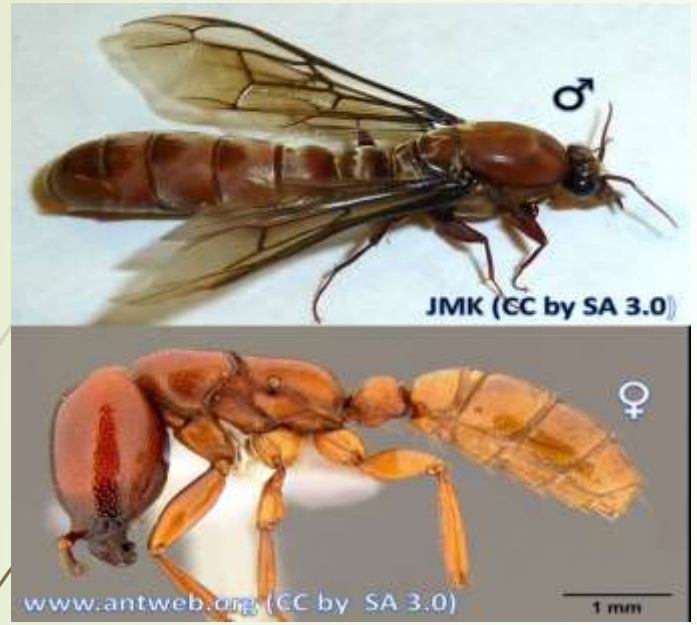
# Principles of Systematics

**Module# 51**

# Principles of Systematics

## Types of Genetic Variation

# Type of Genetic Variation




https://www.google.com/search?q=African+Ant+Dorylus+male+females&tbm=isch&ved=2ahUKEwiajc-y39vsAhUU_RoKHbZ2CW0Q2-cCegQIABAA&oq=African+Ant+Dorylus+male+females&gs_lcp=CgNpbWcQA1DywQFY1egBYJ_qAWgAcAB4AIAB_QOIAfQakgELMC4xLjkuMi4wLjGYAQCgAQGqAQtnd3Mtd2I6LWItZ8ABAQ&sclient=img&ei=ObqbX9qMLpT6a7btpegG&bih=657&biw=1360#imgrc=-C_HA25vlwfHRM&imgdii=kDy2tGctcC0wiM

## Sexual Dimorphism

Striking sexual dimorphism in Hymenoptera.

- Males of African Ant *Dorylus* are so unlike others ants that they were not recognized as ants and considered belong to different family.
- In certain families of wasps small wingless females and large winged males are so different that some taxonomists use different nomenclature for both.

# Type of Genetic Variation

**Sexual Dimorphism**

Some time males and females are so different that it is very difficult to decide which male belong to which female:

- Field study by ecologist
- Which male is attracted to which female

**END**

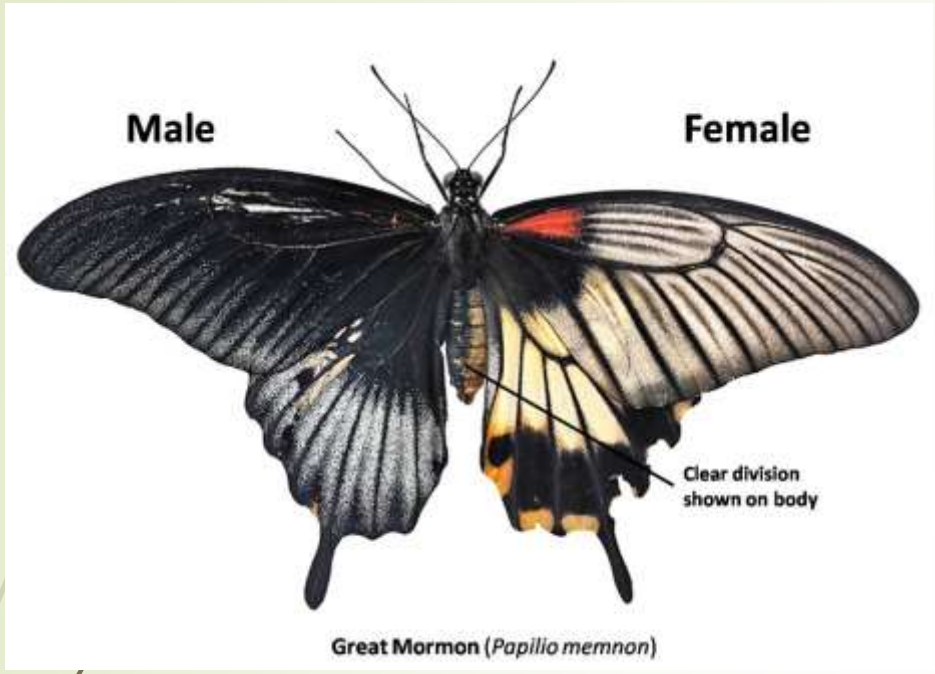
# Principles of Systematics

## Module# 52

# Principles of Systematics

## Types of Genetic Variation

# Type of Genetic Variation




https://www.google.com/search?q=gynandromorph+of+papilo+adardanus&tbm=isch&ved=2ahUKEwiGuK-I5tvsAhWO0YUKHdbFCn4Q2-cCegQIABAA&oq=gynandromorph+of+papilo+adardanus&gs_lcp=CgNpbWcQAzoECAAQQzoCCABQoa0BWImWAmC8mQJoAHAAeAGAAYYJiAGNPpIBEDAuMi4xMS4yLjAuMS4yLjKYAQCgAQGqAQtnd3Mtd2l6LWItZ8ABAQ&sclient=img&ei=dcGbX8apAY6jlwTWi6vwBw&bih=657&biw=1360#imgrc=sCA1PSyDaBTAiM

## Gynandromorphs and intersexes

Individuals that shoe male characters in one part and female characters in other parts of body.

- Two halves of the body may be of opposite sexes.
- The division male be transverse or the sex characters may be scattered in mosaic.
- In later case symmetrical variants may be produced.
- Produced due to unequal

# Type of Genetic Variation



https://www.google.com/search?q=intersexes,+Example:+Lymantria&hl=en&sxsrf=ALeKk01diF3YZHp0xEbXUCJVibS92jJ0-w:1604043782080&source=lnms&tbm=isch&sa=X&ved=2ahUKEwi_udbe6NvsAhXiA2MBHQRuCRYQ_AUoAXoECAsQAw&biw=1360&bih=657#imgrc=_xPgprPy1k57CM&imgdii=Uia9hvBTpb27TM

## Intersexes

Blending of male female characters. body. Intersexes are produced due to:

- Upset in balance between male-tendency and female-tendency genes
- May result from irregularities in fertilization or mitosis
- Physiological disturbances associated with parasitism.
- Example: Gipsy moth, Lymantaria

# Principles of Systematics

## Module# 53

# Principles of Systematics

## Types of Genetic Variation

# Type of Genetic Variation



https://www.google.com/search?q=agemic+generation+gall+wasp&hl=en&sxsrf=ALeKk02oVJis8aalDyE2Y0wVwXJoaodWcQ:1604054434479&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiTu5G2kNzsAhXVh1wKHZ_ZBjoQ_AUoAXoECBkQAw&biw=1360&bih=657#imgrc=SO7OXhMqVG7A9M

## Reproductively different generations

- In many insects thesre is an alternation of generations that is very confusing to the taxonomist.
- In cynips (gall wasp) agemic generation is different from bisexual one.
- In aphids parthenogenetic wingless females are different from winged females of sexual

# Type of Genetic Variation



https://www.google.com/search?q=Slugs,+Arion&hl=en&sxsrf=ALeKk02zw1UrLr9HBnEK7giReGZMJt64pA:1604055019292&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiOuv_MktzsAhWjnVwKHTUaB1MQ_AUoAXoECCUQAw&biw=1360&bih=657#imgrc=AKvOHe8xqdKJVM

## Sexual and uniparental strains

- In some species of slugs (Arion) several strains are found which differ in their sexuality.
- Sexually reproducing strains are highly polymorphic in their enzymes
- While self fertilizing hermaphroditic strains are highly homozygous.

# Type of Genetic Variation

## Discontinuous variation (Polymorphism)

- Certain members of population have certain conspicuous characters.
- Such discontinuous individual variations is called polymorphism.
- Frequently controlled by single gene.
- Follow simple Mendelian inheritance.

# Type of Genetic Variation



https://www.google.com/search?q=laddy+beeles%2C+spotted&tbm=isch&ved=2ahUKEwi0gs_PktzsAhVRlxoKHdwxAYwQ2-

- Spotting in lady beetles
- Industrial melanism in moths
- Understanding of polymorphism is helpful to taxonomists.
- More than 100 birds species names were given to morphs.

# Principles of Systematics

## Module# 54

# Principles of Systematics

## Types of Genetic Variation

# Type of Genetic Variation

## Continuous variations

- Most common type of individual variations is due to slight genetic differences among individuals.
- No two individuals (except monozygotic twins) in a population of sexually reproducing animals are exactly alike
- Morphologically or genetically
- Small difference/special

# Type of Genetic Variation

**Continuous variations**

- Each characters may show a different degree of variability within a single population.
- Different degree of variability among related species.
- Why one species is highly variable while another is NOT?
- Answer this question required lot of background

# Type of Genetic Variation

## Continuous variations

- Some early taxonomists vastly underrated individual variations in many genera of animals.
- Species of the snail genus *Melania* (fresh water and brackish water) have been described largely on the basis of shell characters as presence or absence of spines and diagonal versus spiral ribs.
- Revision showed 114 species

# Principles of Systematics

## Module# 55

# Principles of Systematics

**Systematic Collection**

# Systematic Collection

## Systematic collection

Systematic collection is a method of collecting valuable biological specimens and preserving them along with specimen documentations such as labels and notations. It possess various attributes:

- Raw material of classification
- Document earth's biodiversity
- Initially additional specimens of species—duplicates- exchange purpose

# Systematic Collection

**Systematic collection**

- As there is variability in members of species
- Adequate sample of every population should be collected and preserved.

**What is adequate?**

- More material is needed in s species with strong individual and geographical variation than in uniform species.

# Systematic Collection

**Objective of study?**

Simple study or genetic variability

- When population analysis is attempted dealing with size, proportions, coloration, or polymorphism large samples from numerous locations are needed.
- For example in fishes where each population of a species may vary in some metric characters (fin ray, vertebrae), a single specimen

# Systematic Collection

- Facilitate the process of researching relationships among and within different groups of organisms biodiversity
- We can use systematics to find out how insects are related to the rest of the invertebrates.

# Systematic Collection

**Preservation of collection samples**

The collected samples are preserved by;

- Wet method and
- Dry method

# Systematic Collection

## Wet method

- Wet method of preservation involves the use of liquid preservatives to prevent deterioration.
- Chemicals like alcohols, formalin etc. are commonly used for specimen preservation. Animals belonging to phylum porifera, annelida, nematodaare mostly preserved by wet method.

# Systematic Collection

## Dry method

- In this method, specimens are preserved in dry state is a traditional method to place several kinds of insects in a box for an indefinite period allowing them to become dry.
- Not advisable for soft bodied insects.
- The process of pinning insect specimens is a dry method to preserve and

END

# Principles of Systematics

## Module# 56

# Principles of Systematics

**Systematic Collection**

# Systematic Collection

## Importance of systematic collection

- Safe storage of historical material such as fossils
- Rarely found specimens
- Specimens of geographically inaccessible areas
- Extinct species specimens to place in museums for documentation
- Provide permanent record of faunas and floras from

# Systematic Collection

## Importance of systematic collection

- Help in the verification of original data or for renewed study from new knowledge and technique
- The systematic method assumes the size of the population is available
- Aid in identification of organisms through comparison
- Teaching/Public Outreach:

# Systematic Collection

- Facilitate the process of researching relationships among and within different groups of organisms biodiversity
- We can use systematics to find out how insects are related to the rest of the invertebrates.

END

# Principles of Systematics

## Module# 57

# Principles of Systematics

## Collection Methods

# Collection Methods

## Collection methods

There are various collection methods to collect specimens

- Black light trap
- Sweep net
- Suction device
- De-Vac
- Pit fall method
- Aspirator
- Beating sheet
- Trawling
- Dipper
- Hand picking

# Collection Methods



http://mississippientomologicalmuseum.org.msstate.edu/images/traps/blacklight.sheet.

## Black light trap

- Entomologists often collect night flying insects using a black light and sheet.
- The black light is suspended in front of a white sheet.
- Insects attracted to ultraviolet light fly toward the light, and alight on the sheet.

# Collection Methods







## Black light trap

- **A:** Tying the rope between two trees
- **B:** Sheet and light are hung
- **C:** A quite night of collecting

https://mississippientomologicalmuseum.org.msstate.edu/collecting.preparation.methods/Blacklight.traps.htm

# Collection Methods



https://ffs.ipm-info.org/wp-content/uploads/2015/05/Collecting-insects-with-a-sweep-net.

## Sweep Net

- Sweep nets are used to collect arthropods from long grass and other types of vegetation.
- This net is with round mouth and canvas bag fixed with a long rod.
- Its advantages lie in its simplicity and speed, and its ability to collect relatively dispersed insect species on top of the vegetation.

# Collection Methods




https://www.google.com/search?q=suction+device+for+insect+collection%2F+devac&tbm=isch&ved=2ahUKEwikjtP_hN7sAhUKahoKHTraDt0Q2

## Suction device/De-Vac

- Used for collection of arthropods from vegetation, long grasses, forest, orchards etc.
- Engine operated.
- Large scale insect collection.

# Principles of Systematics

## Module# 58

# Principles of Systematics

## Collection Methods

# Collection Methods



https://upload.wikimedia.org/wikipedia/commons/b/b2/Barber_pitfall_trap

## Pit Fall Method

- In this method, the insect falls into a pit, a container that is buried in the soil.
- The pitfall trap catches ground-dwelling insects.
- Glass jar is buried in soil with its rims at the level of ground.
- Filled with formalin/alcohol
- Detergent drops to beak surface tension
- May be covered

# Collection Methods

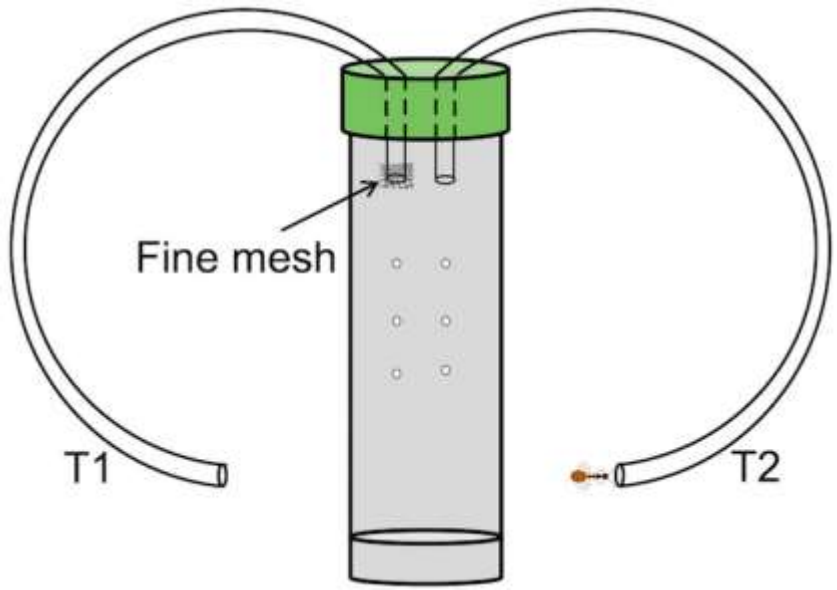


https://bioquipinc.com/wp-content/uploads/2017/05/%C3%92Pooter%C3%93-Style-Aspirator-4

## Aspirator

- Small insects or insects in hard to reach places, can be collected using an aspirator.
- The aspirator is a vial with two pieces of tubing, one with a fine screen material over it.
- By sucking on one tube, you draw the insect into the vial through the other.

# Collection Methods




https://www.google.com/search?q=aspirator%2C+insect+colection&tbm=isch&ved=2ahUKEwiNwOKFiN7sAhUI7xoKHS47DLQQ2-cCegQIABAA#imgrc=7omrJPNCN0eAqM

## Aspirator

- The screen/fine mesh prevents the insect or anything else unpleasant from getting drawn into your mouth.

# Principles of Systematics

## Module# 59

# Principles of Systematics

## Collection Methods

# Collection Methods

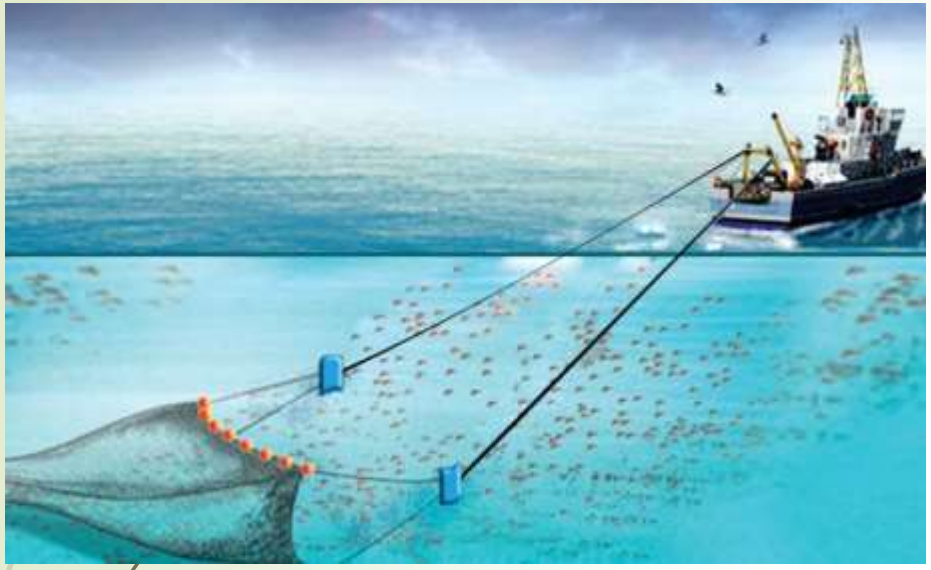


http://www.nzffa.org.nz/images/design/beatingbig.jpg

## Beating Sheet

- To study insects that live on branches and leaves, like caterpillars, a beating sheet is a tool to use.
- Stretch a white or light-colored sheet below the tree branches. With a pole or stick, beat the branches above.
- Insects feeding on the foliage and twigs will fall down onto the sheet, where they can be collected.

# Collection Methods



https://bharatabharati.files.wordpress.com/2015/03/bottom-trawling.jpg?w=460&h=260

## Trawling

- Trawling is a method of fishing that involves pulling a fishing net through the water behind one or more boats.
- The boats that are used for trawling are called trawlers or draggers
- In Trawling baited fishing lines are drawn through the water.
- Trawling is commonly used as a scientific sampling, or survey method for marine

# Collection Methods



https://www.researchgate.net/profile/Richard_Nisbett/publication/265226839/figure/download/fig15/AS:669421387382787@1536613904116/Dipper-for-the-collection-of-mosquito-larvae-41.jpg

## Dipper

- The most commonly used larval collection method is the "standard dipper.
- Estimates of **larval** density can be carried out by counting the number of mosquito **larvae** per **dip**, using a standard 1-pint **dipper**.
- A minimum of three to five dips should be completed at each site.

**E.g.** collection of mosquito larvae

# Principles of Systematics

## Module# 60

# Principles of Systematics

## Curation of Collection

# Curation of Collection

- A **curator** is a person who manages, administers or organizes a collection for exhibition.
- A museum curator is custodian of collection of museum and is responsible for their management.
- **Curation** is process of identification and organization  animal collection.

# Curation of Collection

- All types of museums have at least one curator.
- A large museum may employee several curators
- Each have specialization in particular fields.
- American Museum of Natural History (AMNH), New York
- Housing of collection in fire free buildings that are dust proof
- Air-conditioned Building

# Curation of Collection

## Cataloging

- Museum number
- Original field number
- Scientific Name (at least generic name)
- Exact locality
- Date of collection
- Name of Collector
- Remarks
- Card systems/electronic data processing
- Exchange of materials

END

# Principles of Systematics

## Module# 61

# Important Properties of Texa Names

# Important Properties of Texa Names

**Three important properties**

- Uniqueness
- Universality
- Stability

# Important Properties of Texa Names

**Uniqueness**

- Names of an animal gives immediate access to all known information about that taxon.
- Every names has to be unique
- If several names---clear cut method should be there to determine valid name.
- Principle of priority usually decides in case of conflict.

# Important Properties of Texa Names

**Universality**

- Scientific communication would be very difficult if there were only vernacular names for animals.
- Specialist would have to learn the name of texa in different languages in order to communicate with each other.
- Zoologist use single language

# Important Properties of Texa Names

END

## Stability

- A name should be stable
- Any change of a well-established name is likely to produce confusion and impede information retrieval.
- Even a principle of priority can be set aside by International Commission on Zoological Nomenclature in cases where stability is

# Principles of Systematics

## Module# 62

# Principles of Systematics

## Type Specimen

# Type Specimen

- A type is a specimen (or a group of specimens) which serves to illustrate the defining characters of a species or genus.
- A type is a specimen serve as a reference which can be used by later scholars to verify the identity of species being described.

# Type Specimen

- Precise collecting locality and other relevant data on the labels of specimen.
- Sex
- Developmental stage or form to which the type is referable
- In the case of parasites, the name of host species

# Type Specimen

- Name of collector
- Specimens numbered
- Altitude of the type locality or depth in meters below sea level at which the holotype was taken
- In the case of fossils species, geological horizon

# Type Specimen

- Type designation or fixation should always be completed before publication
- Type designation should be clear and unambiguous
- The deposition and museum number of the type always be recorded

END

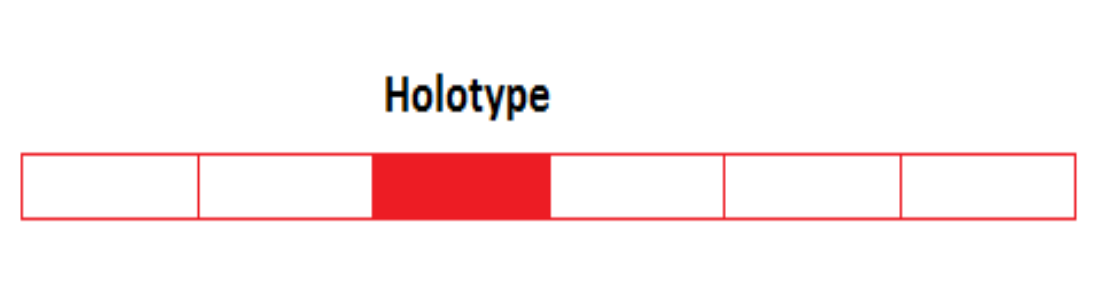
# Principles of Systematics

## Module# 63

# Principles of Systematics

## Kinds of Type Specimen

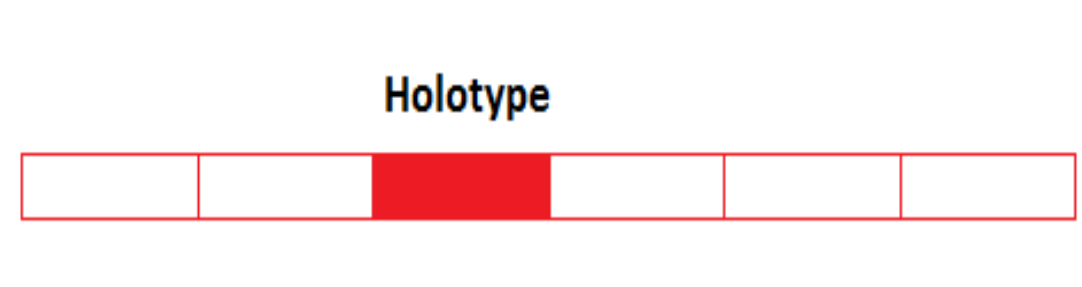
# Kind of Type Specimens




## Holotype

- A single specimen designated or indicated as the type by the original author at the time of publication of the original description of a species”
- If no  specimen designated or indicated as the type by the original author but it is clear that only one specimen was used for description then it will be considered as Holotype.

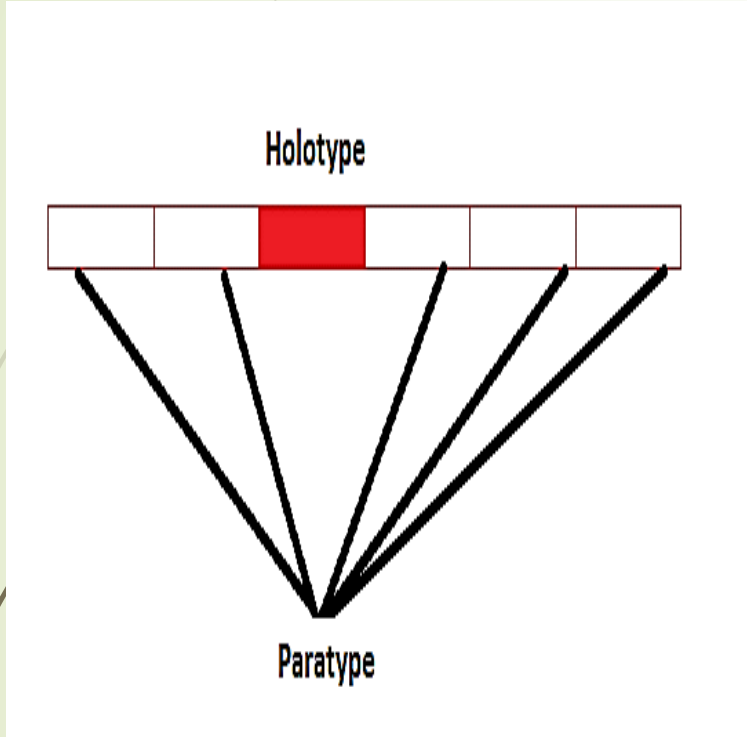
# Kind of Type Specimens




## Holotype

- For a species described after 1999, holotype must be designated.

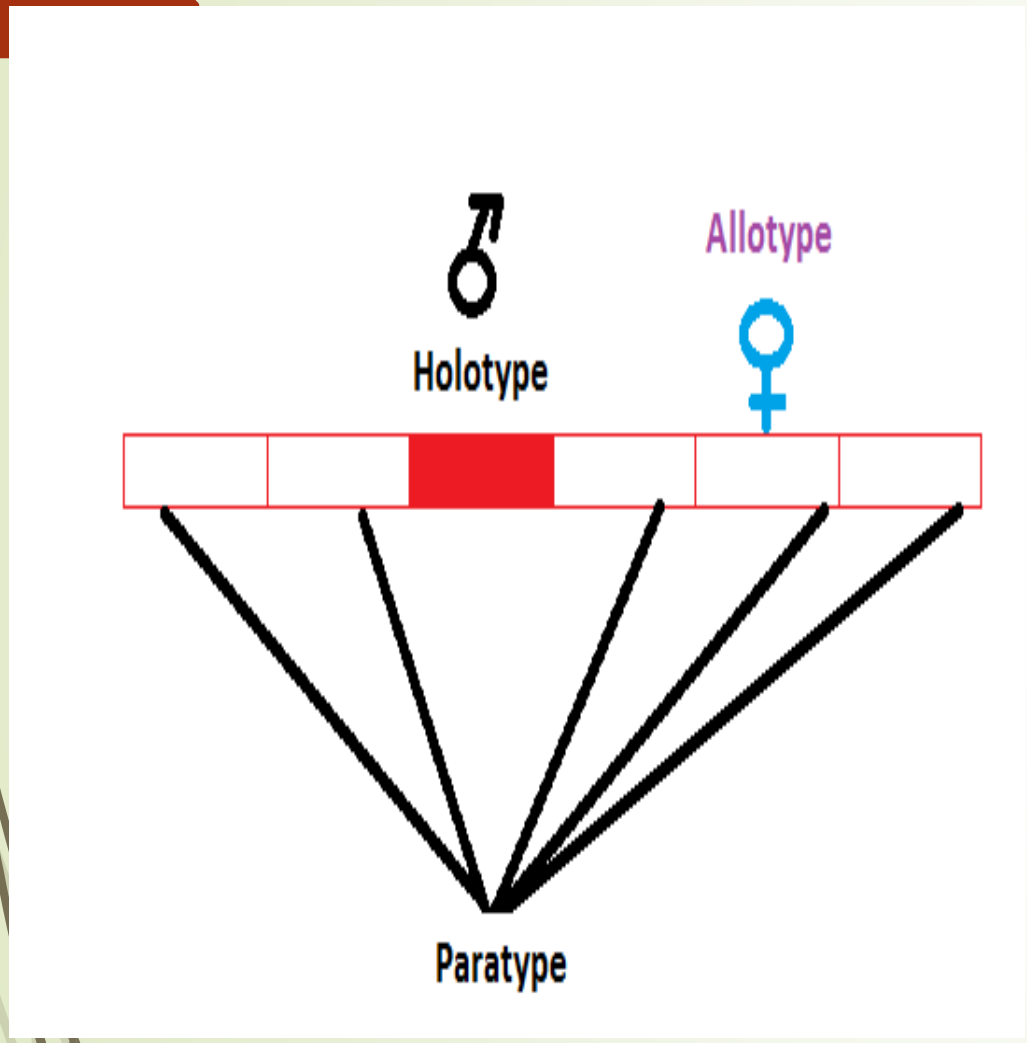
# Kind of Type Specimens




**Paratype**

- The specimen other than the holotype, which has been used in the description.

# Kind of Type Specimens




## Allotype

- A specimen of opposite sex as the holotype, which has been used in the description.
- Its is often used to describe a morphological character not shown in the Holotype

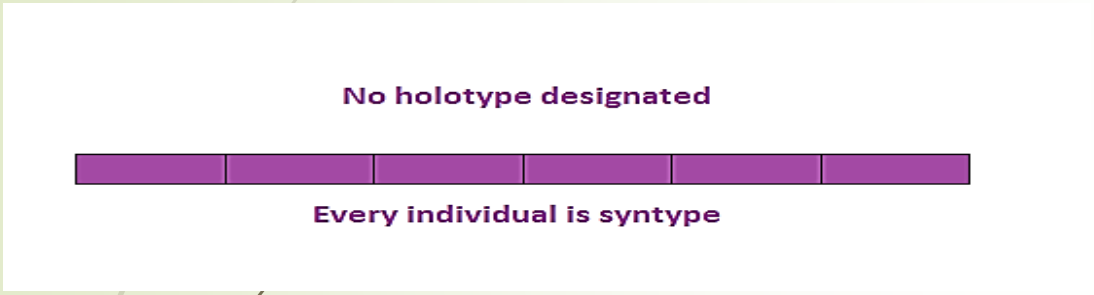
# Principles of Systematics

## Module# 64

# Principles of Systematics

## Kinds of Type Specimen

# Kind of Type Specimens

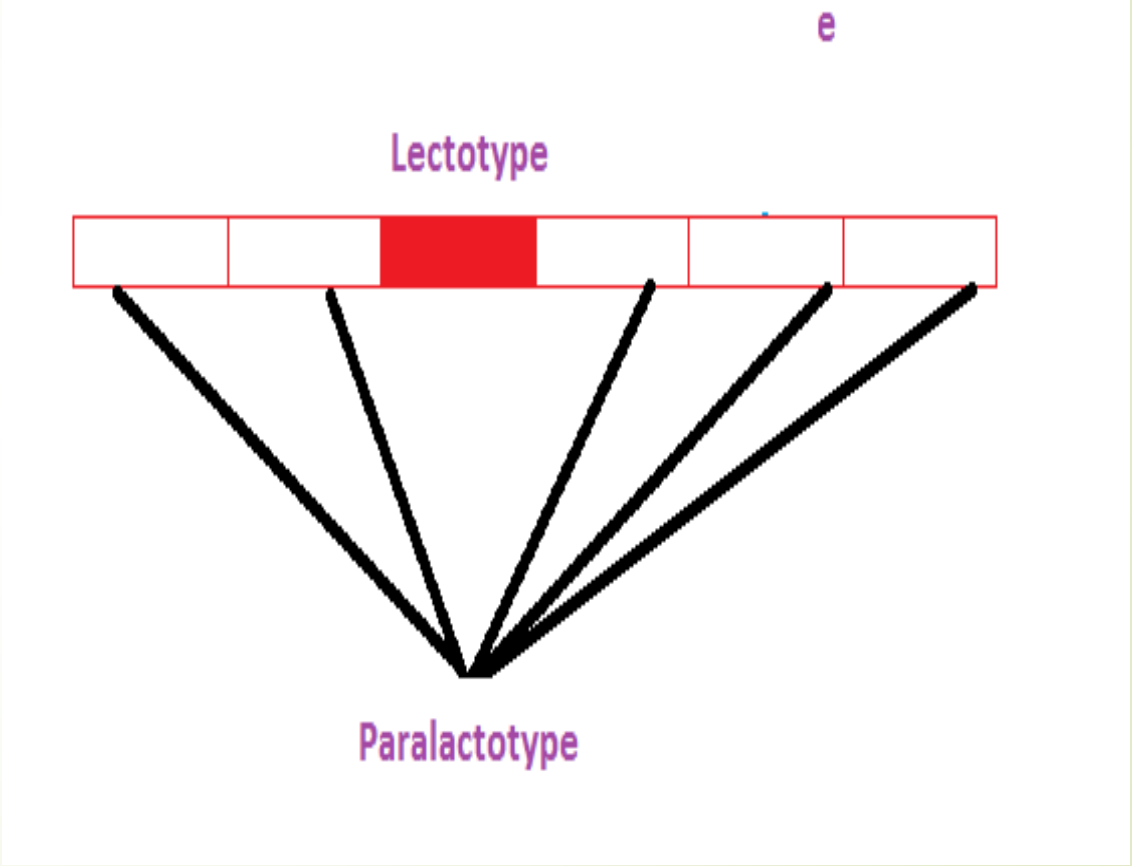



## Syntype

- When description is based on two or more specimens, no holotype was designated, every specimen in a type series will be syntype

# Kind of Type Specimens

## Lectotype/Paralectotype



# Kind of Type Specimens

## Neotype

- A specimen selected as a type subsequent to the original description in the cases where the original types are known to have been destroyed or suppressed by commission

# Kind of Type Specimens

## Isotype

- A specimen that is a duplicate of holotype or very similar to the type specimen collected at the same time but not from same place as holotypes, paratype or syntype and not included in any of these.
- Can be used as a reference specimen if the type specimen is lost.

# Kind of Type Specimens

## Monotype

- When holotype of the species is the only specimen known to science, it is known as monotype.

END

# Principles of Systematics

## Module# 65

# Principles of Systematics

**Type Locality**

# Type Locality

- Place of collection of Holotype and lectotype.
- Paratype can be from different locality to holotype.
- Specimens collected at type locality are called **Topotypes.**
- Population that occur at type locality is called **Topotypical population.**

# Type Locality

- In the case of parasites, the term type host is used to indicate the host organism from which the type specimen was obtained.
- Importance of type locality

END

# Principles of Systematics

## Module# 66

# Principles of Systematics

**Type Locality**

# Type Locality

- According to Article 72h type locality is the place from where type specimen was taken.
- Specimens collected from type locality are called topotype
- Population that occur at type locality is topotypical population

# Type Locality

- Damage holotype
- Information about environment
- Temperature/Humidity/soil type
- Missing type locality
- Article 72H contains advice on designations or restriction
- Designations or restriction not done routinely but by specialists by their revisionary work

END

# Principles of Systematics

## Module# 67

# Principles of Systematics

## Concept of Priority

# Concept of Priority

- Fundamental principal of Zoological nomenclature
- Essentially it is a principle of recognizing first  valid name of a species.
- A principle in taxonomy: the first properly published name of a species or genus takes precedence over any subsequently published
- (Article 19)

# Concept of Priority

- The name which was use first time should be valid name
- Once name has been used, no subsequent publication of that name for other texon.
- There are formal provisions for making exceptions to this principle.

# Concept of Priority

## Limitations

- Priority starts from May 01, 1753
- Apply only to family level
- Not according to Zoological code

# Concept of Priority

- If an archaic or obscure prior name is discovered for an established taxon, the current name can be declared conserved name (zoology), and so conserved against the prior name.
- Conservation can may be avoided in Zoology as these names fall in formal category of ***Nomen oblitum.***

# Concept of Priority

**Nomen oblitum**

- In taxonomy, a forgotten name; i.e. the name of a senior synonym that has not been used in the zoological literature for at least 50 years.
- Such names should not be used if they upset a better-known name, but should preferably be suppressed by the ICZN.

# Concept of Priority

- If a current name for a taxon have archaic or obscure homonym (same name to different taxon), the current name can be declared as ***Nomen protectum*** (available name).

**END**

# Principles of Systematics

## Module# 68

# Principles of Systematics

## Concept of priority

# Concept of Priority



https://www.google.com/search?q=leopard+shark+&tbm=isch&ved=2ahUKEwj5i_2yg_PsAhWU0oUKHUviByEQ2-cCegQIABAA&oq=leopard+shark+&gs_lcp=CgNpbWcQAzICCAAyAggAMgIIADICCAAyAggAMgIIADICCABQq52IEVirnYgRYKmmiBFoAHAAeACAAdgBiAHYAZIBAzItMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=_O6nX_mdCpSIlwTLxJ-IAg&bih=600&biw=1360#imgrc=uTp_hTV4HeDvJM

## What does priority means?

- The first Scientific name applied to the leopard shark was ***Triakis californica***
- Proposed by British Zoologist Edward Gray (1851)
- He did not provided proper description
- In 1854 ***Mustelus felis*** *by william Orville published*
- *In 1855* ***Triakis semifasciata***

# Concept of Priority

**Error in date recording**

- ***Triakis semifasciata***
- ***Nomen protectum*** (available name)
- *Mustelis felis* was invalidated (as a ***nomen oblitum*** ).

# Concept of Priority

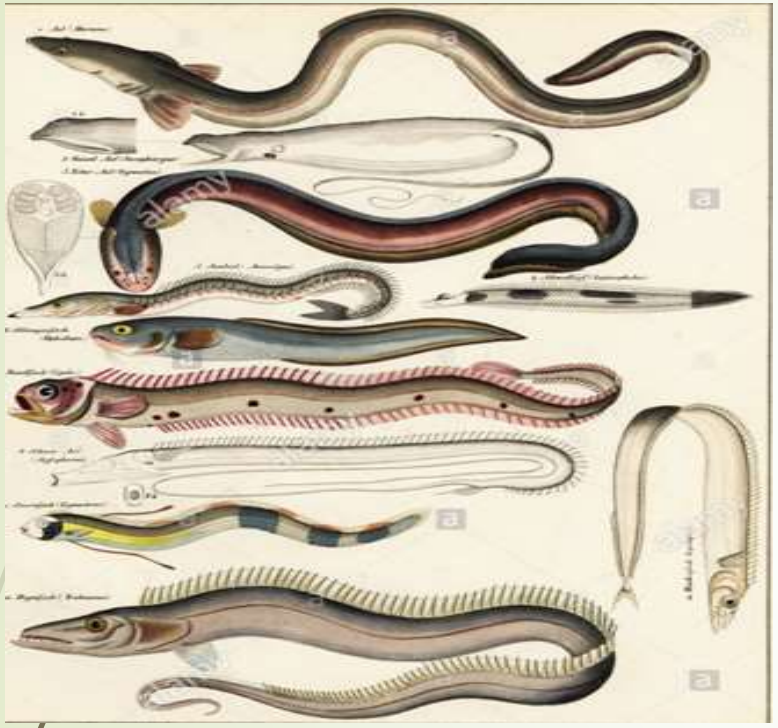


https://www.google.com/search?q=pronghorn&sxsrf=ALeKk016hNi2hIH42LJ1FojE0bD517_3Tw:1604565941448&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj_pNX3gevsAhWMYcAKHUJyCboQ_AUoAXoECDQQAw&biw=1360&bih=600#imgrc=zCJQlNSIekozdM

**Example**

- In 1855 John Edward Gray published the name of Pronghorn *(Antilocarpa anteflexa)*
- Based on pairs of horns
- 1815 George Ordin Published *Antilocarpa americana* for the same species.
- So the older name *Antilocarpa americana* takes the priority over *the*

# Concept of Priority



https://www.google.com/search?q=Muraena+anguilla&tbm=isch&ved=2ahUKEwiv5d7_gevsAhVNZRoKHcT3BYAQ2-cCegQIABAA&oq=Muraena+anguilla&gs_lcp=CgNpbWcQA1CamjVYmpo1YOmgNWgAcAB4AIAB8wOIAfMDkgEDNC0xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=xrujX6_qFs3KacTvl4AI&bih=600&biw=1360#imgrc=EvrSudnM3XTF-M

**Example**

- In 1856 Johann Jakob Kaup published the name *Leptocephalus brevirosteris* for new species of eel.
- In 1893 same name was given to juvenile form of European eel whose name was *Muraena anguilla* given by Carl Linnaeus in 1758.
- Old name was retained.

# Principles of Systematics

## Module# 69

# Principles of Systematics

## Concept of Availability

# Concept of Availability

- Taxonomic name which should be used is called Valid (Accepted, correct) name.
- The name must have been published according to requirements detailed in Articles 8-20.
- An available name can be valid or not
- An unavailable name is never valid.
- Availability criteria vary depending on nomenclature

# Concept of Availability

- The term available means "legitimate"
- A name may be available in this technical sense even though as Junior homonym
- It is automatically invalid at the time of publication (Article 53 and 59a).
- Need to differentiate between **available** and **valid names**

# Concept of Availability

- Available **(properly published).**
- Valid **(not preoccupied by senior synonym or homonym)**
- Article 11-19 describe prerequisite for making a specific name available.
- Particularly article **13a** is very important.

# Concept of Availability

- According to article **13a** names published after 1930 must be
- Accompanied by description that states the characters that are purported to differentiate the Taxon
- It is good taxonomic practice to include in the original description not only such diagnostic information but also differential

**END**

# Principles of Systematics

## Module# 70

# Principles of Systematics

## Concept of Availability

# Concept of Availability

**Nomen nudum**

- A name published without satisfying the condition of availability is called nomen nudum.
- Especially if it fails to satisfy the conditions of Article 12-16.
- A nomen nudum has no standing in Zoological nomenclature

# Concept of Availability

**Nomen nudum**

- If editors of journals were more familiar with the provisions of the Code, they would not permit publications of so many unavailable names because they violate one or another of Article 8-20.

# Concept of Availability

## Differential Diagnosis

- Description of characters after 1930 (Article 13a)
- Comparison with close relative

- Description of diagnostic differences (NOT mandatory)

# Concept of Availability

## Names Given to Hybrids

- Hybrids are normally individuals not population
- Not considered as Texa
- Article 1b (3) states clearly that a name given to an animal that is later found hybrid remain available for the purposes of homonymy, not for purposes of Synonymy
- A name given to hybrid can not be applied to either of parental species.

# Concept of Availability

**Synonyms**

- Different names for same thing
- Earliest published name **senior synonym**
- Later published name **junior synonym**

# Concept of Availability

**Types of Synonyms**

Objective synonym

- The name that objectively refer to the same thing/name based on the same specimen or illustration (at Genera level )

Subjective synonym

- Name based on different type material (*Antelocarpa anteflexa* 1855 vs *Antelocarpa americana*)
- *Antelocarpa anteflexa* was unusual specimen of *Antelocarpa Americana)*

**END**

# Principles of Systematics

## Module# 71

# Principles of Systematics

## International Code for Zoological Nomenclature

# International Code for Zoological Nomenclature

## Zoological Nomenclature

- System of scientific names applied to taxonomic units of animals
- The term nomenclature comes from Latin words
- **Nomen** which means Names
- **Calare** mean to call

- Call by names

# International Code for Zoological Nomenclature

**Important requisite of taxonomic names**

- **Uniqueness**
- **Universality**
- **Stability**

# International Code for Zoological Nomenclature

**ICZC**

- Set of rules for naming animals and resolution of nomenclatural problems
- Authoritative document that contain valid rule of Zoological nomenclature
- Original codes and its revision were the work of International Congress of Zoology

# International Code for Zoological Nomenclature

## ICZC

- Now these congresses are defunct
- Role have been taken over by **General Assemblies of International Union of Biological Sciences** (IUBS)
- International Congresses of Zoology have appointed a judicial body (**International Commission on Zoological Nomenclature**) to interpret the rules and publish opinions on controversial issues.

# International Code for Zoological Nomenclature

## ICZC

- Article 76 through 82 specify the duties of the commission.
- Latest edition of code was published in February 1985
- Can be ordered from International Trust for Zoological Nomenclature 56, Queen Gate London or
- University of California Press, Berkeley and Los Angeles.

# International Code for Zoological Nomenclature

## ICZC

- Code consists of three parts
- The code proper
- Appendices (A section giving additional information at the end of Book)
- Glossary (a list of technical or special words)

# International Code for Zoological Nomenclature

## Code proper

- **Preamble:** Introduction to book or written documents
- 90 articles grouped in 18 chapters
- Each article is composed of one or more mandatory rules
- Followed by recommendations and illustrative examples
- Use of recommendation is not mandatory

# International Code for Zoological Nomenclature

## Appendices

- Appendices are like recommendations not to be followed strictly as article.
- These are five in number and are designated as A-E.

## Glossary

- The terms used in the text are clearly defined in the Glossary.

END

# Principles of Systematics

## Module# 72

# Principles of Systematics

## Molecular Systematics

# Molecular Systematics

- Classification of organisms on the basis of distribution and composition of chemical substances in them.
- Comparisons of Molecular components (DNA, RNA and Proteins) help to establish relationships among different animal texa.
- **Molecular Phylogenetics-** relationships based on

# Molecular Systematics

- First phylogenetic tree based on molecular data 1967 **Fitch and Margoliash**
- Researcher realized that although traditional taxonomic methods are good but Molecular evidences could be final confirmatory
- Phylogenetic studies help to assess the historical processes which affect relationships

# Molecular Systematics

- Phylogeographic studies help to understand the distribution
- Phylogenetic and Phylogeographic studies started with the introduction of mDNA markers in population genetic studies.
- The goal is to provide insight into the history of groups of organisms and the evolutionary processes that

# Molecular Systematics

- The use of amino-acid or nucleotide-sequence data in determining the evolutionary relationships of different organisms.
- Essentially it involves comparing the sequences of functionally homologous molecules from each organism being studied, and determining the number of differences between them.

# Molecular Systematics

- The greater the number of differences, the more distantly related the organisms are likely to be.
- Genetic distances
- Cluster analysis
- Gel electrophoresis
- Agarose/PAGA

END

# Principles of Systematics

## Module# 73

# Principles of Systematics

## Molecular Systematics

# Molecular Systematics

## Applications

- Solving taxonomic problems
- Reorganization of tree of life
- I identifications of immature stages
- I identification of cryptic species
- Identification of disease vectors
- Testing food authenticity
- Monitoring of illegal trade

# Molecular Systematics

## Applications

- Identification of sibling species and sub-species

END

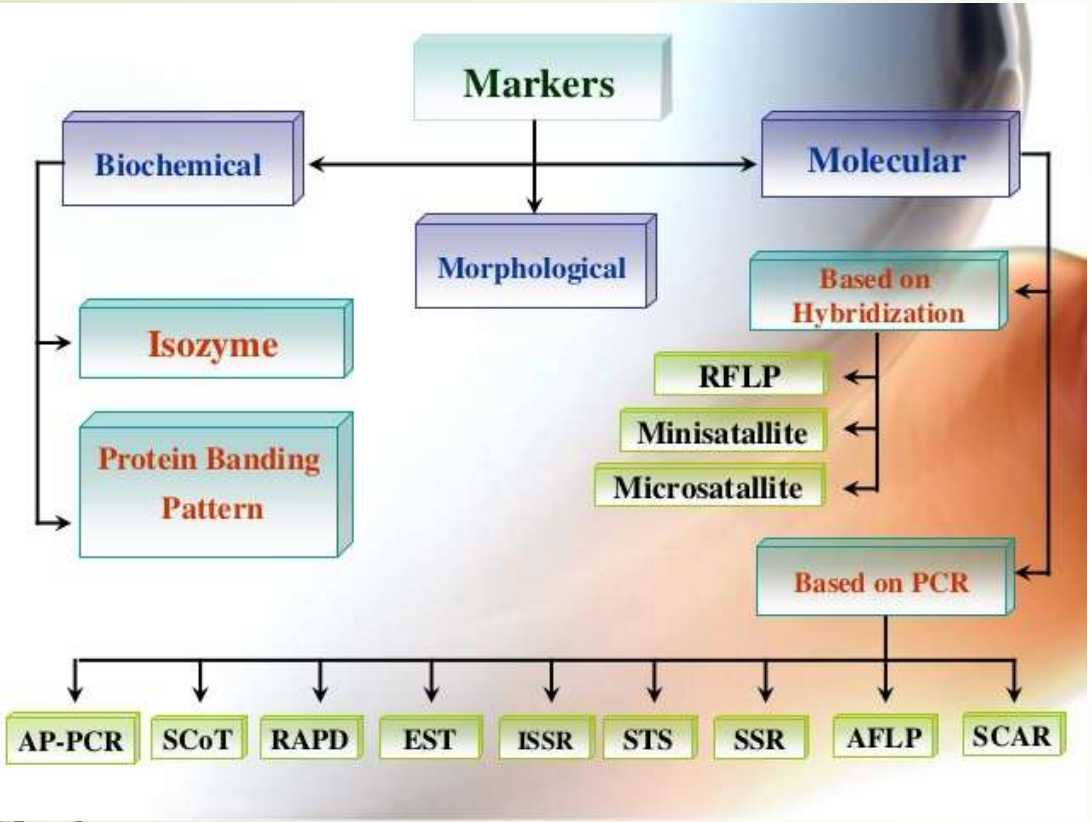
# Principles of Systematics

## Module# 74

# Principles of Systematics

## Molecular Markers

# Molecular Markers

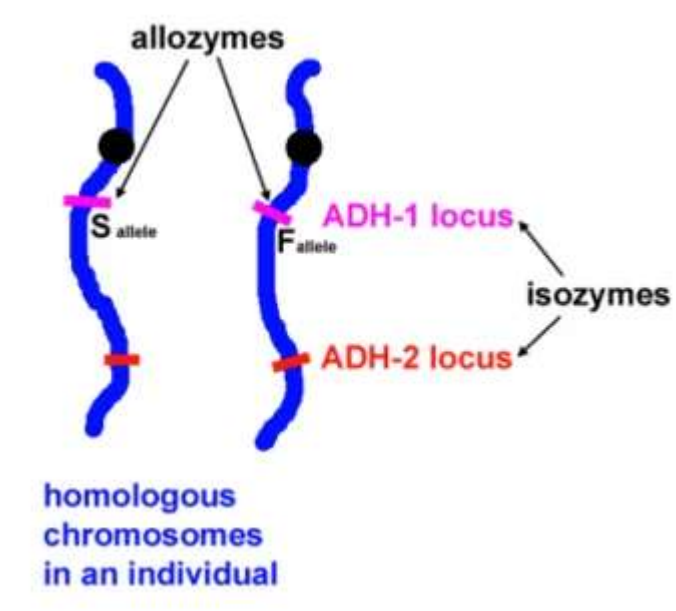



https://www.slideshare.net/FAOoftheUN/molecular-markers-types-and-applications-55688001

## Types of Markers

- **Type I:** Associated with genes of known function. Alloenzymes are type I markers as the protein they encode are associated with genes of known function.
- **Type II:** Associated with genes of unknown function. Microsatellite and other neutral markers are type II markers.

# Molecular Markers




https://www.google.com/search?q=allozyme+markers&sa=X&hl=en&sxsrf=ALeKk03E32wxhpW4rE2JMvw2sHLDlUkIGQ:1604826259215&tbm=isch&source=iu&ictx=1&fir=yKMV0K4mxTe3YM%252C37qgcbgAr7dpfM%252C%252Fm%252F02x3b9m&vet=1&usg=AI4

## Alloenzymes

- Variant form of an enzyme
- Different structurally but not functionally from other alloenzyme coded by different allele present at same locus.
- Codominant markers that express in hetrogygous in MF

## Isoenzymes

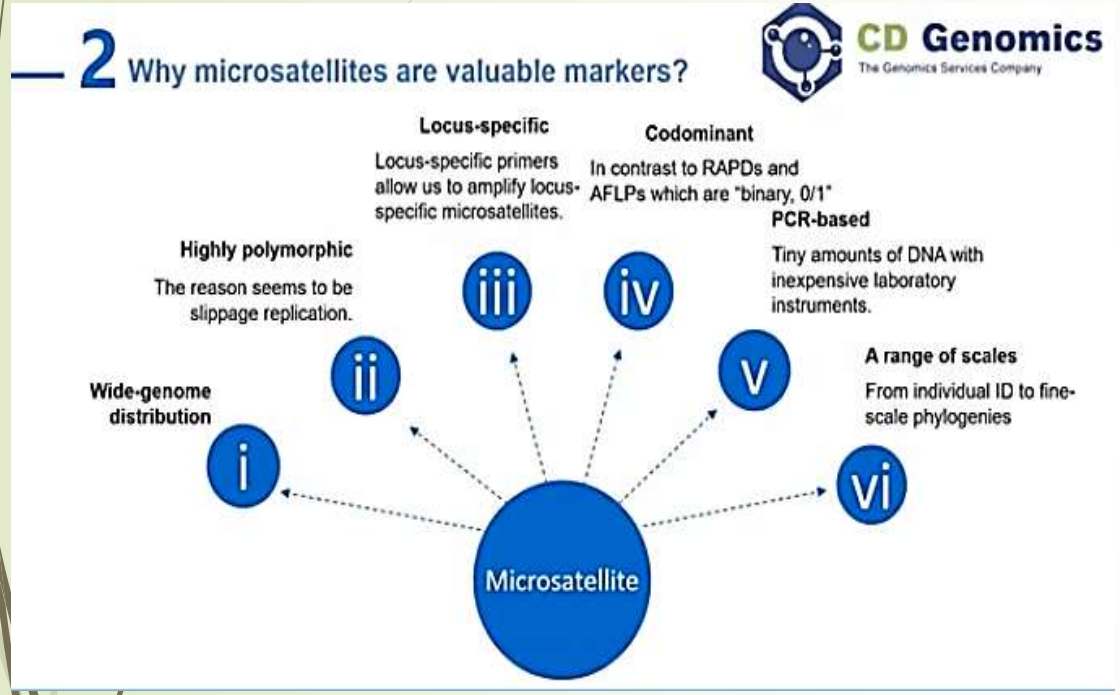
- The enzymes that perform the same function, but which are coded by genes located at different loci

# Molecular Markers

## Mitochondrial DNA Markers

- Non-Nuclear DNA
- Present in Mitochondrial
- Maternally inherited as haploid genome
- Entire genome undergo transcription as single unit
- No recombination
- Homologous markers

# Molecular Markers




https://www.youtube.com/watch?v=r-8SXTHzJFU

## Microsatellite

- Simple DNA sequence
- Repeated several times across various points in DNA of an organism
- 2-6 nucleotides
- 5-50 times
- Present at thousands of locations in an organism
- Microsatellite sequences are composed of non-coding DNA and are not parts of genes.

# Principles of Systematics

## Module# 75

# Principles of Systematics

## Molecular Markers

# Molecular Markers

## SNPs

- Single Nucleotide substitution
- Single Nucleotide insertion/deletion
- These point mutations give rise to various alleles with alternate bases at particular nucleotide position
- Most abundant polymorphism in genome of an organism
- Can be detected by PCR, microchip array or fluorescence technology

# Molecular Markers

## DNA barcoding

- CO1 gene
- 658 base pairs
- Species specific
- Highly conserved
- High mutation rate
- Widely used in identification of arthropods
- Universal primers
- Reliable and quick method
- Hebert Paul 2003

END

# Principles of Systematics

## Module# 76

# Principles of Systematics

## Molecular Markers

# Molecular Markers

## Protein (Biochemical markers)

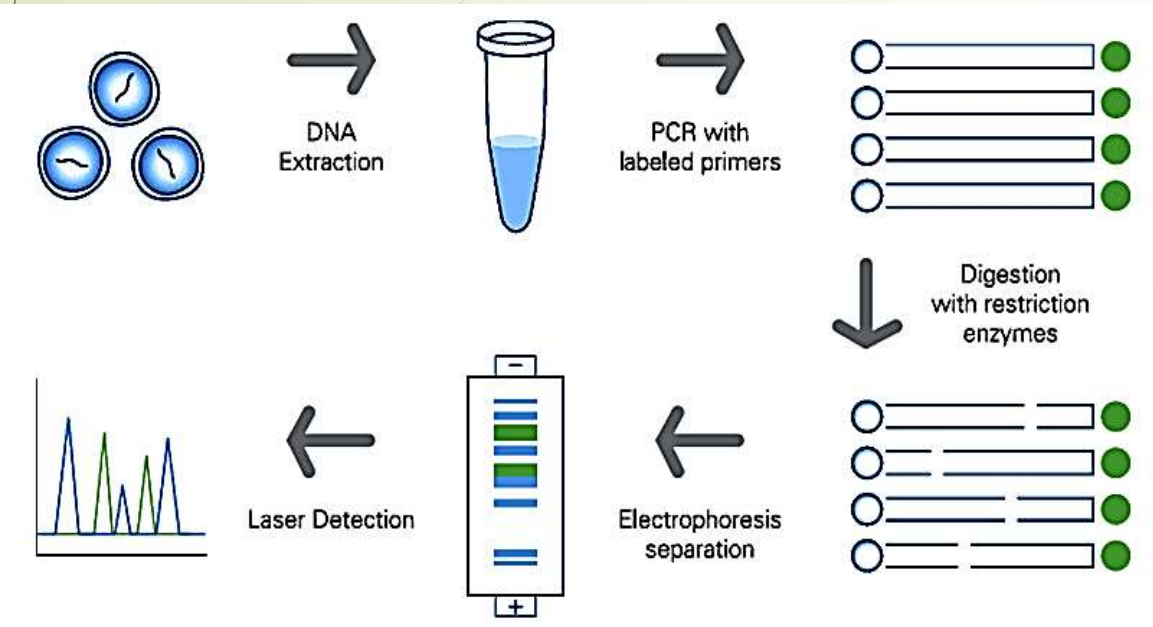
- Available since 1950s
- Related to variation in protein and amino acid bending pattern
- Can be studied using Gel electrophoresis e.g esterase, Acid phosphatase, peroxidase etc.
- Advantage
- Require simple equipment
- A vigorous complement to

# Molecular Markers

**Protein (Biochemical markers) Disadvantages**

- Subject to environmental influences
- Limited in numbers

# Molecular Markers




## Hybridization based techniques (Non-PCR)

- RFLPs (Restriction Fragment Length polymorphism) Analysis
- Introduced by Bostein et al. 1980
- Relatively slow process
- Use of radioisotopes has

# Molecular Markers

## PCR based techniques

- RAPD (Random Amplified polymorphic DNA)
- ISSR (Inter-Simple Sequence Repeat)
- SSR (Simple Sequence Repeat)
- AFLP (Amplified Fragment length Polymorphism)
- EST (Expressed sequence Tags
- SCoT (Stop Codon targeted marker)

END

# Principles of Systematics

## Module# 77

# Principles of Systematics

## Molecular Markers

# Molecular Markers

## Advantages of Molecular Markers

- Ubiquitous
- Stably inherited
- Multiple alleles for each marker
- Devoid of pleiotrpic effects
- Detectable in all tissues, all ages
- Long shelf life of DNA samples

# Molecular Markers

## Application of Molecular Markers

- Fingerprinting
- Diversity studies
- Genetic maps
- Gene tagging
- F1 identification
- Seed testing
- Comparative maps

END

# Principles of Systematics

## Module# 78

# Principles of Systematics

## Methods of Animal Preservation

# Methods of Animal Preservation

## Animal Preservation

- Long term animal preservation in best possible condition
- May be assessed as reference collection in future for scientific purposes
- Many methods are used for preservation of vertebrate and invertebrates

# Methods of Animal Preservation

## Reasons of Animal Preservation

- Taxonomic reasons
- For detail examination
- For morphological examination of particular animal as each and every animals can not be in researcher's vicinity
- For zoological museum record

# Methods of Animal Preservation

## Steps for Animal Preservation

- Killing and relaxing of animals
- Fixation (stop cellular respiration, kill bacteria within organism, a good penetrating ability)
- Storage in bottle, vials, jars, bottles etc.

# Methods of Animal Preservation

## Type of specimens

- Entire fluid preserved specimens (for study of histology and anatomy, entirely change fur colour)
- Study skins with skulls /partial skeleton (Some bones in skin) for studying colour, hair quality and moulting pattern
- Mounted skins
- Entire skeletons (Anatomy, age determination, geographic variations)

END

# Principles of Systematics

## Module# 79

# Principles of Systematics

## Methods of Animal Preservation

# Methods of Animal Preservation

## Formalin as Preservative

- It is used for vertebrate only
- Not used for long term storage
- Acidic and difficult to handle
- Used when colour of animal is important
- Alcohols dissolve most colour immediately
- Penetrate more easily and internal organs remain in better condition

# Methods of Animal Preservation

## Side effects of Formalin

- Fumes of formalin are harmful
- Extreme discomfort of eyes and nose
- Severe irritation if contact with body
- Can cause cancer
- Hand should be rinsed after use
- Should be placed at safe place
- Water-tight

# Methods of Animal Preservation

**Industrial Alcohol (For fixing and storage)**

- Not used for killing and fixing of vertebrates.
- For long term storage
- Colour of sample is lost quickly
- Glycerin in alcohol help to preserve colours and to keep specimens flexible
- 70-75% alcohol is used for long term storage
- 95% of DNA studies

# Methods of Animal Preservation

## Side effects

- Skin irritation with prolong contact.
- Industrial alcohol is toxic and should not be drunk
- Highly flammable
- Rapid evaporation
- Should be kept covered not in open jars.

END

# Principles of Systematics

## Module# 80

# Principles of Systematics

## Methods of Animal Preservation

# Methods of Animal Preservation

## Iropropyl Alcohol

- Cheep
- Easy to obtain
- Available strengths (90%, 70%)
- For use dilute upto 40%
- Can be hard to specimens and specimens tends to brittle over the time
- Can be buffered with few drops of glycerin
- Pinch of calcium carbonate tablet (crush

# Methods of Animal Preservation

## Ethyl Alcohol

- Better solution for long term storage of invertebrate specimens
- 80% ethyl alcohol
- Denatured alcohol
- Painting supplies
- Should be buffered with glycerin/antacid tablets

# Methods of Animal Preservation

## Fish preservation

- After capture fishes are placed in 10% formalin for quick killing.
- This is highly painful
- Fishes dies with its parts well spread off and body straight and well stretched.
- Examination and counting of rays, fins etc. very easy.
- 30 cm fish, Fish in formalin for one week, in water for one day (leech out formalin), use alcohol for long term

END

# Principles of Systematics

## Module# 81

# Principles of Systematics

## Methods of Animal Preservation

# Methods of Animal Preservation

## Herptiles

- Individually kept in plastic jars
- Killed by freezing or chloroform
- Shallow trays are used for herptiles
- Snakes should be coiled
- Frogs are placed on their bellies with their limbs set at right angles to their bodies.
- Small salamanders fixing time 2 days
- Month for alligator, snapping turtle

# Methods of Animal Preservation

## Herptiles

- Add water for day or two
- 75% alcohol for long term storage.

# Methods of Animal Preservation

## Birds an Mamamls

- Skinned
- Skulls or skeletal mounts
- Remove flesh
- Boiling or dermested beetle
- Placed defleshed bones in bleach or hydrogen peroxide solution to whiten
- Allow to dry
- Place in bag with complete label

END

# Principles of Systematics

## Module# 82

# Principles of Systematics

## Methods of Animal Preservation

# Methods of Animal Preservation

## Preservation of invertebrates

- Use of alcohol is the easiest way
- One should be aware about the type of alcohol
- Each specimen type require different concentration of alcohol
- Preserved in jars, tubes or bottles

# Methods of Animal Preservation

**Coelenterates, platyhelminthes and echinoderms**

**Coelenterates** are difficult to preserve

- Preserve in 70% alcohol or 5% formalin
- Hydra can be quickly fixed in warm Bouin's warm not hot.

**Platyhelminthes (flatworms)**

- Bouin solution for fixation
- Paraffin is best long term preserving and storage medium for all flatworms.

# Methods of Animal Preservation

- **Echinoderms** are narcotized by addition of magnesium sulphate  or methanol to the sea water in which they live
- When completely insensitive to stimulus such as pricking, 70% alcohol for preservation and storage

# Methods of Animal Preservation

- **Arthropods** are easy to process as killed immediately and preserved and store in alcohol
- Crabs and prawn may be killed in formalin but renders their joints hard and brittle.
- Larger invertebrate need to be injected 10% formalin to prevent from rotting
- Industrial alcohol is used for most arthropods.

# Methods of Animal Preservation

- Insects, crustaceans and arachnids can be simply dropped in alcohol for immediate preservation.
- Industrial alcohol is used for most arthropods.

END

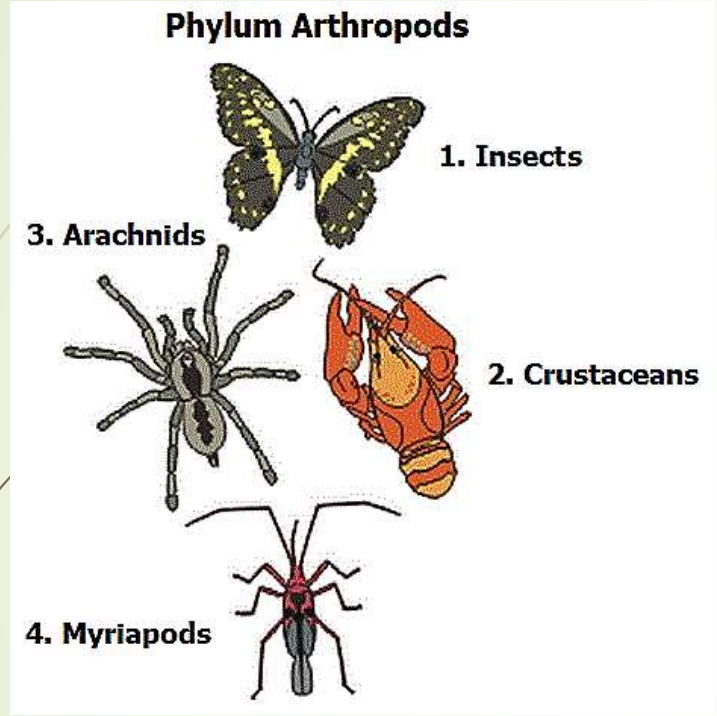
# Principles of Systematics

## Module# 83

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

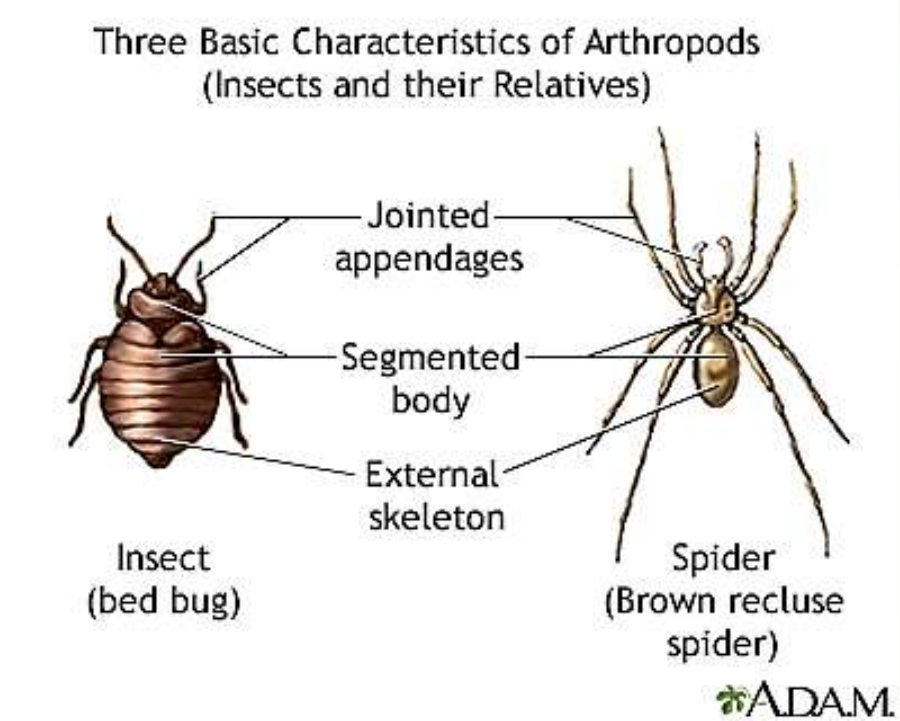



https://www.google.com/search?q=distribution+and+abundance+of+arthrpods&tbm=isch&ved=2ahUKEwiHrMC3i5XtAhU1gc4BHc6LBOEQcCegQIABAA&oq=distribution+and+abundance+of+arthrpods&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoCCAA6BggAEAgQHjoECAAQGDoECAAQHjoGCAAQBRAeULTPCljDtgtgq78LaABwAHgDgAHUCYgB6luSAQ4yLTIzLjIuMC4yLjEuM5gBA6ABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=0Mq5X4eED7WCur4PzpeSiA4&bih=657&biw=1360#imgrc=Wkf1B_slQa0UyM&imgdii=b2K7h2f-v7gNoM

## What is an Arthropod?

- Any member of Phylum Arthropoda
- Include lobster, prawn, mites, insects, spiders, centipedes, millipedes
- Represented in every habitat of earth
- Show great diversity of adaptations

# Morphological Identifications of Model Arthropods




https://www.google.com/search?q=basic+body+characteristics+of+arthropods&sxsrf=ALeKk024Jhkr92CzBt-ISL1YUUtAMZ600A:1606011594113&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiPv8m0i5XtAhVXShUIHV4dBeAQ_AUoAXoECCIQAw&biw=1360&bih=657#imgrc=82cOAzf5eyewuM

- Some are aquatic, others are terrestrial, some are free living, other are parasitic
- All arthropods have a body supported by hard exoskeleton
- Legs, antennae and mouth are jointed
- Tube like heart
- Overall body bilateral symmetrical

# Morphological Identifications of Model Arthropods

- Seven million species
- Arthropods are the most successful of all animal phyla based on diversity, distribution, and numbers.
- Nearly one million species identified so far, mostly insects.
- The exoskeleton, or cuticle, is composed of protein and

# Morphological Identifications of Model Arthropods

- Molting of the cuticle is called ecdysis
- Extensive cephalization
- Open circulatory systems in which a heart pumps hemolymph through short arteries and into open spaces (sinuses)
- Aquatic members- gills for gas exchange; terrestrial members- tracheal system of branched tubes leading from surface throughout body

END

# Principles of Systematics

## Module# 84

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

- Appendages in arthropods are modified for a number of functions:
- Walking legs
- Mouth parts
- Antennae
- Claws
- Organs for mating
- Holding eggs

# Morphological Identifications of Model Arthropods

- Segmentation is an ancestral characteristic that carries over from the Annelids
- Note the similarity of the insect caterpillar to polychaete worms
- The exoskeleton provides rigid support and strong points for muscle attachment
- The exoskeleton also forms a protective shell

# Morphological Identifications of Model Arthropods

- The exoskeleton prevents drying out, but also prevents respiration through the skin
- The exoskeleton provides rigid support, but prevents growth – requiring molting
- ***Striated muscle*** associated with the exoskeleton, and ***smooth muscle*** associated with the gut

# Morphological Identifications of Model Arthropods

- Reduced body cavity, filled with blood, forming an ***open circulatory system***
- Specialized ***excretory glands*** (coxal, antennal or maxillary glands; malpighian tubules)
- Specialized ***respiratory organs*** (gills, tracheae, book lungs)
- Dorsal brain with fused ventral nerve cords

# Morphological Identifications of Model Arthropods

**Four lineages of arthropods**

- Trilobites – extinct
- Chelicerates – horseshoe crabs, spiders
- Uniramians – centipedes, millipedes, insects
- Crustaceans – crabs, lobsters, barnacles

From these lineages arose five major classes of arthropods.

END

# Principles of Systematics

## Module# 85

# Principles of Systematics

## Identification Keys and catalogues

# Identification Keys and Catalogues

- Identifications key aids in the identification of biological identities:
- Plants
- Animals
- Fossils
- Microorganisms
- Pollens

# Identification Keys and Catalogues

- Keys are constructed so that user is presented with series of choices about the characteristics of unknown organisms
- By making correct choice at each step of key user is ultimately let to identity of unknown specimen
- At each step use must answer a question about one of more character of gentility to be identified.

# Identification Keys and Catalogues

- If at each step there are only two alternative, the key is said to be **DICHOTOMOUS ( s**equential key, analytical key, pathway key)

- **POLYTOMUS** key

# Identification Keys and Catalogues

END

## Methods of using taxonomic keys

- Read introductory comments, format of the key, abbreviations etc before using key
- Read both choices at each step of the key
- Use glossary to find out the definition of any term your do not understood
- Use a ruler if measurements are needed, do not guess
- Examine several specimens of

# Principles of Systematics

## Module# 86

# Principles of Systematics

## Identification Keys and catalogues

# Identification Keys and Catalogues



## Dichotomous keys

- If each step in a key has only two alternative the key will be dichotomous
- A dichotomous key is a device for identifying unknown organisms to some taxonomic levels such as species, genus or family level.
- These most commonly published keys were first published by Jean B.

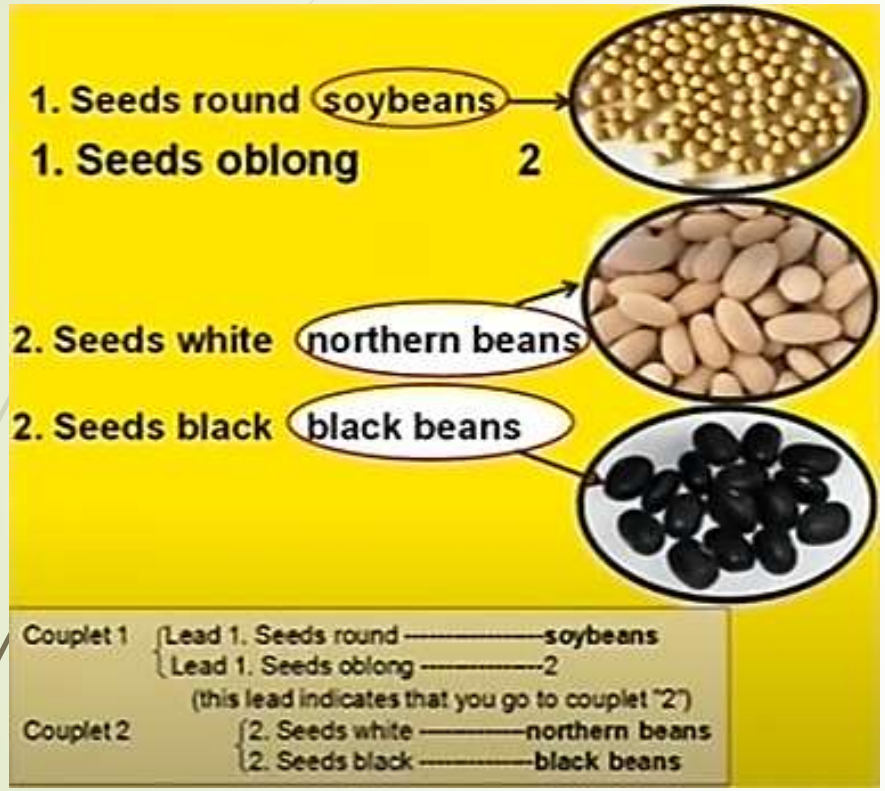
# Identification Keys and Catalogues



## Dichotomous keys

- Consists of series of paired statements, called couplets.
- Couplets describes some features of organism
- Each statements in a couplet is called a lead
- The statements or leads are in contract (Mutually

# Identification Keys and Catalogues

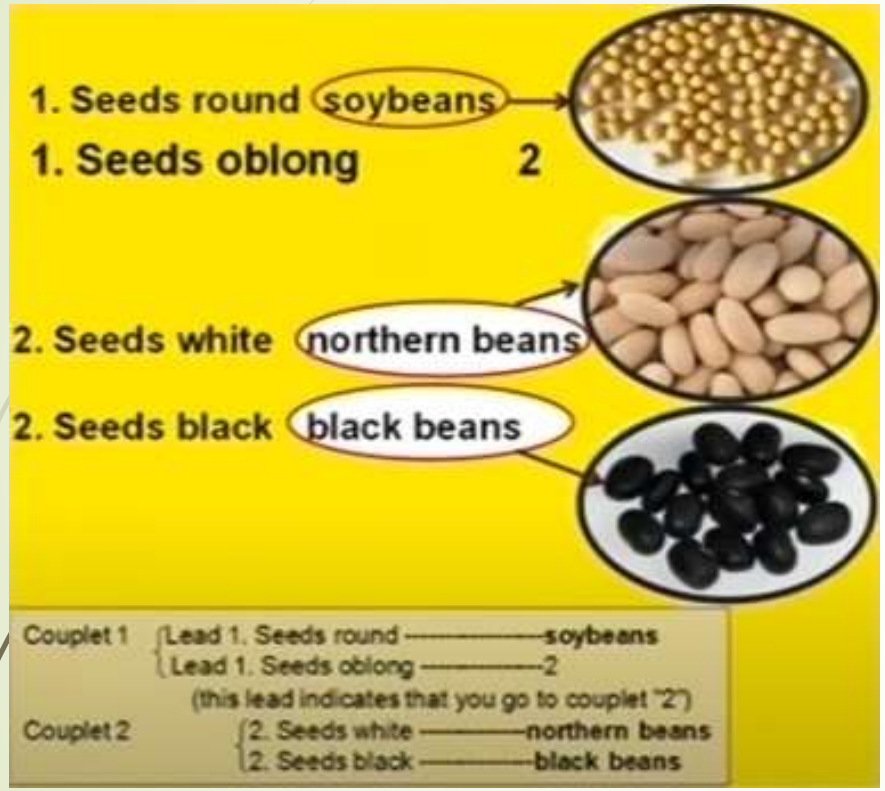



https://www.youtube.com/watch?v=srJ4aab5RYc

## Dichotomous keys

- The use of key begins with first couplet and select the statement which best fit the organism to be identified
- The leads typically begins with board characteristics and become narrower as more choicely are required.
- We move through key- Single choice remains

# Identification Keys and Catalogues




https://www.youtube.com/watch?v=srJ4aab5RYc

## Dichotomous keys

- If we continue to select correct option of couplets for unknown organism, correct identification of animals will be achieved.

# Principles of Systematics

## Module# 87

# Principles of Systematics

## Identification Keys and catalogues

# Identification Keys and Catalogues

## Catalogue

- No study in science is complete until it is made available to others
- The best way is to made it available through circulation
- In taxonomy publication may be in the form of Books, pamphlets, journal articles, catalogues,

# Identification Keys and Catalogues

## Catalogue

- Catalogue is mainly an index to the texa arrange in such a way that as to give a vivid picture of references for both zoological and nomenclatural purpose.
- The **Catalogue** currently compiles data from 168 peer-reviewed **taxonomic** datab ases, that are maintained by specialist institutions

# Identification Keys and Catalogues

## Catalogue of life

- The **Catalogue of Life** is an online database
- It provides the most comprehensive and authoritative index of known species of organisms

.

- The Catalogue interface is available in twelve languages and is used by research scientists, citizen scientists, educators, and

# Identification Keys and Catalogues

## Catalogue of life

- June 2001; 19 years ago
- Current status is active
- Registration not required
- Not commercial
- Available in English

# Identification Keys and Catalogues

## Catalogue of life

- The Catalogue listed 300,000 species by 2003, 500,000 species by 2005, and over 800,000 species by 2006. As of 2019, the Catalogue lists 1.9M extant and extinct species.
- There are an estimated 14M mainly unpublished species.

END

# Principles of Systematics

## Module# 88

# Principles of Systematics

## Construction of Identification Key

# Construction of Identification Key

## Key /Taxonomic aids

- Key is a list of statements
- Used for identification
- Similarities and dissimilarities
- Used pair of contrasting characters "couplet"
- Each statement in couplet is "Lead"

# Construction of Identification Key

- Taxonomist have different opinions about the design and format of the key
- Some use numbers to separate the couplets
- Other use letters
- Some taxonomist place each couplets together
- Others may seprate couplets

# Construction of Identification Key

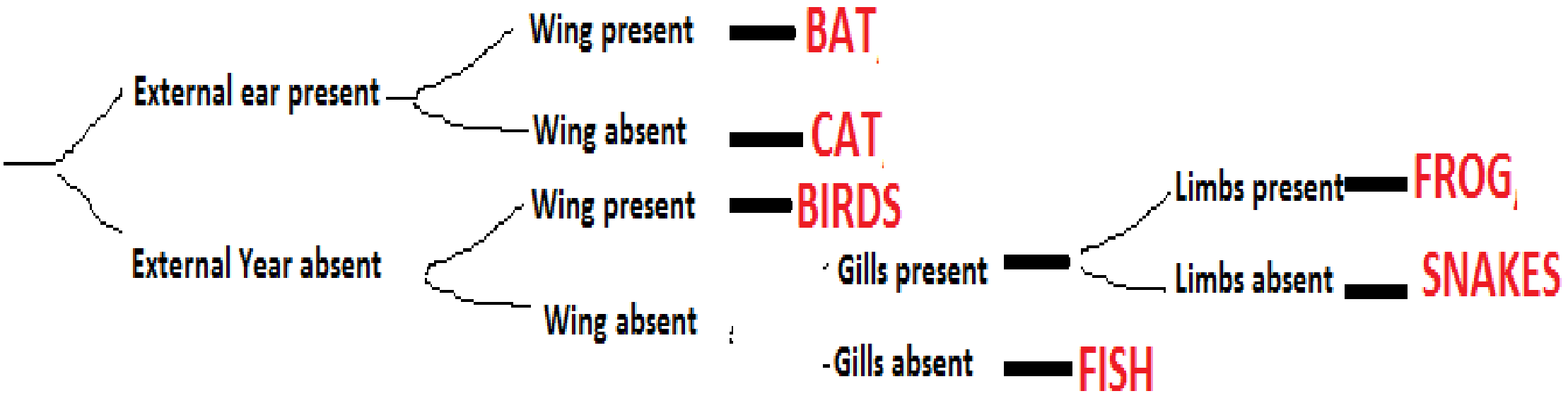
## Types of Keys

- Indented Key
- Bracket Key
- Pictoral Key
- Polyclave Key

# Construction of Identification Key

In indentd key we have sequences of choices between two or more statements

For example: BAT, CAT, FISH, FROG, SNAKES, BIRDS



# Principles of Systematics

## Module# 89

# Principles of Systematics

## Construction of Identification Key

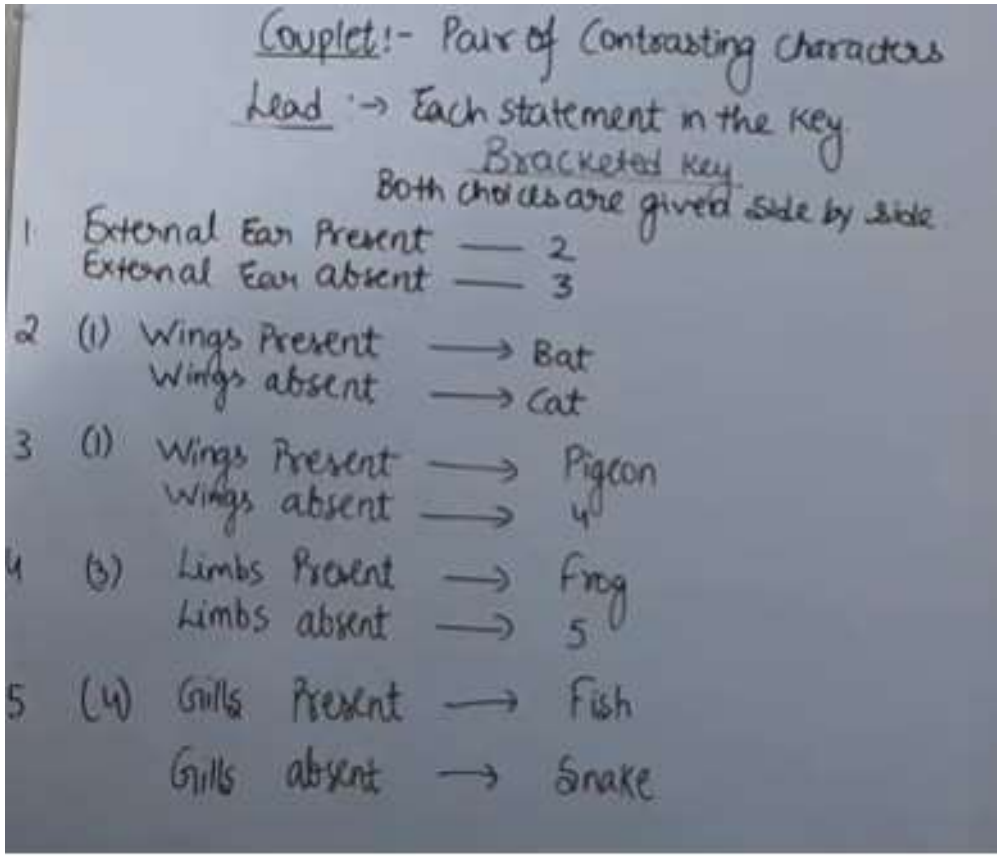
# Identification Keys and Catalogues

| Characters | Vertebrates |
|---|---|
| (1) External ear present | 2 |
| (1) External ear absent | 3 |
| (2) Wings present | BAT |
| (2) Wings absent | CAT |
| (3) Wings present | BIRDS |
| (3) Wing absent | 4 |
| (4) Gills present | FISH |
| (4) Gills absent | 5 |
| (5) Limbs Present | FROG |
| (5) Limbs absent | SNAKE |

## Bracketed Key

- Pair of contrasting characters are taken in brackets
- Simple to use
- It is most commonly used in the filed of taxonomic study.

# Identification Keys and Catalogues




https://www.youtube.com/watch?v=Uqbm-8ZIqwo

## Bracketed Key

- Pair of contrasting characters are taken in brackets
- Simple to use
- It is most commonly used in the filed of taxonomic study.

# Identification Keys and Catalogues

## Pectoral Keys

- For non-scientists
- Simple to use
- Key with pictures
- Jungle or desert/easy for identification of plants and animals

# Identification Keys and Catalogues

## Polyclave Key

- Based on computer program
- Simple to use
- Plant identification
- Take photos/Google/identification

END

# Principles of Systematics

## Module# 90

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

- Spiders (Arthropoda: Arachnida)Most
- Most abundant natural predators
- Present in all types of habitat
- Highly diverse groups of Arthropod
- More than 46000 species
- Ranked 7$^{th}$ in animal Kingdom

# Morphological Identifications of Model Arthropods

- Present on every continent
- Excellent predators
- Types of spiders
- Can survive without food for several months
- Low metabolic rate
- Prey switching ability

# Morphological Identifications of Model Arthropods

- Prey switching
- Four pairs of legs
- Two body parts
- No antennae

END

# Principles of Systematics

**Module# 91**

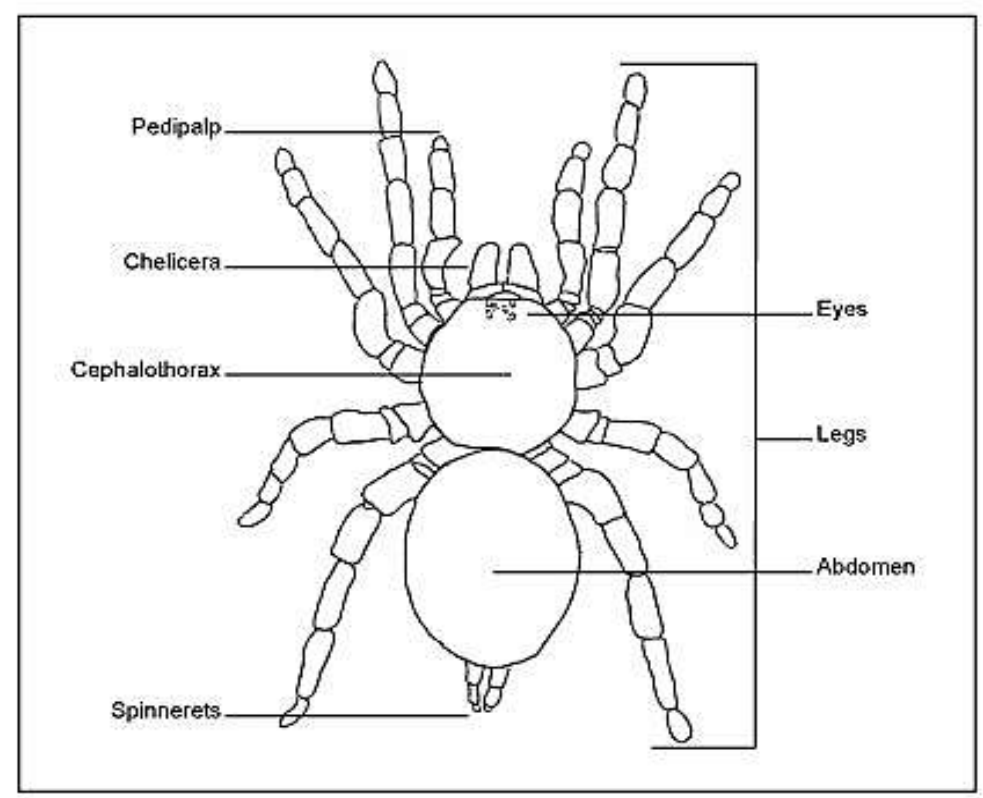
# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

- Spiders have two body segments
- Front segment is called **Cephalothorax**
- It contain eyes, mouth parts, fangs, brain, venom gland etc.
- Legs are also connected to this parts
- Four pairs of legs/Chelicera/padipalp

# Morphological Identifications of Model Arthropods



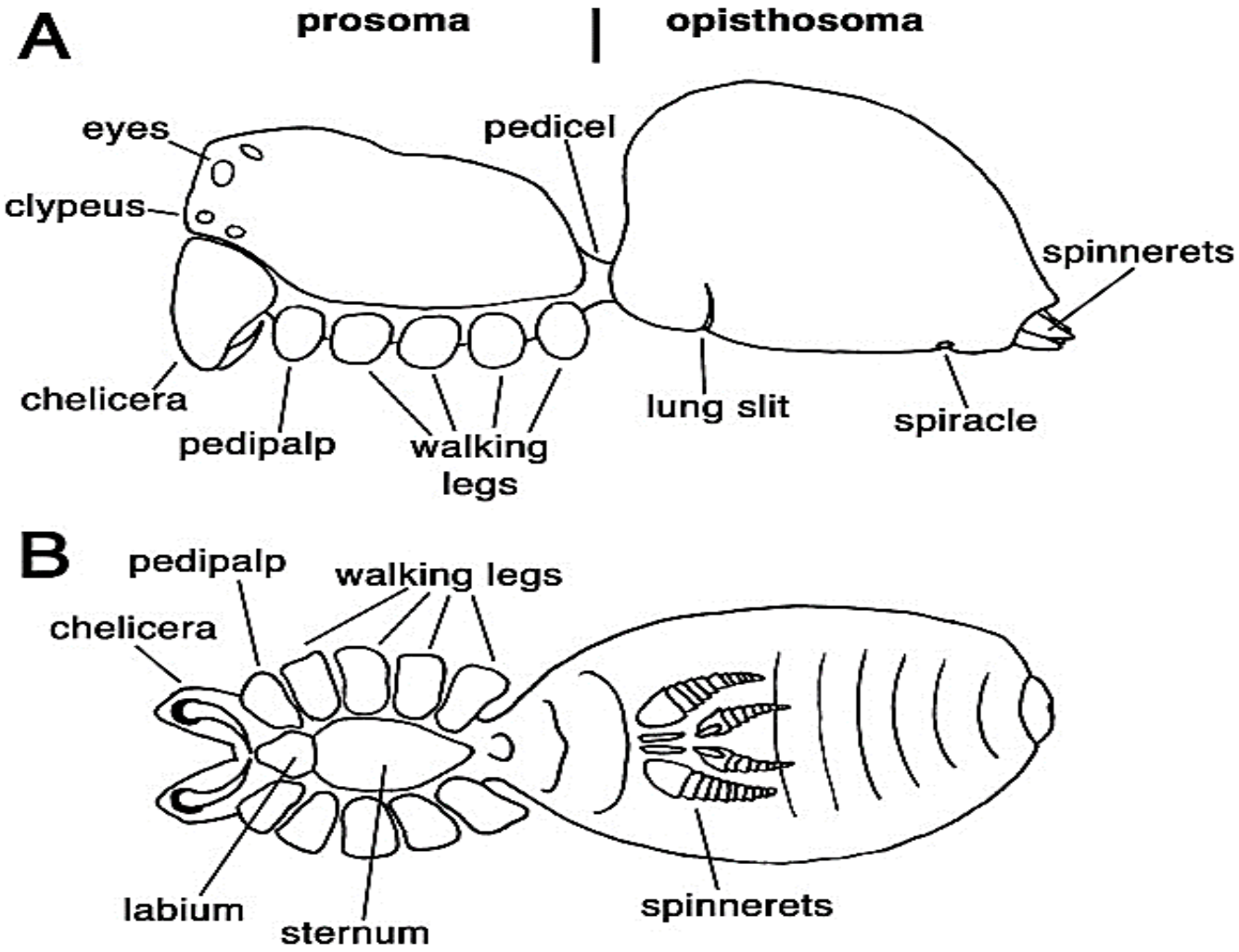

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- 2nd part is abdomen
- Spinneret are attached at the end of this part
- Silk glands
- Legs are covered with many hairs
- Hairs pick up vibrations and smell from air
- Claws 2-3 at the end of legs

# Morphological Identifications of Model Arthropods

- 2nd part is abdomen
- No skeleton inside their bodies
- Hard outer shell called exoskeleton
- Molting
- Males are smaller than females

# Morphological Identifications of Model Arthropods




https://www.google.com/search?q=spiders+morphology+ventral+side&tbm=isch&ved=2ahUKEwjYsIm0m5TtAhVX_IUKHblUAhMQ2-cCegQIABAA&oq=spiders+morphology+ventral+side&gs_lcp=CgNpbWcQAzoECAAQHlC-xRZYgecWYInpFmgAcAB4AIABpwOIAfwckgEHMi03LjQuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=WFW5X9ikEdf4lwS5qYmYAQ&bih=657&biw=1360#imgrc=wI6EJC0oSRchlM&imgdii=g-x_VzORBM7K5M

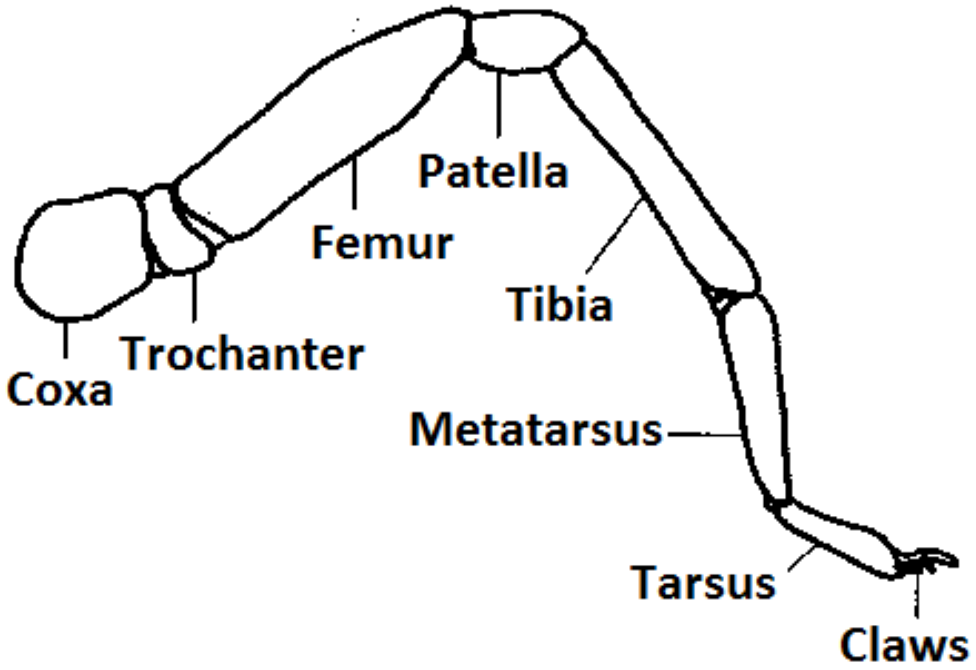
# Principles of Systematics

## Module# 92

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

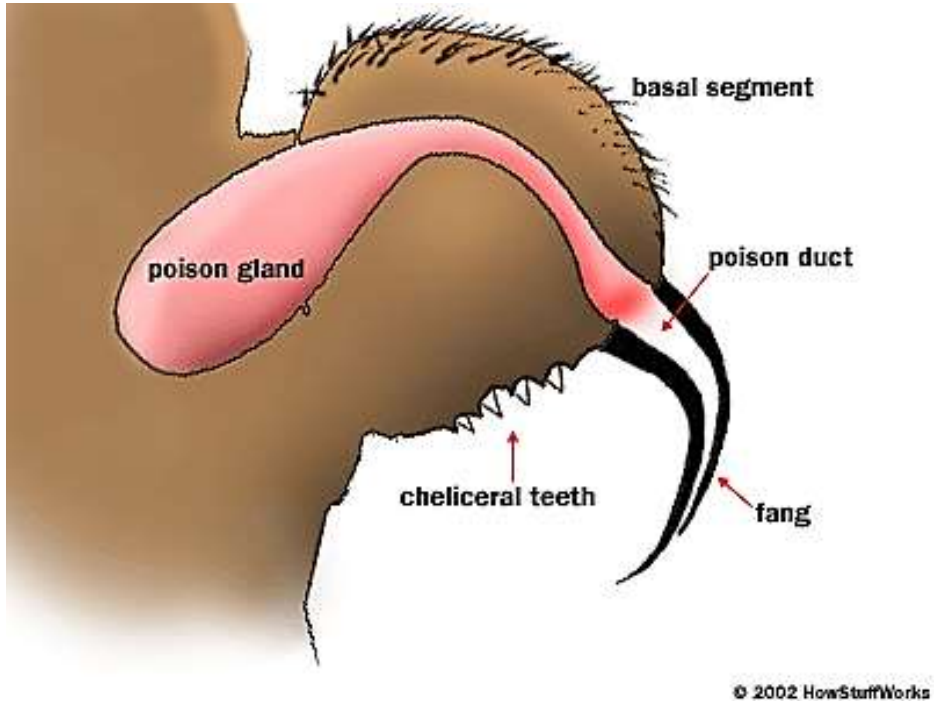



https://www.google.com/search?q=spiders++leg+morphology+&tbm=isch&ved=2ahUKEwjGosznnJTtAhXk1-AKHZdzAx0Q2-cCegQIABAA&oq=spiders++leg+morphology+&gs_lcp=CgNpbWcQA1C6zApY2fgKYNf6CmgAcAB4AIAB9gaIAaQUkgELMi0yLjAuMS4wLjKYAQCgAQGqAQtnd3Mtd2l6LWItZ8ABAQ&sclient=img&ei=0Fa5X8b6LuSvgweX543oAQ&bih=657&biw=1360#imgrc=0J_63uMCGI3QzM&imgdii=PYJrvMBoGcmOnM

## Parts of Legs

- Coxa
- Traochanter
- Femur
- Patella
- Tibia
- Metatarsus
- Tarsus
- Claws

# Morphological Identifications of Model Arthropods

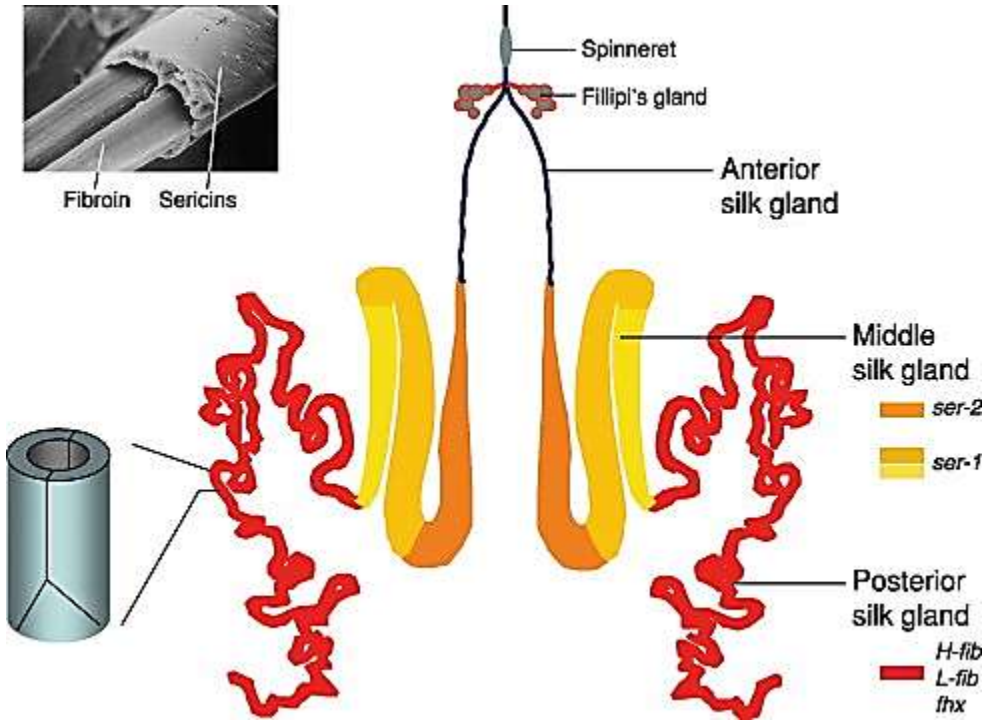



https://www.google.com/search?q=venom+gland+of+spiders&sxsrf=ALeKk03L0hhd-Hg2f_BL_MYyBTdPVy7X5w:1606355341110&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjWm_X7i5_tAhWCThUIHVHSDmEQ_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=JnmQ_pY9IUUA_M

## Parts of Venom gland and use

- Poison gland
- Poison Duct
- Fang
- Absent in the family Uloboridae. The **glands** are located either in the chelicerae or under the carapace.

# Morphological Identifications of Model Arthropods




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## Parts of Silk gland and use

- For capturing prey
- For food storage
- Travelling
- Ballooning

# Principles of Systematics

## Module# 93

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

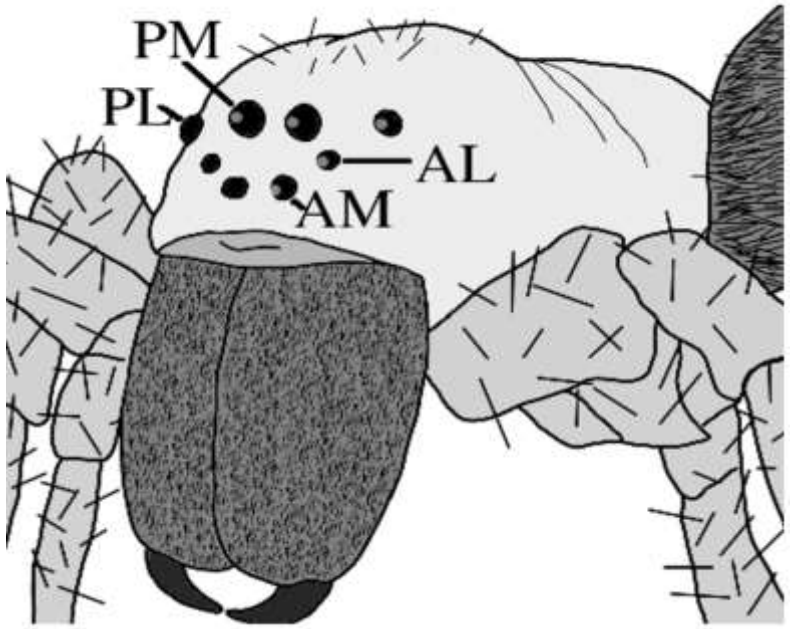


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## Eyes in spiders

- Most spiders have eight eyes. Some species have six or fewer eyes, but they always come in an even number.
- Some species of spiders, such as those that live in caves or under the soil, have no eyes at all. Even those species with eight eyes don't usually see very well.

# Morphological Identifications of Model Arthropods

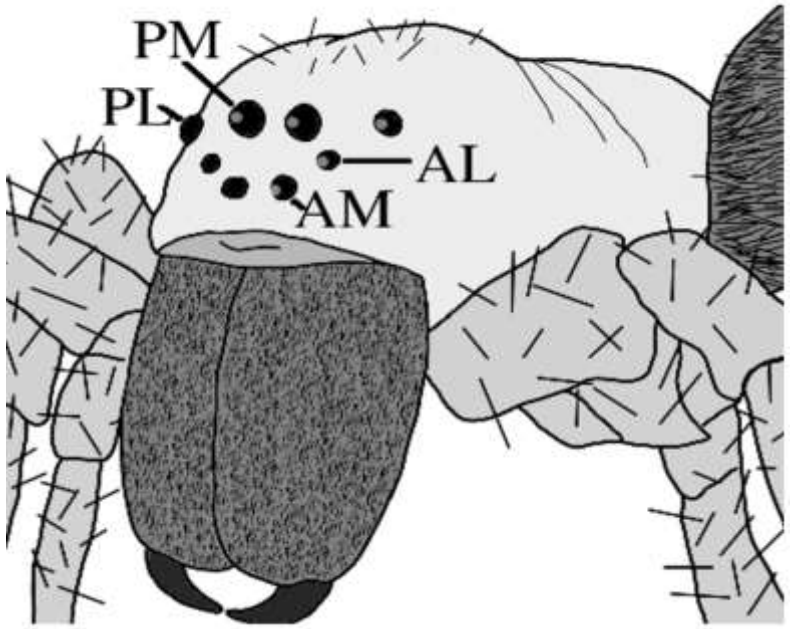



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## Eyes in spiders

- Important Taxonomic character
- Eye arrangement in different families is unique

# Morphological Identifications of Model Arthropods

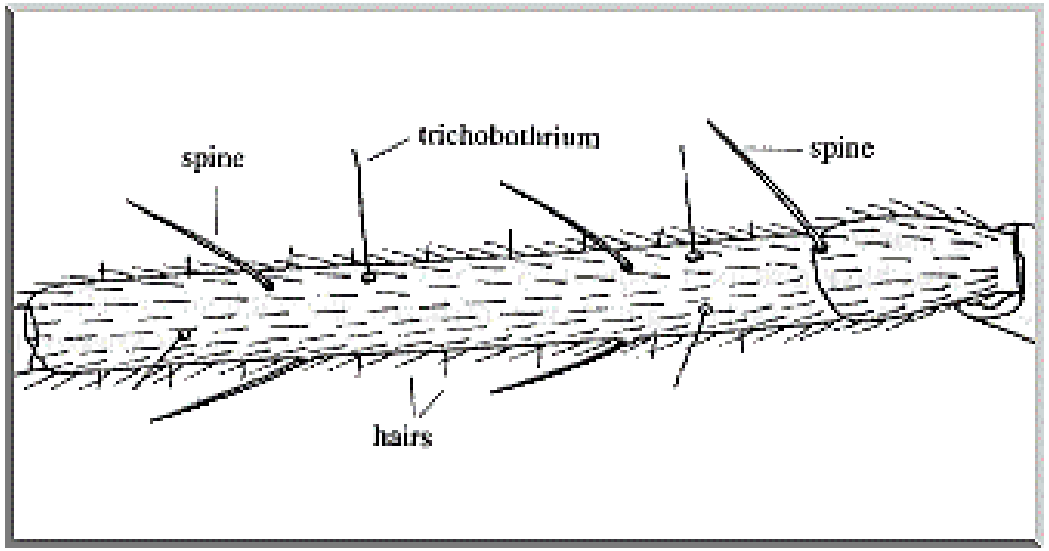



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## Eyes in spiders

- 99% of all spiders have 8 eyes
- 1% nearly all have 6, but there are a few exceptions.
- Varying number of eyes in the same spider family.
- For example, there are spiders in Cybaeidae that have eight eyes,

# Morphological Identifications of Model Arthropods




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## Sensory structures

- **Spiders**, in fact, do taste, and also smell, through special **sensory organs** on their legs, as well as on their pedipalps.
- And they hear – or, more specifically, they sense vibrations – through hairs and tiny slits distributed over much of their body.
- A **spider's** sensitivity to vibrations is finely tuned.

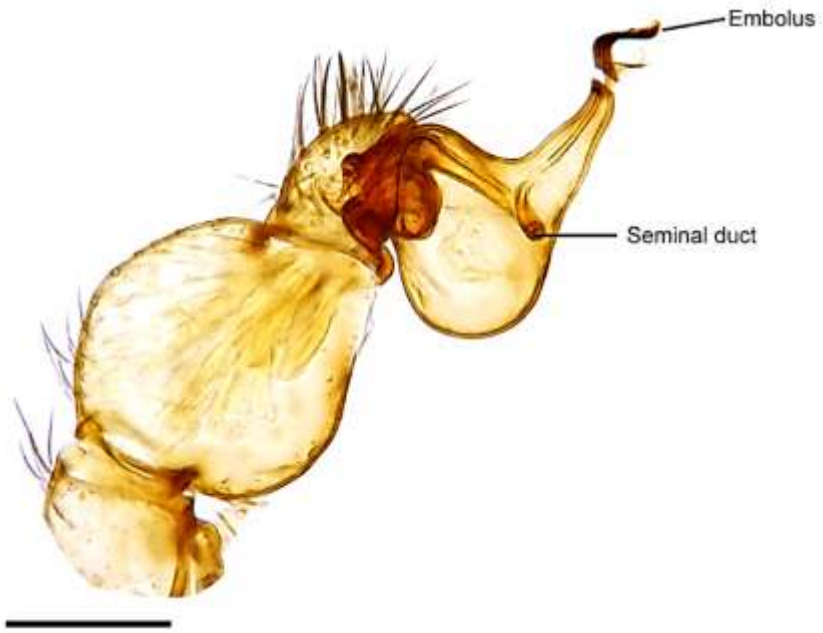
# Principles of Systematics

## Module# 94

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods




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## Padipalp/Epigyne

- **Pedipalps** (commonly shortened to palps or palpi) are the second pair of appendages of chelicerates
- E**pigyne** or epigynum is the external genital structure of female spiders.

# Morphological Identifications of Model Arthropods




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## Padipalp/Epigyne

- As the **epigyne** varies greatly in form in different species, even in closely related ones.
- Distinctive characteristic for recognizing species.

# Morphological Identifications of Model Arthropods




https://www.google.com/search?q=eyes+of+spiders&hl=en&sxsrf=ALeKk03MPwHBA8hK0MNcLPNSAuHBGFOVoQ:1606355925079&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj89K-Sjp_tAhU1rHEKHfiCALgQ_AUoAXoECBEQAw&biw=1360&bih=657

## Reproduction in Spiders

- Male find a female
- A male spider deposits his sperm into a sperm web, holding it in his palps.
- Female spiders produce either one egg sac containing several to a thousand eggs or several egg sacs each with successively fewer eggs. Females of many species die after producing the

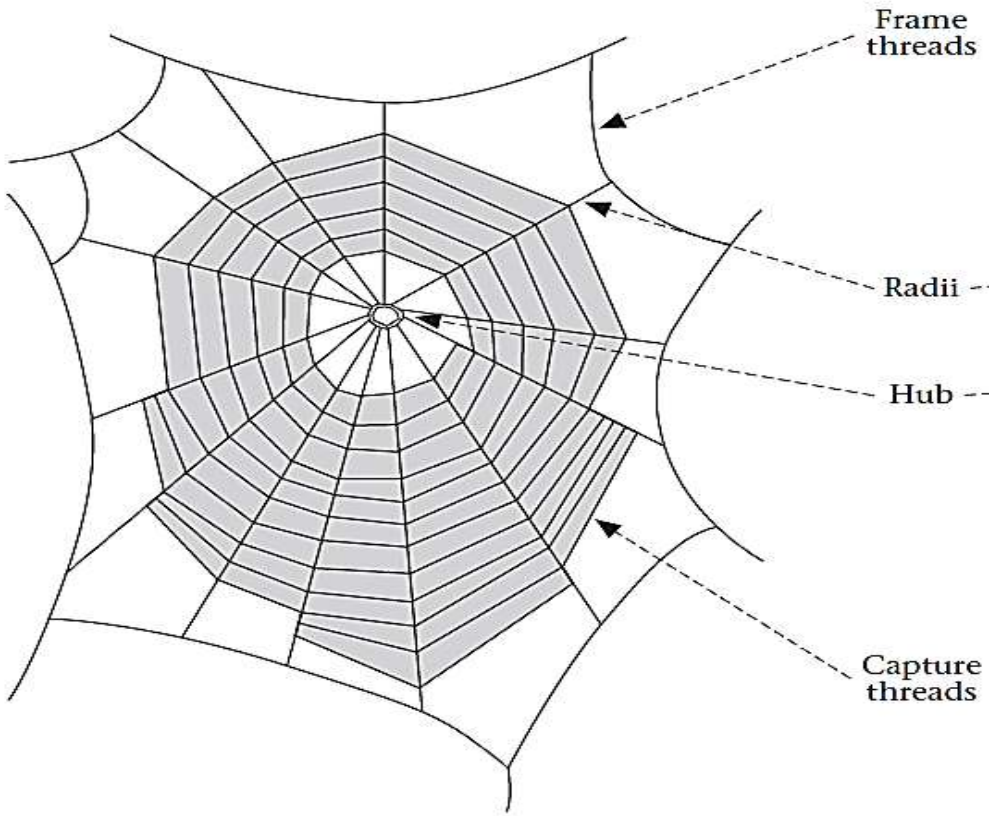
# Principles of Systematics

## Module# 95

# Principles of Systematics

## Morphological Identifications of Model Arthropod

# Morphological Identifications of Model Arthropods

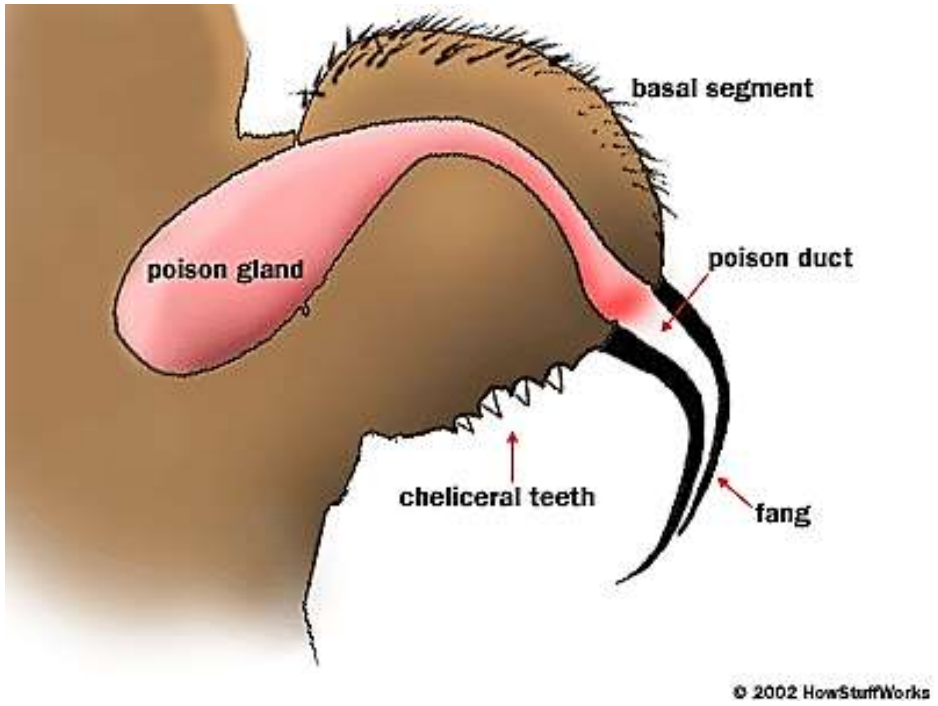



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## Spider web

- Orb-web spiders
- Hunters
- Design of web
- Web/prey size
- Wind director/prey density
- Repair/vibrations/self-eating/sticky nature/quality of silk

# Morphological Identifications of Model Arthropods




https://www.google.com/search?q=venom+gland+of+spiders&sxsrf=ALeKk03L0hhd-Hg2f_BL_MYyBTdPVy7X5w:1606355341110&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjWm_X7i5_tAhWCThUIHVHSDmEQ_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=JnmQ_pY9IUUA_M

# Morphological Identifications of Model Arthropods

- 2nd part is abdomen
- No skeleton inside their bodies
- Hard outer shell called exoskeleton
- Molting
- Males are smaller than females

END

# Principles of Systematics

**Module# 96**

# Principles of Systematics

## Identifications of Family Salticidae

# Identification of Family Salticidae



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- As of 2019, it contained over 600 described genera
- Over 6000 described species.
- Largest family of spiders
- 13% of all species

# Identification of Family Salticidae



https://www.google.com/search?q=salticidae+family+eyes&tbm=isch&ved=2ahUKEwjbw9fQnK3tAhVE3hoKHZygBI4Q2-cCegQIABAA&oq=salticidae+family+eyes&gs_lcp=CgNpbWcQAzoGCAAQCBAeUKuKAVi9IQFg9JgBaABwAHgBgAG5BogB8g-SAQkzLTIuMS4wLjGYAQCgAQGqAQtnd3Mtd2l6LWItZ8ABAQ&sclient=img&ei=EnLGX9vyCsS8a5zBmvAF&bih=600&biw=1360

- Jumping spiders are generally recognized by their eye pattern.
- All jumping spiders have four pairs of eyes, with the anterior median pair being particularly large.
- Best vision among arthropods

# Identification of Family Salticidae

- The legs are not covered with any very prominent spines.
- Their front four legs generally are larger than the hind four
- Not held in the outstretched-arms attitude characteristic of the Thomisidae.
- The body length from 1 to 25 mm (0.04–0.98 in).

# Identification of Family Salticidae



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- Prior to jumping, the spider attaches a silk thread to the surface beneath it, so it can climb quickly back to its perch if needed.
- Jumping spiders hunt and feed on small insects.
- All are carnivorous, but a few species also eat pollen and nectar.

# Identification of Family Salticidae



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- Female jumping spiders build a silk case around their eggs and often stand guard over them until they hatch.
- Young jumping spiders emerge from the egg sac looking like miniature versions of their parents.
- They molt and grow into adulthood.

# Principles of Systematics

**Module# 97**

# Principles of Systematics

## Identifications of Family Salticidae

# Identification of Family Salticidae

- Jumping spider can jump quite far, achieving distances more than 50 times its body length.
- Legs not strong or muscular in appearance.
- Salticids are able to quickly increase the blood pressure in their

# Identification of Family Salticidae

- Some jumping spiders mimic other insects such as ants.
- Others are able to camouflage themselves to blend into their surroundings, helping them sneak up on prey.
- Salticids hunt diurnally as a rule, which is consistent with their

# Identification of Family Salticidae



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- Although jumping spiders are generally carnivorous.
- Many species have been known to include nectars in their diets and one species, *Bagheera kiplingi* feeds primarily on plant matter

# Identification of Family Salticidae



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- During courtship "dance“ colored or iridescent parts of the body are displayed.
- In addition to the display of colors, jumping spiders perform complex sliding, vibrational, or zigzag movements to attract females.
- Many males have

# Identification of Family Salticidae

- These amplified sounds presented to the females resemble buzzes or drum rolls.
- Species vary greatly in visual and vibratory components of courtship.
- If receptive to the male, the female assumes a passive, crouching position.
- In some species, the

# Identification of Family Salticidae



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- The male then extends his front legs towards the female to touch her.
- If the female remains receptive, the male climbs on her back and inseminates her with his palps.

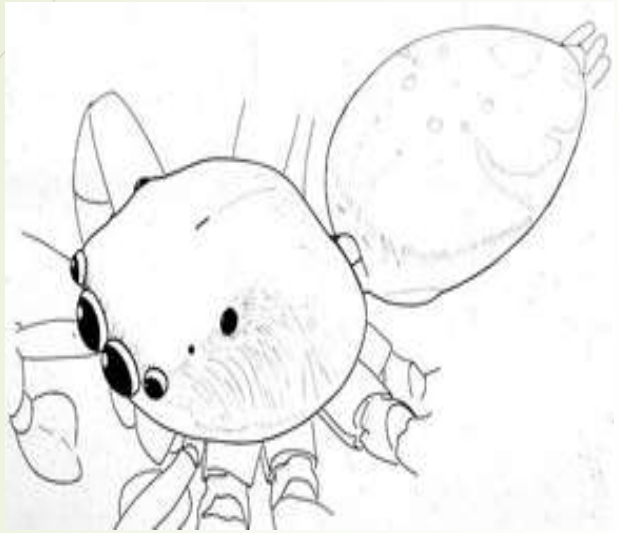
# Principles of Systematics

## Module# 98

# Principles of Systematics

## Identifications of Family Salticidae

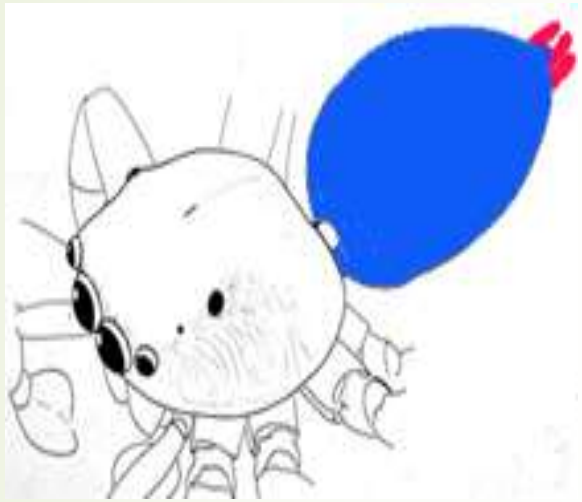
# Identification of Family Salticidae



- Note that there are eight eyes, four in front and four on top.

http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

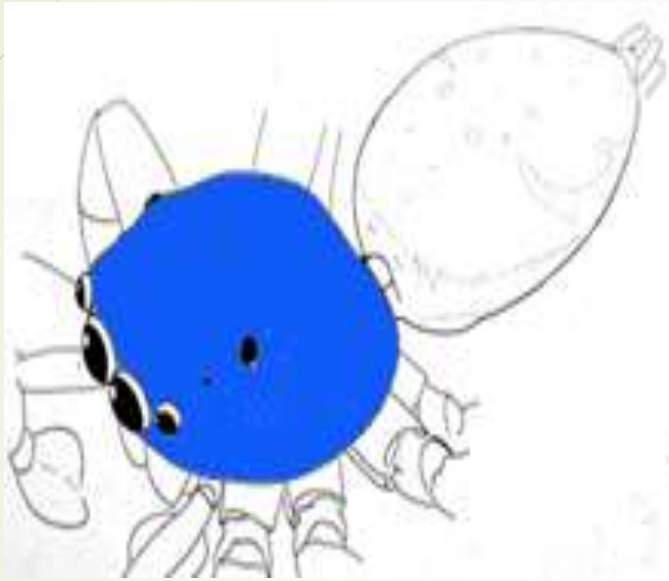
# Identification of Family Salticidae



http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

- The body is divided into two main parts, the posterior of which is the opisthosoma or abdomen, shown in blue:
- The spinnerets are shown in red. These are the silk-producing appendages.
- Jumping spiders, although they do not usually use silk to catch

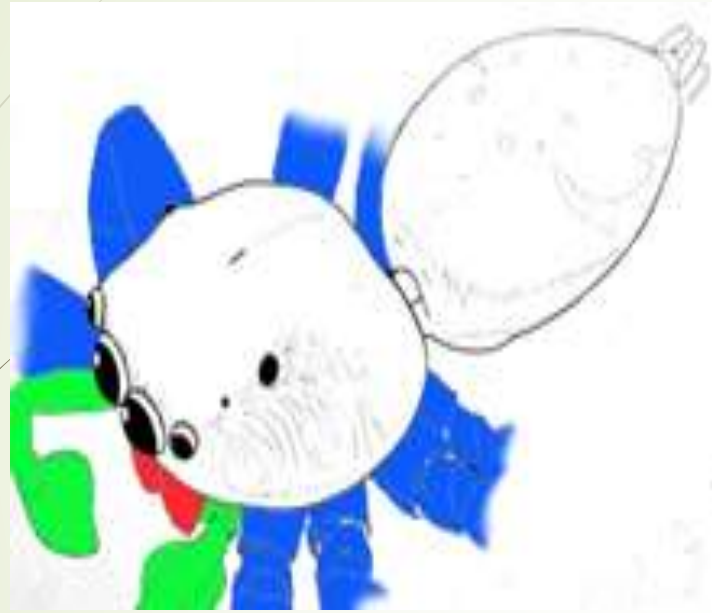
# Identification of Family Salticidae



http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

- The anterior body part is the prosoma or cephalothorax.
- It is covered above by a shell or carapace:

# Identification of Family Salticidae



http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

- Belonging to the prosoma are the main appendages:
- In this diagram the chelicerae (jaws with fangs) are shown in red, the palpi in green, and the legs in blue.

# Principles of Systematics

## Module# 99

# Principles of Systematics

## Identifications of Family Oxyopidae

# Identification of Family Oxyopidae



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- Lynx spiders
- The common name refers to their quickness and agility; lynx-like in nature.
- Family Oxyopidae Thorell, 1870
- 9 genera worldwide, 419 species (WSC)

# Identification of Family Oxyopidae



https://www.google.com/search?q=lynx+spider&tbm=isch&ved=2ahUKEwiMyJWp07ftAhWO0YUKHcLSCQoQ2-cCegQIABAA&oq=lynx+spider&gs_lcp=CgNpbWcQA1DijCdY4ownYJCPJ2gAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=mOnLX4ycJ46jlwTCpadQ&bih=657&biw=1360#imgrc=Zf71CWvaO3PBXM&imgdii=gpTVaNJ4dzXzsM

- Oxyopid body length ranges from 3-25mm.
- Primarily tropical and subtropical in distribution, but also in temperate zones
- Primarily vegetation dwellers found on grasses, shrubs, and trees.

# Identification of Family Oxyopidae



https://www.google.com/search?q=lynx+spider&tbm=isch&ved=2ahUKEwiMyJWp07ftAhWO0YUKHcLSCQoQ2-cCegQIABAA&oq=lynx+spider&gs_lcp=CgNpbWcQA1DijCdY4ownYJCPJ2gAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=mOnLX4ycJ46jlwTCpadQ&bih=657&biw=1360#imgrc=Zf71CWvaO3PBXM&imgdii=gpTVaNJ4dzXzsM

- Feed on Insects and spiders
- Oxyopids are active hunters during both day and night.
- They use their vision to detect and legs to catch prey, sometimes jumping up to 2cm into the air to catch insects in flight

# Identification of Family Oxyopidae



https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=IC6Pj2HMuL3D0M%252CIuWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=IC6Pj2HMuL3D0M

- Some lynx's also capture prey by jumping from a stationary location much like a Salticid.
- An interesting member of *Tapinillus* is a web building, social lynx spider.
- Eight eyes

# Identification of Family Oxyopidae



https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=IC6Pj2HMuL3D0M%252CluWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=IC6Pj2HMuL3D0M

- Most species make little use of webs
- Instead spending their lives as hunting spiders on plants.
- Many species frequently visit flowers in particular, ambushing pollinators, much as crab spiders do
- Most of them have

# Identification of Family Oxyopidae



https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=IC6Pj2HMuL3D0M%252CluWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=IC6Pj2HMuL3D0M

- Oxyopidae in general rely on keen eyesight in stalking, chasing, or ambushing prey, and also in avoiding enemies.
- Active hunters/Chase prey
- Some members of the genus *Oxyopes* are abundant enough to be

# Principles of Systematics

**Module#**
**100**

# Principles of Systematics

## Identifications of Family Oxyopidae

# Identification of Family Oxyopidae

- High carapaces
- Distinctive eyes
- Numerous spines on their legs
- Bright colors in some species
- Some have cryptic coloration.

# Identification of Family Oxyopidae



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- The abdomen is usually oval shaped and tapers to a point.
- The long legged hunting spiders are capable of running very fast and jump on their prey like a cat. This is why their common name is lynx spider.
- Although their eyesight is not as good as that of the jumping spiders

# Identification of Family Oxyopidae



https://www.google.com/search?q=oxyopidae&sxsrf=ALeKk02Iear-tKcnT56n8kqrbe2HRLd-Ow:1607225520796&source=lnms&tbm=isch&sa=X&ved=2ahUKEwizofbRtbjtAhXBRBUIHRosD1gQ_AUoAXoECAUQAw&biw=1360&bih=657#imgrc=SOcXM8zGlxgFCM&imgdii=AhcO2CW_UUh5OM

- The male performs a dance before the female using it legs, abdomen and especially his palps.
- The female spider spins a cocoon that she attaches to a plant.
- She will guard the eggs until they hatch.
- Identified by the very

# Identification of Family Oxyopidae

END

- Their abdomen is like a pencil ending in a point.
- Some oxyopids may produce several egg sacs, while others only one.
- Lynx's are very protective of their egg sacs, guarding them avidly.

- Many will not eat while

# Principles of Systematics

**Module#**
**101**

# Principles of Systematics

## Identifications of Family Oxyopidae

# Identification of Family Oxyopidae

- Generalist predator that feeds on a wide variety of insect prey, including several economically important insect pests.

- This family is one of the most abundant beneficial spider species in gardens, yards, and agricultural fields throughout world

# Identification of Family Oxyopidae

- To capture prey, the striped lynx spider will sit and wait in grassy areas and herbaceous vegetation for small insects to land close enough for them to stalk and capture.

- This spider can detect odors given off by insect prey and will spend time in areas

# Identification of Family Oxyopidae

- There is evidence that spiderlings can learn specific odors during foraging and retain specific prey preferences from these experiences later in life.

- Newly hatched spiderlings can feed on nectar, which can help them live longer and reduce cannibalism

# Identification of Family Oxyopidae

- Other sources of sugars, including honeydew, may be potentially important to immature spiders

- The striped lynx spider is considered a major predator of bollworm and tobacco budworm larvae in cotton fields and the southern green stink bug in soybean

# Identification of Family Oxyopidae

END

- The striped lynx spider is significantly affected by insecticide use that causes decreased survival and other sublethal effects, such as causing males to decrease the amount of time they spend courting and mating with females

- Predatory potential and

# Principles of Systematics

**Module#**
**102**

# Principles of Systematics

## Identifications of Family Araneidae

# Identification of Family Araneidae

- The family Araneidae is a member of the order Araneae (Foelix, 1982).
- The Araneidae include the spiders who spin the elaborate webs most often seen.
- Araneids are one of the most successful of the spider families; they have over 3,500 species

# Identification of Family Araneidae

- The spiders vary greatly in size as well, some species of *Singa* are about 2 mm in length while some of the *Nephila* can grow to over 50 mm, even killing small birds.
- Because of this great variety of species a great variety of orb web designs exists.

# Identification of Family Araneidae

- Eight eyes
- Eyes are arranged in two horizontal rows
- Males are usually smaller than females
- Spiders in the family Araneidae will demonstrate several different strategies of predation.

# Identification of Family Araneidae

- Some will immediately wrap their prey in the web then bite
- Some wrap, move, then bite
- Some just seize their prey with their chelicerae, move them to the hub of the web, and then wrap up their prey.

# Identification of Family Araneidae

- Spiders are found in woods, caves, grasses and almost any other habitat imaginable
- Being bitten by an orb weaver is very uncommon
- Only bite if they feel threatened and trapped without a chance for escape
- A bite is often compared to a bee sting, and for most people, is nothing serious. Should be observe in their environments

END

# Principles of Systematics

**Module#**
**103**

# Principles of Systematics

## Identifications of Family Araneidae

# Identification of Family Araneidae

- Present anywhere with abundant prey and structures to support the web
- Most noticeable in late summer in fall, when webs and adults reach their largest size.
- Food sources vary, but typically any small insects they catch in

# Identification of Family Araneidae

- Some of the bigger orb weavers (Argiope genus) have been observed eating small frogs and humming birds (only if ensnared within the web).

- The adult males will wander in search of a mate, and as such, typically do not stay on a web for long, if they

# Identification of Family Araneidae

- The most commonly noticed orb weaver is female, since she sits on her web, feeding and eventually waiting for the males to find her.
- Toward the end of fall, the females will lay their last clutch of eggs, and then die at the first frost.
- The eggs can survive

# Identification of Family Araneidae

- The warmth of the spring will cause the eggs to mature, releasing a new generation of orb weavers.
- A single egg sac can contain upwards of several hundred eggs
- Juvenile orb weavers will spin "perfect" orb

# Identification of Family Araneidae

- Orb weavers are very docile, non-aggressive spiders that will flee at the first sign of a threat (typically they will run or drop off the web).

- They **are not dangerous** to people & pets, and are actually quite beneficial because they will catch and eat a lot of pest-

# Identification of Family Araneidae

- Feed on medium-sized insect in the web
- Spider bite the prey, wrap it in silk, wait for it to die, then begin to eat.
- As a first step in eating, the spider will literally vomit digestive fluid over the prey.
- The prey is chewed with the "jaws" (chelicerae), and the

# Identification of Family Araneidae

- The spider repeats this process as often as necessary to digest, and ingest, all but the inedible hard parts.
- What is discarded afterwards is a small ball of residue.

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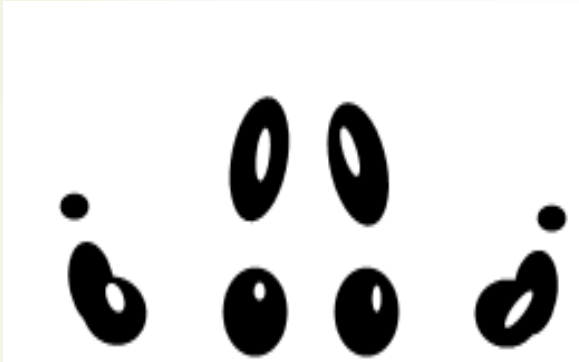
# Principles of Systematics

**Module#**
**104**

# Principles of Systematics

## Identifications of Family Araneidae

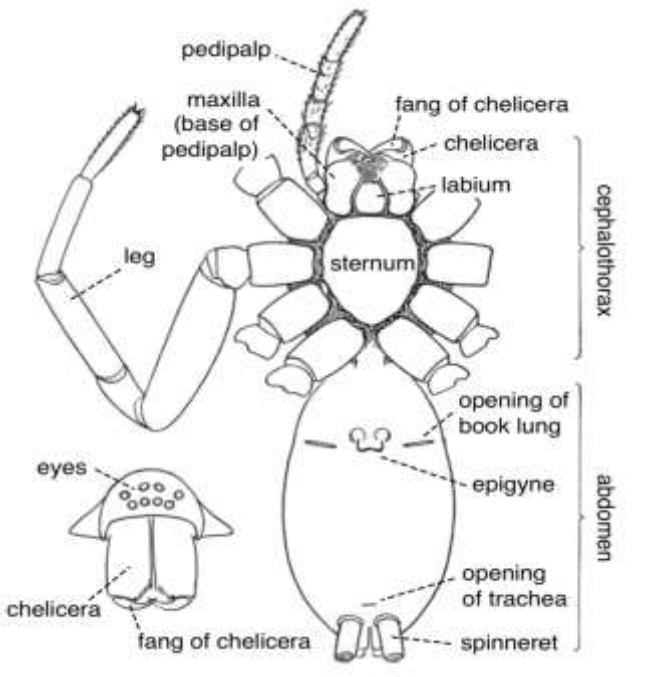
# Identification of Family Araneidae



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- Eight eyes
- Three tarsal claws
- The eyes in 2 rows of 4 with the lateral eyes widely separated from the medial eyes.
- The carapace is often flat with a distinct head region.

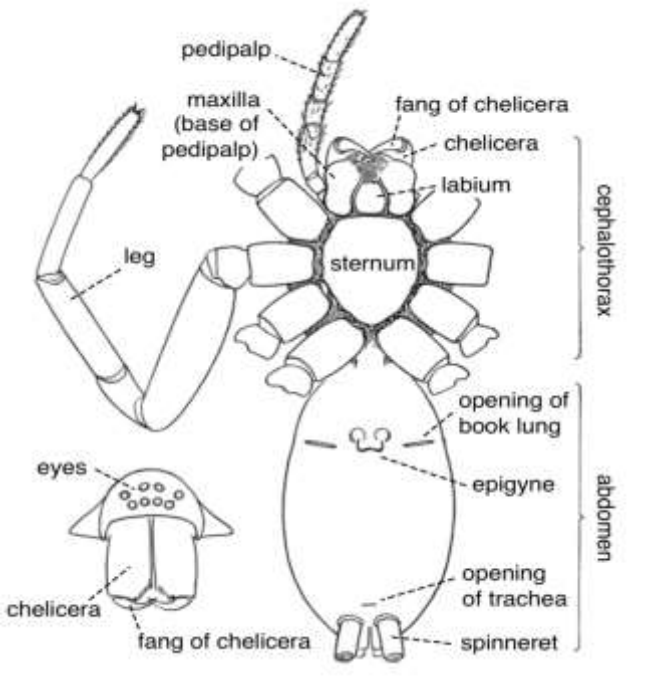
# Identification of Family Araneidae




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- Labium is wider than long and swollen at anterior edge
- Maxilla are widest anteriorly
- Legs with 3 claws and often furnished with strong spines and trichobothria on all segments except tarsi.

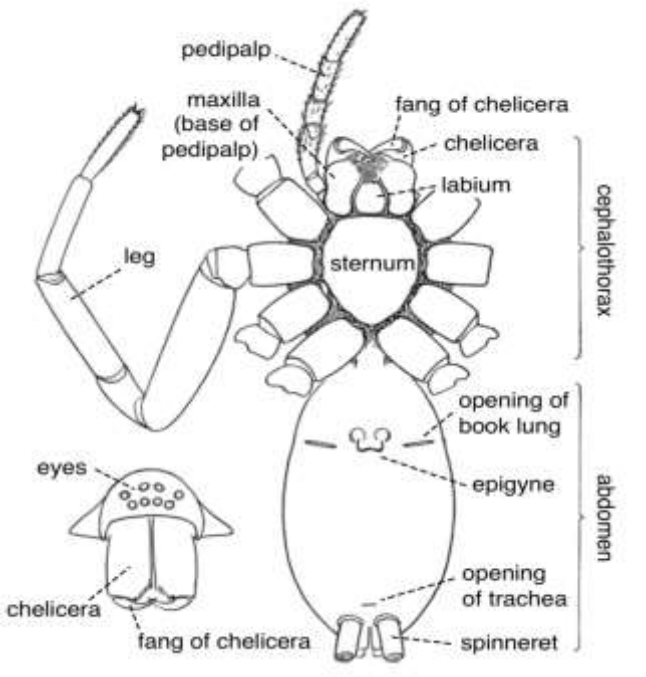
# Identification of Family Araneidae




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- Often legs are clearly annulated.
- The abdomen is usually globose and nearly always with species-specific often bright colour patterns.
- The tracheal spiracle is situated close to spinnerets.

# Identification of Family Araneidae




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- Neoscona
- Tetragnathidae
- Regular webs/Time of web making
- Repair/new web
- Common in agro-ecosystems
- Biocontrol agent

# Principles of Systematics

**Module#**
**105**

# Principles of Systematics

## Identifications of Family Lycosidae

# Identification of Family Lycosidae

- Wolf Spiders
- Active hunters
- Ground spiders
- Some species clime plant to chase prey
- Widespread from the Arctic to the Subtropics.

# Identification of Family Lycosidae

- Preferred habitat varies between species but includes open grasslands, suburban lawns, deciduous forests, deserts, coastal dunes, sandy soil and wet terrain such as marshes and swamps

# Identification of Family Lycosidae

- Preferred habitat varies between species but includes open grasslands, suburban lawns, deciduous forests, deserts, coastal dunes, sandy soil and wet terrain such as marshes and swamps

- Mainly feed on insects

# Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- Eight dark eyes of unequal size arranged in three rows, the first having four eyes.
- The abdomen and the cephalothorax are usually as long as wide.
- The long legs have three microscopic claws at each tip.

# Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- Nocturnal
- Their dark mottled colors help camouflage them among the leaves.
- Do not spin webs
- Some dig burrows in the ground, others make holes under rocks, and many have

# Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- They are robust and agile hunters with excellent eyesight.
- Do not spin webs
- Some dig burrows in the ground, others make holes under rocks, and many have no retreat at all.
- Live and hunt alone

# Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- The abdomen must be held in a raised position to keep the egg case from dragging on the ground.
- However, despite this handicap, they are still capable of hunting
- Another aspect unique to wolf spiders is their

# Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- Spiderlings emerge from their protective silken case
- Climbed up their mother's legs and crowd onto the dorsal side of her abdomen.
- The mother carries the spiderlings for several weeks before they are

# Identification of Family Lycosidae



https://bugguide.net/node/view/89986

- Wolf spiders inject venom if continually provoked.
- Symptoms of their bites include swelling, mild pain, and itching.
- American Wolf spider
- Australian wolf spiders have also been associated with necrotic wounds, but careful study has

# Principles of Systematics

**Module#**
**106**

# Principles of Systematics

## Identifications of Family Lycosidae

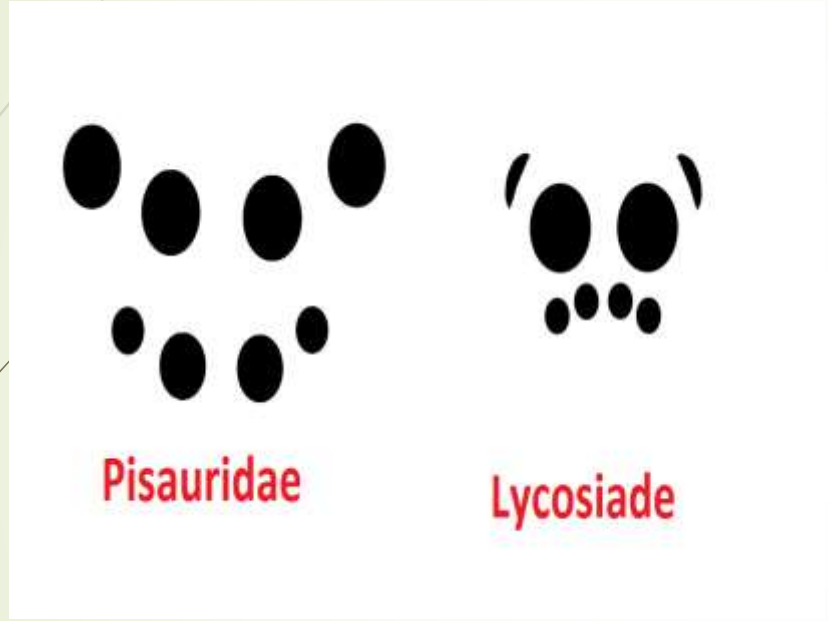
# Identification of Family Lycosidae



https://www.biodiversityexplorer.info/arachnids/spiders/lycosidae/index.htm

- They are robust and agile hunters with excellent eyesight
- They live mostly in solitude and hunt alone
- Wolf spiders resemble Nursery Web Spiders (family Pisauridae), but wolf spiders carry their egg sacs by attaching them to their spinnerts, while the Pisauridae carry their egg sacs with Chelicerae and

# Identification of Family Lycosidae




https://bugguide.net/node/view/89936

- Two of the wolf spider's eight eyes are large and prominent, which distinguishes them from the nursery web spiders whose eyes are all of roughly equal size.
- 10-35 mm size
- Unlike most other arachnids, which are generally blind or have poor vision, wolf spiders have excellent eyesight.

# Identification of Family Lycosidae

- *Hogna* is the genus with the largest of the wolf spiders
- Found everywhere
- Body colour is brown and dull so inconspicuous on ground
- Collector notice them when spiders are disturbed by ground vibration

# Identification of Family Lycosidae

- Do not spin web
- *Hippasa* and *Sosippus* are seen build funnel retreat, outside of which expended into sheet web
- These spiders wait for prey
- Prey into sheet/spiders come out/bite/kill/suck prey

# Identification of Family Lycosidae

- No particular location of these webs on ground
- Danger/Extreme end of tube
- Morning travelling on vehicle/Train
- *Hippasa* webs look very conspicuous due to deposition of dew drops on them.

**END**

# Principles of Systematics

**Module#**
**107**

# Principles of Systematics

## Identifications of Family Lycosidae

# Identification of Family Lycosidae

- Members of genus *Lycosa* inhabit varied habitat
- Found on dry ground
- Under stones/Logs/ in crevices
- Seen in large numbers near heaps of cow dung, kept for drying

# Identification of Family Lycosidae

**Why near cow dung:**

- For preying on insects like Diptera which come to lay their eggs in cow dung
- Other insects are attracted by smell.
- Seen at night chasing their prey and occasionally coming in human habitations for food search.

# Identification of Family Lycosidae

- Some members of Lycosa prefer
- Cool, moist cavities under stone, where they lie in day time.
- *Paradosa* and *Aractosa* prefer humid atmosphere
- Frequently found at the edes of ponds, lakes, rivers etc.

# Identification of Family Lycosidae

- *Paradosa birmanica. Paradosa samatrana* common near water source.
- When disturb, they hide under stones and soil crevices
- Run on water surface
- 
- Jump in water to chase prey

END

# Principles of Systematics

**Module#**
**108**

# Principles of Systematics

## Identifications of Family Lycosidae

# Identification of Family Lycosidae

- *Lycosids* mainly feed on insects
- Diptera
- Lepidoptera
- Hymeniptera
- Coleoptera
- Collembola
- Dermaptera etc

# Identification of Family Lycosidae

- May consume young spiders of their on species
- As well as different species
- Cannibalistic/importanc e
- Feed at night
- Feeding mechanism

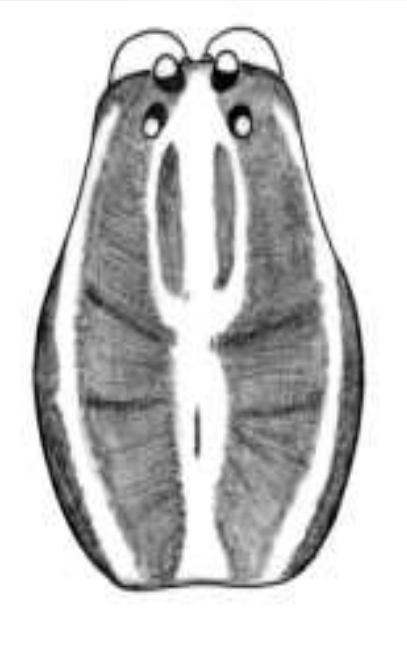
# Identification of Family Lycosidae

- The family name and common name are derived from the Greek word "lycosa" meaning "wolf" due to the spiders' hunting method of ambushing and running down its prey.
- Research has shown that the Lycosidae are important in agriculture, as they are efficient controlling agents of insect pests. They are harmless to man.

# Identification of Family Lycosidae

- Lycosids are often parasitized by wasps probably because they are free roaming and do not enjoy the protection of a web.

# Identification of Family Lycosidae



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- Centre of thoracic region is provided with fovea.
- Male has enlarged swollen palp
- Female genetalia is covered externally by heavey sclerotized plate called epigynal plate

# Identification of Family Lycosidae

- Centre of thoracic region is provided with fovea.
- Male has enlarged swollen papl
- Female genetalia is covered externally by heavey sclerotized plate called epigynal plate
- Structure of genetalia is species specific.

# Principles of Systematics

**Module#**
**109**

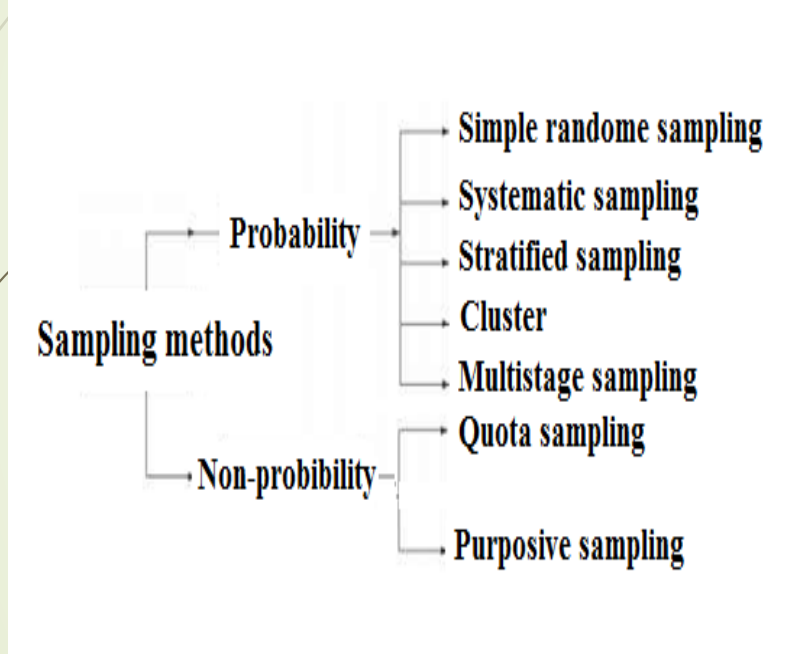
# Principles of Systematics

## Sampling Methods

# Sampling Methods

- Probability Sampling
- Non-Probability Sampling

# Sampling Methods




## Probability Sampling

- Random Sampling
- Systematic Sampling
- Stratified sampling
- Cluster Sampling
- Multistage Sampling

# Sampling Methods

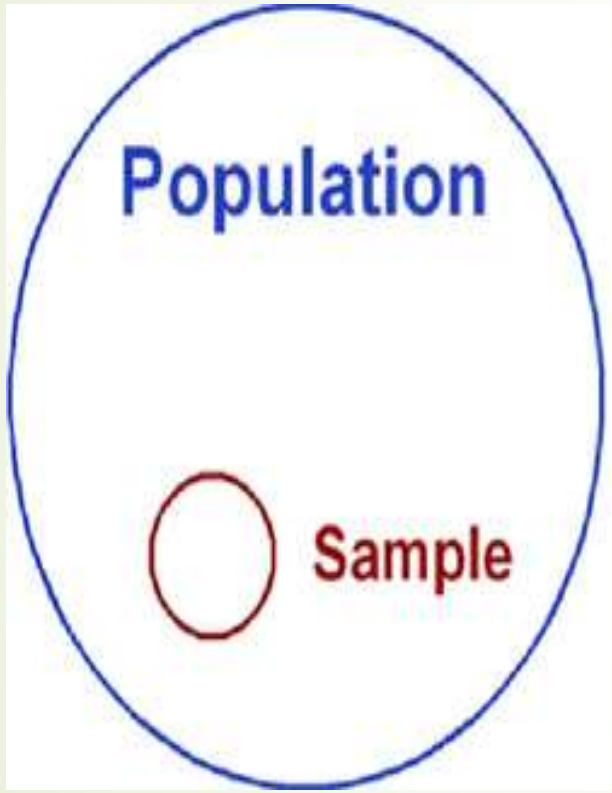
## Non-Probability Sampling

- Purposive sampling
- Quota Sampling

# Sampling Methods

- **Sampling:** It is the process of selecting a representative group from the population. The target population may be as board as all humans but it may be a smaller group as people who are diabetics or only the teenagers.

# Sampling Methods




- **Sample:** A sample is a group of units selected from a population by a defined procedure or simply a part of a statistical population.

- The sample should be representative of the general population. 5-10% portion of the sample size of total population is used for research.

# Sampling Methods

- **Population:**
  Population is the total group of individuals from which the sample might be drawn.
- All members of population possess the same basic and defined characteristics, but varying in amount or quality from one member to another. For example, rice plants in field.

END